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ESG/CSR MEASURES IMPACTING FINANCIAL PERFORMANCE: AN ANALYSIS OF MODERATING VARIABLES

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Thesis submitted to the Graduate Program in Business Administration at Mackenzie Presbyterian University, as a partial requirement for the attainment of the Doctor of Business Administration degree.

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"A good theory is one that holds together long enough to get you to a better one."

— Merton Miller

ABSTRACT

This thesis explores the complex relationship between Environmental, Social, and Governance (ESG) and Corporate Social Responsibility (CSR) initiatives and Corporate Financial Performance (CFP), combining a comprehensive systematic review with empirical analysis in emerging markets. The first article conducts a bibliometric and systematic review of 108 peer-reviewed studies published between 2019 and 2023, aiming to identify the primary moderating variables, as well as those that remain underexplored, within the ESG/CSR–corporate financial performance (CFP) nexus. Additionally, the study examines financial performance metrics, theoretical foundations, ESG and CSR indicators, and econometric methodologies, highlighting critical gaps that inform a future research agenda. Building upon these insights, the second article investigates how ESG disclosure and performance affect financial outcomes in 2,987 firm-year observations from BRICS economies between 2016 and 2023. Using a Two-Stage Least Squares approach to address endogeneity, the results indicate that integrated ESG disclosure enhances profitability and investor confidence, while excessive or selective engagement may diminish firm value. The findings suggest that in emerging markets, sustainability-driven financial performance depends more on effective signaling than on operational depth, raising questions about authenticity, greenwashing, and strategic alignment. Together, the two studies contribute to a nuanced understanding of how sustainability initiatives shape financial outcomes across varying institutional and economic contexts, mostly in emerging countries.

RESUMO

Esta tese investiga a complexa relação entre iniciativas de meio ambiente, social e governança (ESG) e responsabilidade social corporativa (CSR) e o desempenho financeiro corporativo (CFP), por meio de uma revisão sistemática abrangente e de uma análise empírica aplicada a mercados emergentes. O primeiro artigo realiza uma revisão bibliométrica e sistemática de 108 estudos revisados por pares, publicados entre 2019 e 2023, com o objetivo de identificar as principais variáveis moderadoras, bem como aquelas ainda pouco exploradas, na relação entre ESG/CSR e o desempenho financeiro corporativo. Adicionalmente, o estudo examina métricas de desempenho financeiro, fundamentação teórica, indicadores de ESG e CSR, e metodologias econométricas, destacando lacunas críticas que fundamentam uma agenda futura de pesquisa. Com base nessas lacunas, o segundo artigo investiga como a divulgação e o desempenho em ESG afetam os resultados financeiros em 2.987 observações de empresas dos países do BRICS entre 2016 e 2023. Utilizando o método de Mínimos Quadrados em Dois Estágios (2SLS) para tratar a endogeneidade, os resultados indicam que a divulgação integrada de ESG melhora o retorno do acionista e o valor da firma, enquanto o engajamento excessivo ou seletivo pode reduzir o valor da empresa. Os achados sugerem que, em mercados emergentes, o desempenho financeiro orientado pela sustentabilidade depende mais de um sinal eficaz ao mercado do que de uma eficiência em práticas sustentáveis concretas, levantando questões sobre autenticidade, greenwashing e alinhamento estratégico. Em conjunto, os dois estudos contribuem para uma compreensão de como iniciativas de sustentabilidade moldam os resultados financeiros em diferentes contextos institucionais e econômicos, sobretudo em países emergentes.

LIST OF FIGURES

Figure 1 - PRISMA flow diagram for selection of studies. 41
Figure 2 - Systematic literature review graphical abstract. 53
Figure 3 - Annual scientific production and average citation 54
Figure 4 - Graphic resume of Bradford's Law 57
Figure 5 - Bibliographic coupling maps by sources (Journals) 58
Figure 6 - Author productivity through Lotka's Law 60
Figure 7 - Bibliographic coupling articles map 61
Figure 8 - Keyword Map 65
Figure 9 - Moderating variables frequency count 69
Figure 10 - Percentage distribution of financial performance categories 75
Figure 11 - ESG and CSR measures gaps 79
Figure 12 - Percentage distribution of moderating variables categories 81
Figure 13 - Percentage distribution of theoretical framework categories 85
Figure 14 - Percentage distribution of suggestions for future research categories 87
Figure 15 - Percentage distribution of statistical modeling categories 89
Figure 16 - Percentage distribution of data sources of ESG and CSR categories 92
Figure 17 - Graphic resume of the suggestions for a research agenda 94
Figure 18 - Conceptual Distinction between Holistic and Selective ESG Strategies 108
Figure 19 - Curvilinear Relationship between ESG Disclosure and Financial Performance (Models 3.1–3.4) 127
Figure 20 - Curvilinear Relationship between ESG Performance and Financial Performance (Models 3.5–3.8) 128
Figure 21 - Curvilinear Interaction between ESG Performance² and ESG Disclosure² (Models 6.5–6.8) 135

LIST OF TABLES

Table 1 - Classification and Coding Framework 48
Table 2 - Annual Scientific Production over the Years and Average Citations per Year..... 54
Table 3 - Bradford's Law Zone Distribution 58
Table 4 - Annual Scientific Production and Average Citations Per Year 58
Table 5 - Author´s Productivity through Lotka's Law 59
Table 6 - Moderating Variable Classification 67
Table 7 - Categorized Articles based on the Classification and Coding Framework 70
Table 8 - Resume of the Gaps identified 93
Table 9 - Industry Sample Distribution 112
Table 10 - Country Sample Distribution 113
Table 11 - Descriptive Statistics 114
Table 12 - Variable Description 115
Table 13 - Instrumental Variable Diagnostics and Validity Assessment 118
Table 14 - Impact of Environmental, Social and Governance Score on Financial Performance - 2SLS Regression Results. (Model 1.1 to 1.8). 121
Table 15 - Impact of Environmental, Social and Governance Disclosure on Financial Performance (Model 2.1 to 2.8). 123
Table 16 - Curvilinear Effect of ESG Disclosure and ESG Performance on Financial Performance (Model 3.1 to 3.8). 125
Table 17 - Impact of Selective Environmental, Social and Governance Disclosure on Financial Performance - 2SLS Regression Results. (Model 4.1 to 4.12). 130
Table 18 - Impact of Selective Environmental, Social and Governance Performance on Financial Performance - 2SLS Regression Results. (Model 5.1 to 5.12). 131
Table 19 - Interact Effect (Linear and Curvilinear) of ESG Disclosure and ESG Performance on Financial Performance (Model 6.1 to 6.8). 133
Table A1 - Results of the Durbin–Wu–Hausman Endogeneity Test for ESG Disclosure and Performance Models 195
Table A2 - Reverse Causality Regressions: Financial Performance → ESG Performance and ESG Disclosure 195
Table A3 - Results of Fisher-Type Panel Unit Root Tests (ADF) Using Pm Statistic. 197

LIST OF ABBREVIATIONS AND ACRONYMS

BRICS: Brazil, Russia, India, China, and South Africa

CDP: Carbon Disclosure Project

CFP: Corporate Financial Performance

CSP: Corporate Social Performance

CSR: Corporate Social Responsibility

CSRD: Corporate Social Responsibility Disclosure

DDF: Directional Distance Function

DEA: Data Envelopment Analysis

DID: Difference-in-Differences

ESG: Environmental, Social, and Governance

FDI: Foreign Direct Investment

GMM: Generalized Method of Moments

GRI: Global Reporting Initiative

ICW: Internal Control Weaknesses

IV: Instrumental Variable

MSCI: Morgan Stanley Capital International

OECD: Organization for Economic Cooperation and Development

OLS: Ordinary Least Squares

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

R&D: Research and Development

RBV: Resource-Based View

ROA: Return on Assets

ROE: Return on Equity

SASB: Sustainability Accounting Standards Board

TABLE OF CONTENTS

1 GENERAL INTRODUCTION 17

2 ARTICLE 1 - REVISITING KNOWLEDGE ON ESG/CSR AND FINANCIAL PERFORMANCE: A BIBLIOMETRIC AND SYSTEMATIC REVIEW OF MODERATING VARIABLES 18

2.1 ABSTRACT 18

2.2 INTRODUCTION 19

2.3 THEORETICAL FRAMEWORK 21

2.3.1 Shareholder theory 21

2.3.2 Stakeholder theory 24

2.3.3 Agency theory 27

2.3.4 Resource-based view theory 30

2.3.5 Legitimacy theory 32

2.3.6 Institutional theory 34

2.4 METHODOLOGY 37

2.4.1 Sample selection 38

2.4.2 Bibliometric analysis 40

2.4.2.1 Trend analysis 42

2.4.2.2 Bradford's law and bibliographic coupling analysis 42

2.4.2.3 Lotka's law 45

2.4.2.4 Keyword analysis 46

2.4.3 Systematic review 46

2.5 RESULTS AND DISCUSSION 54

2.5.1 Trend analysis 54

2.5.2 Bradford's law and bibliographic coupling analysis 56

2.5.3 Keyword analysis (Co-Occurrence analysis) 63

2.5.4 Moderating variables analysis 65

2.6 SYSTEMATIC REVIEW 70

2.6.1 Financial performance research gaps 75

2.6.2 ESG and CSR Measures 77

2.6.3 Moderating variables category analysis 80

2.6.4 Theoretical framework research gaps 83

2.6.5 Significant gaps to be explored in the ESG/CSR-CFP research 86

2.6.6 Statistical modeling research gaps 88

2.6.7 Data Sources research gaps 90

2.7 LIMITATIONS, IMPLICATIONS, AND CONCLUSION 94
2.7.1 Research limitations 94
2.7.2 Theoretical implications 95
2.7.3 Practical implications 96
2.7.4 Social implications 96
2.8 CONCLUSION 97
2.9 COMBINING RESEARCH GAPS FOR EMPIRICAL EXPLORATION 100
3 ARTICLE 2 - THE ESG DISCLOSURE–PERFORMANCE PARADOX IN BRICS: EVIDENCE FROM SHAREHOLDER RETURNS AND FIRM VALUE 101
3.1 ABSTRACT 101
3.2 INTRODUCTION 102
3.3 THEORETICAL BACKGROUND AND HYPOTHESES DEVELOPMENT 105
3.3.1 Theoretical Perspectives on Sustainability, Disclosure, and Performance . 105
3.3.2 Corporate Sustainability Framework 107
3.3.3 The Financial Effects of Holistic versus Isolated ESG Strategies 109
3.3.4 The Financial Impact of Selective ESG Strategies 110
3.3.5 Curvilinear Effects of ESG Engagement 110
3.3.6 ESG Disclosure–Performance Interaction 111
3.4 METHODOLOGY 112
3.4.1 Sample and data sources 112
3.4.2 Measurements 113
3.4.3 Variable construction and transformation 114
3.4.4 Econometric model 116
3.5 RESULTS 120
3.5.1 ESG Performance and Financial Metrics 120
3.5.2 ESG Disclosure and Financial Performance 122
3.5.3 Curvilinear Effect of ESG on Financial Metrics 124
3.5.4 Selective ESG Dimensions and Financial Performance 128
3.5.5 Interaction Between ESG Disclosure and Performance 132
3.6 DISCUSSION AND CONCLUSION 136
3.6.1 Synthesis of the Findings 136
3.6.2 Contributions to Theory 139
3.6.3 Contributions to Practice 140
3.6.4 Research Limitations and Future Research Agenda 142
4 FINAL CONSIDERATIONS 143

REFERENCES **146**
APPENDIX A – ADDITIONAL STATISTICAL TABLES **182**

1 GENERAL INTRODUCTION

Over the past two decades, corporate sustainability has transitioned from a voluntary initiative to a strategic imperative. The adoption of Environmental, Social, and Governance (ESG) practices and Corporate Social Responsibility (CSR) initiatives has become essential for firms seeking to ensure long-term resilience, meet regulatory expectations, and strengthen competitive positioning. This shift reflects growing societal and investor pressure for transparency and accountability, alongside recognition that sustainability is integral to financial and operational performance.

Despite this progress, academic literature remains divided regarding the relationship between ESG/CSR and corporate financial performance (CFP). Some studies report positive associations, while others highlight neutral or even negative effects, due in part to methodological inconsistencies, the multidimensional nature of financial outcomes, and the influence of contextual variables such as governance, industry, and geography. Moreover, standardization of sustainability metrics remains a challenge, complicating efforts to compare results across studies and regions.

This thesis addresses these gaps through two complementary articles. The first article presents a systematic and bibliometric review of the ESG/CSR–CFP literature, with a particular focus on moderating variables that shape this relationship. It identifies key research clusters, theoretical foundations, and significant gaps that constrain the development of a cohesive knowledge base. Among the most salient challenges are the lack of standardized ESG indicators, limited attention to longitudinal effects, and the underexplored role of contextual moderators such as governance quality, regulatory frameworks, and economic conditions.

Building upon these insights, the second article empirically investigates the interplay between sustainability disclosure and performance in BRICS economies—Brazil, Russia, India, China, and South Africa. These countries represent a relevant context due to their institutional heterogeneity, regulatory fragmentation, and increasing exposure to global sustainability norms. Drawing on agency, stakeholder, legitimacy, and signaling theories, the article examines whether sustainability efforts—both disclosed and implemented—enhance firm-level financial outcomes, and whether excessive disclosure may backfire in weak institutional environments.

By combining a broad, theory-driven review with a context-specific empirical analysis, this thesis contributes to a more nuanced understanding of the ESG/CSR–CFP relationship. It highlights the importance of distinguishing between symbolic and substantive sustainability strategies, especially in emerging markets where institutional voids often distort incentives.

Furthermore, it offers insights for scholars and practitioners interested in designing sustainability strategies that are not only aligned with stakeholder expectations but also capable of generating durable economic value.

The thesis is structured as follows. Chapter 1 provides this general introduction. Chapter 2 presents the first article—a systematic and bibliometric review of ESG/CSR and financial performance literature. Chapter 3 presents the second article—an empirical investigation of the disclosure–performance paradox in BRICS economies. The final chapter offers general conclusions, theoretical and practical implications, and recommendations for future research.

2 ARTICLE 1 - REVISITING KNOWLEDGE ON ESG/CSR AND FINANCIAL PERFORMANCE: A BIBLIOMETRIC AND SYSTEMATIC REVIEW OF MODERATING VARIABLES

2.1 ABSTRACT

This study conducts a comprehensive bibliometric analysis and systematic literature review to examine the moderating variables that influence the relationship between Environmental, Social, and Governance (ESG), Corporate Social Responsibility (CSR), and Corporate Financial Performance (CFP). Based on a review of 108 peer-reviewed articles published between 2019 and 2023 in the Web of Science and Scopus databases, the study identifies 26 categories of moderating variables. The findings reveal a predominant focus on internal governance mechanisms and firm-level attributes. The study identifies twenty key research gaps (G1–G20), highlighting the need for future investigations to incorporate overlooked financial and intangible indicators, disaggregated ESG components, and underexplored contextual moderators. It calls for broader theoretical integration—drawing from economic, innovation, and behavioral frameworks—and the use of more robust methodologies, including advanced econometrics, content analysis, and diverse data sources. This review offers a structured agenda to advance research on the ESG/CSR–CFP nexus and enrich the discourse on corporate sustainability and financial performance.

Keywords: Environmental, social, and governance, Corporate social responsibility, Firm value, Corporate financial performance, Moderating variables, Systematic review, Bibliometric analysis, Research agenda.

2.2 INTRODUCTION

Sustainability has evolved from a peripheral concern to a pivotal element shaping corporate strategies in today's business landscape. Environmental, Social, and Governance (ESG) disclosures and Corporate Social Responsibility (CSR) engagement — now considered essential — play a critical role in ensuring transparency, enhancing financial performance, and securing long-term competitive advantages (Ellili, 2022; Sanchez García, 2018). Moreover, companies are increasingly adopting green finance to comply with regulatory requirements and boost their competitiveness and resilience in a rapidly changing global environment (Khan et al., 2024; Lyulyov et al., 2024). In tandem, sustainable business models enable firms to better adapt to market dynamics and global challenges, offering strategic advantages and fostering innovation (Bashir et al., 2022). As sustainability becomes embedded in core operations, improvements in operational efficiency, risk management, and stakeholder relationships drive long-term profitability and growth (Saini et al., 2023; Coelho et al., 2023).

The widespread adoption of ESG principles and CSR initiatives signifies a critical shift in how corporations address global challenges such as climate change, social inequality, and ethical governance. These frameworks, once considered optional, are now integral to corporate strategies, with firms recognizing their importance in building resilience, maintaining a competitive edge, and ensuring financial sustainability while addressing societal and environmental concerns (Sun et al., 2022). Research exploring the relationship between ESG/CSR initiatives and corporate financial performance (CFP), however, faces numerous challenges due to the inconsistencies and contradictory findings across studies. Some findings highlight the positive effects of ESG/CSR on financial outcomes, while others report neutral or even negative impacts (Barnett & Salomon, 2012; Friede et al., 2015).

One significant challenge in this research field is the lack of standardization in ESG/CSR metrics. Different data providers (e.g., MSCI and Sustainalytics) employ varying methodologies, making cross- comparisons difficult (Gillan et al., 2021). Compounding these challenges is the multi-dimensional nature of financial performance, which includes profitability, market performance, and risk management, each of which may be differently affected by sustainability efforts (Orlitzky et al., 2003; Alshehhi et al., 2018). Furthermore, the indirect benefits of ESG and CSR, such as improved stakeholder trust or corporate reputation, may take years to materialize and are difficult to quantify, complicating the assessment of the true value of these initiatives (Bar- nett & Salomon, 2012; Lins et al., 2017).

The underexplored role of moderating variables further complicates the analysis. Governance structures, industry characteristics, and regional factors can significantly influence the relationship between ESG/CSR and financial performance (Ye et al., 2021; Gillan et al., 2021). For example, companies with strong governance frameworks are more likely to successfully integrate sustainability into their core strategies, thereby enhancing financial outcomes (Chen et al., 2024). Additionally, economic conditions play a critical role in shaping the financial impacts of ESG efforts, with varying results depending on whether firms operate in stable or volatile markets (Flammer, 2018; Lins et al., 2017). The interaction between these moderating factors and ESG/CSR initiatives highlights the need for more refined studies that consider context-specific conditions (Raman et al., 2024).

In addition, the lack of long-term studies and the over-reliance on cross-sectional data limit our understanding of the evolving impacts of ESG on financial performance (Orlitzky et al., 2003; Margolis et al., 2009). Consequently, longitudinal research is essential to capture the lasting effects of sustainability efforts and better understand how these initiatives influence firm resilience and financial outcomes over time (Eccles et al., 2014; Brammer et al., 2006; Gillan et al., 2021). Standardized metrics, advanced methodologies, and deeper exploration of moderating variables are crucial to overcoming these challenges and fully grasping the complexities of ESG/CSR's impact on financial performance.

The debate around ESG and CSR's financial effects is primarily anchored in theoretical frameworks such as Stakeholder Theory and Agency Theory, which provide insights into how companies balance competing interests (Freeman, 1984; Jensen & Meckling, 1976). While Stakeholder Theory underscores the importance of considering the interests of employees, customers, and communities, it falls short of fully explaining how cultural, social, and institutional factors moderate the relationship between ESG and financial performance (Laique et al., 2023). These complexities are further compounded by varying levels of regulation and public scrutiny across industries and regions (Lyulyov et al., 2024), which affect how firms prioritize sustainability.

The importance of financial sustainability must also be contextualized within broader discussions around innovation and technology. As global challenges such as climate change, social inequality, and resource depletion intensify, businesses are increasingly pressured to balance financial goals with their social and environmental responsibilities (Orlitzky et al., 2003). Firms that successfully integrate sustainability into their core business strategies, especially through technological innovations, are better positioned to create long-term value for both shareholders and stakeholders (Yang et al., 2024). This alignment be- tween financial

sustainability and social/environmental objectives is crucial for ensuring resilience and competitiveness in the face of global challenges (Raman et al., 2024).

While ESG and CSR frameworks offer substantial potential to enhance financial performance, their impact is shaped by a complex web of direct and indirect effects moderated by governance, industry, and market-specific factors (Ye et al., 2021). The full scope of these effects is further complicated by the lack of standardized ESG metrics, the underexplored role of moderating variables, and the multi-dimensional nature of financial performance. To address these challenges, this paper conducts a systematic review and bibliometric analysis, highlighting how ESG and CSR initiatives influence financial outcomes and identifying key research gaps.

2.3 THEORETICAL FRAMEWORK

The relationship between ESG and CSR factors converting into CFP is complex and multifaceted. Numerous empirical studies demonstrate varying positive, neutral, or negative outcomes depending on a range of moderating variables, including CEO characteristics, governance structures, industry type, and market conditions. Understanding the role of these moderating variables is essential to explain the conflicting results found in the literature This theoretical framework will explore several fundamental theories, such as Shareholder Theory, Stakeholder Theory, Agency Theory, Resource-Based View (RBV), and Institutional Theory, while emphasizing the critical role of moderating variables in shaping the impact of ESG and CSR initiatives on financial performance.

2.3.1 Shareholder theory

Milton Friedman's (1970) Shareholder Theory asserts that the foremost duty of a corporation is to maximize shareholder value, casting initiatives such as ESG and CSR as secondary unless they directly contribute to profitability. This perspective — rooted in prioritizing short-term financial gains — traditionally regards sustainability efforts with skepticism, suggesting they impose unnecessary costs. However, as the corporate landscape evolves, recent empirical studies offer a more nuanced view, revealing that under certain circumstances, ESG initiatives can align with the long-term interests of shareholders. The theory's interpretation is, therefore, shifting as moderating factors such as governance structures and market conditions redefine the potential value of ESG activities within this framework.

According to Shareholder Theory, ESG and CSR initiatives are often viewed as diversions from the core purpose of maximizing profit. Critics argue that such activities can

impose unnecessary costs on firms, reducing their competitive edge. For instance, investing in costly sustainability projects can reduce short-term profits and create in- efficiencies. Barnea and Rubin (2010)) argue that CSR initiatives are often driven by managerial motives such as reputation building rather than genuine financial incentives, leading to agency costs and misalignment between managers and shareholders. Additionally, Hart and Zingales (2017) suggest that in firms with concentrated ownership, major shareholders exert pressure to prioritize short-term profits over long-term sustainability goals, disincentivizing ESG initiatives unless they provide immediate financial returns.

Furthermore, evidence shows that ESG initiatives do not always align with financial performance. Albuquerque et al. (2020) have found that during periods of economic downturn, firms often deprioritize ESG activities to preserve financial performance, suggesting that in volatile markets, ESG may not be seen as a value-enhancing strategy. Similarly, Peloza and Shang (2010) argue that the effectiveness of CSR in creating value varies widely across different industries and stakeholder groups, with some firms seeing little to no financial return from these initiatives. In certain cases, ESG and CSR activities may even reduce financial performance by increasing operational costs and diverting resources from core profit-generating activities. In highly competitive industries with low profit margins, the additional costs of ESG compliance can strain financial resources, leading to lower profitability. Companies with substantial environmental or social obligations may find themselves at a disadvantage compared to competitors that do not engage in such practices. This view is supported by empirical findings that suggest ESG's impact on financial performance is not universally positive (Khan et al., 2024), and in industries with low public visibility, the financial benefits of ESG are often minimal.

However, recent studies also provide evidence that ESG initiatives can align with shareholder interests under specific conditions. For example, Ghosh and Gupta (2023) demonstrated that decarbonization strategies, as part of broader ESG efforts, contribute positively to financial performance, particularly in industries under significant regulatory scrutiny. This suggests that ESG activities, when aligned with external pressures such as regulation or consumer expectations, can serve shareholder interests by mitigating risks and enhancing reputation. Similarly, Saini et al. (2023) argue that ESG practices can lead to lower capital costs, improved risk management, and enhanced operational efficiency, but these benefits vary significantly across industries and firm sizes, reinforcing the notion that the financial impact of ESG is context-dependent.

Ownership structure plays a crucial role in moderating the financial outcomes of ESG and CSR initiatives. In firms with dispersed ownership and stronger governance mechanisms, ESG initiatives are more likely to be embraced as part of a long-term strategy for value creation (Coelho et al., 2023). Such firms, driven by long-term institutional investors, may be more willing to invest in sustainability projects, recognizing that these initiatives can enhance reputation and build customer loyalty over time, thus, benefiting shareholder value in the long run.

Similarly, CEO characteristics play a significant role in shaping how ESG initiatives align with shareholder value. CEOs with a background in sustainability or those who personally champion long-term goals are more likely to integrate ESG practices into their firm's strategic vision, leading to better alignment with shareholder value over time (Shen et al., 2019). Rousseau et al. (2023) have found that firms with sustainability-oriented CEOs are better able to balance short-term profitability with long-term ESG benefits, suggesting that leadership commitment is essential for driving the success of ESG initiatives. In contrast, CEOs who prioritize immediate financial returns may resist ESG investments, viewing them as costly distractions from profit-maximizing activities.

While Shareholder Theory traditionally prioritizes profit maximization, conflicting evidence suggests that the relationship between ESG and financial performance is contingent on several moderating factors. Market conditions, for example, play a significant role in shaping this relationship. During periods of market growth, firms that invest in ESG may see positive returns as consumers and investors increasingly value sustainability. Conversely, in periods of economic downturn, ESG initiatives may be viewed as non-essential, leading to reduced financial commitment (Enciso-Alfaro & García-Sanchez, 2024).

Board diversity is another critical moderating factor. Enciso-Alfaro and García-Sanchez (2024) have found that firms with a higher pro- portion of female executive directors are more proactive in driving climate innovation, which can enhance financial performance. However, they also identified a threshold effect, where the positive impact of female representation diminishes after a certain point, suggesting that governance structures must be carefully balanced to optimize the financial returns from ESG activities.

Industry-specific dynamics further complicate the relationship between ESG and financial performance. Firms in industries with high public visibility and regulatory oversight, such as consumer goods or finance, are more likely to benefit from ESG activities (Ellili, 2022). In these sectors, ESG practices serve not only to mitigate risks but also to enhance legitimacy, aligning with shareholder interests by managing reputational risks and maintaining public trust.

In contrast, firms in industries with less exposure to environmental or social risks may not experience the same financial benefits, as ESG activities may not be as integral to their strategic objectives (Khan et al., 2024).

While Shareholder Theory has traditionally been critical of ESG and CSR initiatives, empirical evidence presents a more nuanced view. Under certain conditions, such as strong governance, leadership commitment, and industry-specific pressures, ESG activities can align with shareholder interests and enhance financial performance. However, the financial benefits of ESG are highly contingent on contextual factors such as market conditions, ownership structure, and regulatory environment. These conflicting findings highlight the importance of understanding the moderating variables that shape the relationship between ESG initiatives and financial performance, suggesting that the success of such initiatives is far from universal and often depends on the broader market and governance context.

2.3.2 Stakeholder theory

Stakeholder Theory, introduced by Freeman (1984), expands corporate responsibility beyond shareholders to include various stake- holders such as employees, customers, suppliers, communities, and the environment. This theory posits that creating value for all stakeholders, rather than focusing solely on shareholders, leads to long-term financial success. ESG and CSR initiatives naturally align with this framework as they seek to address the needs and concerns of a wide array of stake- holder groups. However, the financial impact of these initiatives is influenced by several moderating variables, leading to diverse outcomes across different contexts.

Recent empirical evidence supports the positive link between ESG practices and financial performance, particularly when addressing stakeholder needs. Saini et al. (2023) have found that ESG practices are associated with lower capital costs and improved risk management, especially in firms that actively engage with diverse stakeholder groups. This aligns with the idea that addressing stakeholder concerns can reduce risks and enhance operational efficiencies, leading to better financial outcomes. However, the study also highlights that these benefits are not uniform across all industries, suggesting that the influence of ESG on financial performance depends heavily on industry-specific dynamics.

Board composition is a critical moderating variable affecting the relationship between ESG activities and financial performance through Stakeholder Theory. Firms with more diverse boards — particularly those with higher levels of gender diversity — tend to manage

stakeholder relationships more effectively, resulting in improved financial outcomes (Franco et al., 2020).

Enciso-Alfaro and García-Sanchez (2024) reinforce this view, finding that companies with higher female representation on boards are more proactive in addressing climate change and sustainability issues, strengthening stakeholder relationships and enhancing long-term financial performance. This mirrors earlier findings on the role of diversity such as those by Enciso-Alfaro and García-Sanchez (2023), who argue that female leadership, particularly among executive directors, significantly enhances the transition toward circular business models. This transition not only drives environmental sustainability but also improves operational efficiency and financial outcomes. However, both studies note a diminishing return when board diversity exceeds certain thresholds, suggesting that diversity has limits in terms of its financial impact.

CEO leadership is another significant moderating variable. CEOs who are personally committed to sustainability and social responsibility are more likely to integrate ESG initiatives into the company's core strategies, ensuring that stakeholder needs are prioritized (Shen, 2019). For example, Rousseau et al. (2023) show that companies led by CEOs who actively champion sustainability tend to exhibit stronger alignment between ESG practices and financial performance. In contrast, CEOs who focus on short-term financial results may deprioritize ESG activities, potentially damaging stakeholder relationships and long-term profitability. Similarly, Yang et al. (2024) emphasize the importance of leadership in aligning CSR with technological innovation, noting that responsible leadership can enhance innovation while balancing stakeholder needs. However, excessive CSR can draw focus away from innovation by consuming limited resources, further complicating financial outcomes.

Stakeholder pressure also plays a crucial role in shaping the financial outcomes of CSR activities. Firms operating in consumer-facing industries, where customers and activist groups are vocal about sustainability issues, are likelier to adopt ESG practices to maintain their reputation and market share (Rangan et al., 2021). In such sectors, failing to meet stakeholder expectations regarding social and environmental responsibility can lead to reputational damage and revenue loss. Hossain et al. (2024) have highlighted that companies under high stakeholder pressure, whether from customers, investors, or regulators, tend to exhibit better financial outcomes when prioritizing ESG initiatives. This reinforces the argument that addressing stakeholder needs through ESG efforts can improve financial performance, especially in industries with high public visibility.

Ownership structure also influences the effectiveness of ESG practices in improving financial performance. Coelho et al. (2023) found that firms with dispersed ownership are more likely to prioritize ESG initiatives, as these firms tend to cater to a broader range of stakeholders, including institutional investors who often demand higher standards of sustainability and governance. In contrast, companies with concentrated ownership may face pressure to prioritize short-term financial gains, which can hinder the implementation of ESG initiatives that focus on long-term stakeholder value. Chen et al. (2024) also underscore the role of governance, showing that financial technology-driven sustainability efforts are more effective in firms with strong governance structures. These firms experience enhanced carbon emission reductions and improved financial performance, further proving the moderating role of governance in ESG success.

Additionally, industry-specific factors moderate the relationship between ESG practices and financial outcomes. For example, in industries with significant environmental impact, such as energy or manufacturing, firms that engage in proactive ESG initiatives are more likely to maintain legitimacy and manage risks effectively, thereby, ensuring long-term financial success (Khan et al., 2024). Conversely, companies in less-regulated industries may not experience the same benefits from ESG activities, as the pressure to address stakeholder concerns is less pronounced. Lyulyov et al. (2024) demonstrate how green branding within the context of Sustainable Development Goals (SDGs) enhances national competitiveness in the EU, suggesting that firms within more regulated environments are likely to see greater financial returns from their sustainability initiatives.

Despite the positive correlation between stakeholder engagement through ESG practices and financial performance, some studies report conflicting results. For instance, Peloza and Shang (2010) have found that while CSR activities can create value by influencing stakeholder perceptions and behaviors, their effectiveness varies depending on the type of activity and the stakeholder group targeted. Philanthropic efforts may yield less tangible financial benefits than product-related initiatives that directly impact consumer behavior. This suggests that the type of CSR activity and the stakeholder group being engaged serve as critical moderating variables in determining the financial impact of ESG practices.

In general, Stakeholder Theory provides a compelling rationale for adopting ESG and CSR activities, with the expectation that firms can achieve long-term financial success by creating value for all stakeholders. However, the financial outcomes of these initiatives are influenced by several moderating variables, including board composition, CEO leadership, stakeholder pressure, ownership structure, and industry-specific dynamics. Firms that

successfully balance these factors are more likely to realize the financial benefits of ESG practices, while others may experience neutral or even negative financial impacts depending on their specific context. Integrating new insights from studies on gender diversity, digital integration, and national sustainability initiatives further highlights stakeholder engagement's complex but promising role in shaping long-term financial outcomes.

2.3.3 Agency theory

Agency Theory, developed by Jensen and Meckling (1976), focuses on the potential conflicts of interest between principals (shareholders) and agents (managers). Managers may pursue CSR and ESG activities for personal or reputational gains, even when these initiatives do not directly benefit shareholders, leading to agency costs. The presence of robust governance structures (e.g., performance-based executive compensation and independent boards) can mitigate these agency conflicts by aligning managerial actions with shareholder interests, ensuring that ESG initiatives contribute to long-term financial performance. The latest empirical evidence indicates that the correlation between ESG activities and financial outcomes, as per Agency Theory, is frequently influenced by various moderating factors.

One critical moderating variable is governance structure, particularly the presence of independent directors and the design of executive compensation schemes. Rousseau et al. (2023) demonstrate that firms with well-structured executive compensation schemes tied to ESG performance tend to outperform those without. These firms can align managerial incentives with shareholder interests by ensuring that executives are rewarded not just for short-term financial results but also for achieving long-term sustainability goals. In contrast, firms with weaker governance structures are more susceptible to agency problems, where managers may pursue ESG initiatives that enhance their personal reputation or social standing but fail to deliver financial returns for shareholders (Cohen et al., 2023).

Recent research from Sun et al. (2022) supports these findings, showing that board independence, CEO duality, and the adoption of integrated reporting standards significantly influence the integration of CSR disclosures. However, the study found that board independence and gender diversity did not directly impact CSR integration levels, highlighting that governance mechanisms must be specifically aligned with sustainability goals to mitigate agency conflicts effectively. EncisoAlfaro and García-Sanchez (2023) further argue that female executive di- rectors, more so than independent directors, are more likely to drive meaningful ESG changes, emphasizing that governance structures must be carefully tailored to ensure that managerial ESG actions align with shareholder interests.

Ownership concentration also significantly determines how ESG activities align with shareholder interests. In firms with dispersed ownership, shareholders have less direct control over managerial decisions, which can exacerbate agency problems and lead to ESG initiatives that may not align with financial performance goals (Coelho et al., 2023). Conversely, in firms with concentrated ownership, large shareholders can exert greater influence over managerial decisions, ensuring that ESG initiatives are more closely aligned with the company's financial objectives. This dynamic is particularly evident in the study by Saini et al. (2023), which has found that concentrated ownership structures tend to lead to more focused ESG initiatives, improving operational efficiency and risk management.

Chen et al. (2024) also demonstrate that governance characteristics such as board independence and firm size moderate the impact of digital integration on corporate sustainability. Their findings align with Agency Theory, suggesting that firms with stronger governance structures are better positioned to ensure that digital sustainability efforts translate into financial benefits rather than becoming a source of agency costs.

CEO characteristics are another moderating factor that influences the effectiveness of ESG activities under Agency Theory. Shen (2019) argues that CEOs who are personally committed to sustainability and long-term value creation are more likely to integrate ESG practices into the firm's strategic vision. This can help reduce agency conflicts by aligning managerial actions with shareholder interests. For example, firms led by sustainability-oriented CEOs tend to balance short-term profitability with long-term ESG goals, as demonstrated in the findings of Rousseau et al. (2023). However, CEOs who prioritize short-term financial performance may resist ESG initiatives, viewing them as a distraction from immediate financial gains. Yang et al. (2024) further note that while responsible leadership enhances innovation in ESG practices, excessive focus on CSR can crowd out innovation by consuming limited resources, thereby, complicating the financial outcomes.

Recent empirical evidence also highlights the role of board diversity in mitigating agency problems. Enciso-Alfaro and García-Sanchez (2024) have found that firms with higher levels of gender diversity on their boards tend to adopt more comprehensive ESG initiatives, which not only enhance corporate governance but also align more closely with shareholder interests. This is particularly true for firms with female executive directors, who were found to play a stronger role in driving climate innovation and long-term sustainability strategies. However, the study also notes that the positive effects of gender diversity diminish after reaching a certain threshold, suggesting that there are limits to the governance benefits provided by diverse boards. EncisoAlfaro and García-Sanchez (2023) further highlight how female

leadership drives the transition to circular economy models, suggesting that when aligned with governance, leadership characteristics play a critical role in the success of ESG initiatives.

The role of executive compensation in driving ESG performance is well-documented in the literature. Firms that tie managerial compensation to ESG outcomes are more likely to see improved financial performance, as managers are incentivized to focus on both short-term profitability and long-term sustainability goals (Rousseau et al., 2023). However, the effectiveness of this approach depends on the specific design of the compensation package. For instance, firms that offer long-term stock options linked to ESG performance tend to perform better financially, as managers are encouraged to adopt a longer-term perspective (Coelho et al., 2023). In contrast, firms with short-term bonuses tied only to financial results may experience weaker alignment between managerial decisions and shareholder value.

Market conditions also moderate the impact of ESG initiatives under Agency Theory. During market uncertainty or economic downturns, managers may deprioritize ESG activities to focus on short-term financial performance, exacerbating agency conflicts. Lins et al. (2017) and Flammer (2015) have found that firms facing financial instability are more likely to reduce ESG initiatives, which can lead to reputational risks and long-term value destruction. However, firms that maintain their ESG commitments during difficult market conditions tend to build stronger relationships with stakeholders, ultimately improving their financial performance in the long run. Chen et al. (2024) add that fintech-driven sustainability initiatives (e.g., carbon emission re- ductions) can help firms weather economic downturns by enhancing efficiency and resilience, further proving the role of market conditions in shaping the financial outcomes of ESG initiatives.

Despite the general alignment of ESG initiatives with shareholder interests when robust governance structures are in place, conflicting findings in the literature suggest that not all ESG activities lead to financial benefits. For example, Ye et al. (2021) have found that the effectiveness of ESG initiatives depends heavily on the presence of mediating and moderating variables, such as governance quality, CEO characteristics, and industry-specific dynamics. In some cases, ESG activities may create agency costs, particularly when managers pursue these initiatives for personal gain rather than shareholder value. Enci- so-Alfaro and García-Sanchez (2024) also highlight the risks of over-investing in ESG activities without clear financial returns, noting that firms can experience diminishing financial benefits if ESG initiatives are not carefully aligned with core business strategies.

Agency Theory offers a more cautious perspective on ESG and CSR activities, highlighting the potential for agency costs when managerial actions do not align with

shareholder interests. Managers may pursue ESG initiatives for personal or reputational reasons, leading to in- efficiencies and potential conflicts of interest. Nonetheless, strong governance structures, such as independent boards and executive compensation schemes tied to ESG performance, can mitigate these agency problems. The effectiveness of ESG activities under Agency Theory is influenced by several moderating factors, including ownership concentration, board diversity, and market conditions. While firms with robust governance mechanisms tend to experience financial benefits from ESG efforts, those with weaker governance may face significant agency costs. Thus, the relationship between ESG activities and financial performance is contingent on the firm's ability to align managerial decisions with shareholder value through effective governance practices.

2.3.4 Resource-based view theory

The Resource-Based View (RBV), as articulated by Barney (1991), posits that a firm's competitive advantage stems from its unique resources and capabilities. ESG and CSR initiatives, under this framework, can be viewed as strategic resources that help firms achieve long-term financial success. Through improved reputation, enhanced innovation, and operational efficiencies, ESG practices can differentiate firms in the marketplace. However, the ability to convert ESG activities into competitive advantage is heavily moderated by factors such as industry dynamics, R&D intensity, innovation capacity, and governance structures, leading to diverse financial outcomes across firms and sectors.

Empirical research strongly supports the notion that ESG activities, when integrated into a firm's resource base, can enhance financial performance by strengthening key capabilities. For instance, Ghosh and Gupta (2023) found that companies investing in decarbonization and sustainable innovation exhibit superior financial performance compared to those that do not. These firms leverage ESG initiatives as strategic resources, differentiating themselves and responding to consumer demand for environmentally responsible products. Similarly, Saini et al. (2023) demonstrated that ESG practices improve risk management and reduce capital costs, reinforcing the idea that ESG can act as a valuable resource for strengthening a firm's financial position.

However, the financial benefits of ESG practices are not uniform. Industry dynamics play a pivotal role in moderating the financial impact of ESG activities. Firms in industries with substantial environmental footprints, such as energy and manufacturing, are more likely to benefit from proactive ESG strategies, due to regulatory pressures and growing demand for sustainable practices (Khan et al., 2024). In contrast, firms in less-regulated sectors, where

external pressures to adopt sustainability initiatives are lower, may not experience similar financial returns from ESG activities. This divergence underscores the importance of understanding the specific industry context when evaluating the financial impact of ESG initiatives.

R&D intensity and innovation capacity further influence how effectively firms can capitalize on ESG efforts. Bartolacci et al. (2019) have found that firms with strong R&D capabilities are better positioned to integrate ESG principles into their product development processes, which in turn improves their competitive advantage and financial outcomes. Companies with higher R&D intensity can innovate around sustainability challenges, developing new products that align with the growing demand for environmentally friendly solutions. Similarly, Chen et al. (2024) emphasized the role of fintech in reducing carbon emissions, particularly in high-carbon regions, highlighting how innovation capacity can translate ESG activities into financial gains. The ability of firms to harness technological innovation for environmental improvement underscores the importance of integrating ESG into the firm's innovation strategy.

In addition to innovation, governance structures play a critical role in the financial success of ESG initiatives. Saini et al. (2023) found that larger firms and those with strong governance frameworks are better able to turn ESG initiatives into financial gains. These firms typically have more resources to invest in comprehensive sustainability strategies and the governance mechanisms to ensure that ESG activities align with long-term financial goals. In contrast, smaller firms or those with weaker governance structures may struggle to achieve the same financial benefits, as they often lack the resources and oversight to manage sustainability initiatives effectively.

Despite the generally positive relationship between ESG activities and financial performance suggested by the RBV, some studies report mixed results. For instance, Verma and Mukhtaruddin (2023) have highlighted that the financial impact of environmental responsibilities can vary significantly based on geographical and regulatory differences. Firms in developed markets tend to experience more positive financial outcomes from proactive environmental practices, while firms in emerging economies may face challenges in translating ESG activities into financial gains due to weaker regulatory frameworks and less demand for sustainability from consumers. This geographical variation underscores the role of institutional support in shaping the financial benefits of ESG activities.

Additionally, the long-term nature of ESG investments poses challenges for firms focused on short-term financial performance. While the RBV emphasizes that ESG activities contribute to long-term competitive advantage, the financial benefits may not always be

immediate. Bos et al. (2017) have found that some firms, particularly in sectors such as healthcare, improved short-term financial performance through cost-cutting measures that negatively impacted ESG-related factors (e.g., employee well-being), suggesting a trade-off between short-term financial gains and long-term sustainability goals.

New evidence from Quttainah and Ayadi (2024) reinforces the role of digital integration as an enabler of ESG-driven competitive advantage. They have found that digital technologies significantly enhance emissions reduction, environmental innovation, and resource efficiency, particularly in firms with lower initial sustainability performance. This highlights how technology, as a unique resource, allows firms to achieve both environmental and financial goals, further supporting RBV's premise that valuable and rare resources (e.g., ESG-driven technological innovation) can lead to competitive advantage.

The RBV emphasizes the strategic value of unique resources and capabilities in generating competitive advantage. From this perspective, ESG initiatives can be seen as valuable intangible assets that enhance a firm's reputation, foster innovation, and improve operational efficiencies. Firms that invest in sustainable practices are better positioned to meet evolving market demands and regulatory standards, thereby improving their financial performance in the long term. However, the financial benefits of ESG initiatives depend on a firm's ability to leverage these activities as strategic resources. Companies in industries with high environmental impact, such as energy or manufacturing, are more likely to benefit from sustainability efforts, while firms in less environmentally focused sectors may see fewer direct financial gains. The RBV underscores the importance of firm-specific capabilities in translating ESG initiatives into financial success, with industry-specific factors serving as critical moderators.

2.3.5 Legitimacy theory

Legitimacy Theory, as articulated by Suchman (1995), suggests that organizations seek to align their actions with societal norms and expectations to gain, maintain, or restore legitimacy in the eyes of stake- holders. In the context of ESG and CSR, firms engage in sustainability initiatives not only to secure financial returns but also to enhance their legitimacy, especially under public scrutiny or regulatory pressure. However, the financial outcomes of these actions are moderated by several factors, including regulatory environments, stakeholder scrutiny, and public visibility, leading to mixed empirical results in literature.

Firms increasingly view ESG and CSR practices as mechanisms for aligning corporate behavior with societal expectations. Companies operating in highly scrutinized industries, such

as consumer goods or pharmaceuticals, are particularly inclined to adopt ESG practices to avoid reputational damage and preserve stakeholder trust. For instance, Deegan (2002) emphasizes that firms in highly visible sectors are more likely to engage in CSR to maintain legitimacy in the eyes of the public. Additionally, firms facing adverse publicity or operating in regulated industries are under continuous pressure to demonstrate ethical behavior, further reinforcing the relevance of Legitimacy Theory in understanding the relationship between ESG and financial performance. Moderating variables such as regulatory pressures, public visibility, and stakeholder scrutiny are critical in shaping how effectively firms maintain legitimacy through ESG and CSR initiatives. Companies in heavily regulated industries such as energy or finance often adopt ESG practices to comply with regulatory requirements and reduce legal risks (Khan et al., 2024). In these sectors, regulatory frameworks compel firms to align their operations with environmental and social standards, enhancing their legitimacy among stakeholders. Failing to meet regulatory expectations can result in penalties, reputational damage, and loss of market share, demonstrating the importance of legitimacy in these industries.

Public visibility is another significant moderating factor in the financial impact of ESG activities. Firms that operate in industries under constant public scrutiny, particularly consumer-facing businesses, are more likely to implement comprehensive ESG strategies to avoid negative publicity and maintain stakeholder confidence. Peloza and Shang (2010) have found that companies in highly visible industries are more inclined to engage in robust ESG efforts to prevent reputational damage, which can, in turn, lead to financial gains as consumers and investors increasingly favor businesses with strong sustainability profiles.

Stakeholder scrutiny is also crucial in moderating the financial out- comes of legitimacy-driven ESG activities. Companies closely monitored by investors, NGOs, or consumer groups are more likely to adopt transparent ESG strategies, which can lead to enhanced financial performance. Ellili et al. (2022) have found that firms with strong stakeholder engagement adopt ESG practices that not only meet societal expectations but also contribute to financial objectives. In industries with high stakeholder pressure, companies that fail to engage in ESG and CSR activities risk reputational damage and financial underperformance due to declining consumer trust and investor confidence.

While Legitimacy Theory offers a robust theoretical explanation for why firms adopt ESG and CSR practices, the empirical evidence is mixed. In highly visible industries, ESG practices often lead to positive financial outcomes. Saini et al. (2023) have demonstrated that ESG practices reduce capital costs and improve risk management, particularly in industries subject to stringent

regulatory oversight. These firms benefit financially from their ESG efforts, as they are better positioned to meet regulatory and societal expectations.

However, the financial benefits of ESG initiatives are not equally distributed across industries. Firms in less visible sectors or those facing fewer regulatory pressures may not experience the same financial gains from adopting ESG practices. Verma and Mukhtaruddin (2023) have found that the financial impact of environmental responsibilities is mixed, particularly in industries with lower public scrutiny or weaker regulatory frameworks. Companies in these sectors may engage in ESG activities to maintain legitimacy but struggle to convert these actions into financial returns.

Additionally, the pursuit of legitimacy can sometimes lead to short- term, symbolic actions that fail to generate long-term financial benefits. Delmas and Montes-Sancho (2011) note that firms may adopt ESG initiatives in response to external pressures rather than integrating sustainability into their core business strategies. Such companies may engage in superficial activities, such as “greenwashing” to maintain legitimacy, which can harm long-term financial performance if stake- holders perceive these initiatives as insincere. This can lead to a loss of trust and undermine the firm’s ability to achieve sustained financial success.

Legitimacy Theory suggests that firms engage in ESG and CSR activities to maintain or enhance their legitimacy in the eyes of stake- holders. These efforts are often driven by external pressures, such as regulatory requirements, societal expectations, and public scrutiny, rather than direct financial incentives. While ESG activities can help firms secure their legitimacy and manage reputational risks, their financial impact is less straightforward. Firms operating in industries with high public visibility are more likely to benefit financially from ESG efforts, as these activities help them maintain stakeholder trust and comply with regulatory standards. However, for firms in less scrutinized sectors, the financial returns on ESG initiatives may be minimal. Therefore, while Legitimacy Theory provides a strong rationale for adopting ESG practices, the financial outcomes depend heavily on the level of external pressure and stakeholder expectations.

2.3.6 Institutional theory

Institutional Theory, as introduced by DiMaggio and Powell (1983), emphasizes both the formal and informal roles of external pressures in shaping corporate behavior. In the context of ESG and CSR, firms often adopt sustainable practices in response to institutional pressures such as regulatory requirements, industry standards, or normative societal expectations. With increasing demands to integrate sustainability, particularly through global initiatives such as the

SDGs, Institutional Theory provides a framework to understand how these external factors influence corporate actions. However, the financial outcomes of ESG and CSR efforts vary based on the regulatory and normative environments in which companies operate.

Empirical evidence supports the significant influence of institutional pressures on ESG and CSR activities, particularly in regions with stringent regulatory frameworks or high normative expectations. For instance, Khan et al. (2024) have found that companies in areas with strong environmental regulations are more likely to implement robust ESG strategies, which, in turn, improve financial performance. These companies adopt ESG practices not only to comply with legal requirements but also to gain competitive advantages by aligning with societal expectations. Regulatory pressures, thus, act as a key moderating variable, ensuring that ESG practices contribute to both legitimacy and financial success.

Normative pressures, such as industry-specific sustainability standards, also shape corporate behavior. In industries such as energy, chemicals, and automotive, where sustainability norms are well- established, firms that adopt ESG initiatives often perform better financially. For example, Yu (2022) demonstrates that companies in such industries align their operations with both regulatory and market expectations, leading to improved financial outcomes. These normative pressures compel companies to exceed baseline regulatory requirements and adopt best practices, which enhance their competitiveness.

However, the effects of institutional pressures are not uniform across all industries or regions. Firms in less-regulated industries or emerging markets face weaker institutional pressures, resulting in less comprehensive ESG initiatives.

Javed et al. (2016) highlights that companies in non-Western markets often experience fewer regulatory and normative pressures to adopt ESG practices, leading to inconsistent financial outcomes. While these firms may implement ESG initiatives to gain legitimacy in global markets, they often struggle to attain immediate financial benefits due to weaker institutional support in their domestic contexts. Regional variations further complicate the relationship between institutional pressures and ESG performance. Firms in developed markets, such as the European Union, benefit from stronger institutional support for sustainability initiatives. For instance, Lyulyov et al. (2024) have found that European companies, especially in countries such as France, Germany, and Sweden, have been at the forefront of green branding and environmental performance, largely due to the alignment of national policies with SDGs. Conversely, firms in emerging markets such as Ukraine face challenges in achieving similar results, given weaker regulatory frameworks and less robust institutional support for sustainability.

Another important factor is mimetic pressure within industries. According to DiMaggio and Powell (1983), firms often imitate the practices of their peers to maintain legitimacy and competitive parity. This is particularly true in sectors where ESG practices have become the norm. For example, Enciso-Alfaro and García-Sanchez (2024) have found that firms with greater female representation on their boards are more pro- active in climate change innovation, partly because of normative pressures to align with best practices in corporate governance. Such mimetic pressures push firms to adopt ESG strategies to maintain legitimacy and meet investor and stakeholder expectations.

Despite the strong theoretical foundation of Institutional Theory, empirical findings on the financial benefits of ESG practices remain mixed. Some studies, such as those by Quttainah and Ayadi (2024), show that digital integration and environmental innovation drive financial success, particularly for firms with low initial sustainability performance. However, other studies, such as Verma and Mukhtaruddin (2023), suggest that the financial benefits of environmental responsibilities are contingent on industry and regional factors, with firms in emerging markets facing more challenges in translating ESG activities into financial gains due to weaker institutional frameworks.

Institutional Theory emphasizes the role of external pressures such as regulations, norms, and societal expectations in shaping corporate behavior. Firms adopt ESG and CSR practices not necessarily for direct financial returns but to conform to these institutional demands and secure legitimacy. The financial outcomes of ESG activities under this theory are influenced by the strength of regulatory frameworks and normative pressures within specific industries or regions. Firms operating in markets with strong regulatory oversight or high societal expectations for sustainability tend to experience better financial outcomes from ESG initiatives. However, in regions with weaker regu- latory environments, firms may find it more challenging to translate ESG activities into financial success. Institutional Theory highlights the importance of aligning ESG practices with external institutional demands to achieve both legitimacy and financial performance.

In summary, the theoretical perspectives of Shareholder Theory, Stakeholder Theory, Agency Theory, RBV, Legitimacy Theory, and Institutional Theory each provide unique lenses through which the relationship between ESG/CSR initiatives and CFP can be understood. These theories highlight the critical role of moderating variables such as governance structures, industry type, ownership concentration, market conditions, and institutional pressures in shaping the financial outcomes of ESG and CSR activities. However, despite the extensive theoretical exploration of ESG/CSR and CFP relationships, significant research gaps remain,

particularly regarding the nuanced role of moderating variables. The lack of consensus in empirical findings, driven by variations in these moderating variables, underscores the need for a more targeted analysis that synthesizes recent trends in literature.

This study aims to address these gaps by exploring the role of moderating variables in the ESG/CSR-CFP relationship through a systematic review and bibliometric analysis of research published from 2019 to 2023. Specifically, it seeks to answer the following research questions:

- What are the most frequently studied moderating variables in the relationship between ESG/CSR and CFP in the literature from 2019 to 2023?
- How has the focus on specific moderating variables in the ESG/CSR and CFP literature evolved from 2019 to 2023?
- What primary theoretical frameworks are employed in recent ESG/ CSR-CFP relationship studies?
- Are there any notable gaps or under-researched areas in the ESG/CSR and CFP literature regarding moderating variables?
- What are the most influential studies, journals, or authors in ESG/ CSR and CFP research concerning moderating variables?

By addressing these questions, this study aims to provide a comprehensive overview of the evolving role of moderating variables in shaping the ESG/CSR-CFP relationship, identify trends and gaps in the current literature, and offer guidance for future research in this critical area.

To ensure a robust examination of the relationship between ESG/CSR initiatives and CFP, with a particular emphasis on moderating variables, it is essential to adopt a methodological approach synthesizing the wide range of existing research. In the following section, we will outline the methodology employed in this study, including systematic review and bibliometric analysis. This approach allows for a comprehensive literature assessment, providing insights into the most frequently studied moderating variables, theoretical frameworks, and research trends from 2019 to 2023. The methodology will detail the steps taken to ensure the rigor and reliability of the analysis, guiding the investigation of the key research questions introduced above.

2.4 METHODOLOGY

Based on the vast number of articles that presented conflicting results, the justification for new research in ESG-CSR-CFP relationships, with a particular focus on moderating

variables, is compelling. Addressing the inconsistencies and gaps identified in existing studies, exploring underrepresented moderating factors, and adopting more rigorous and standardized methodologies are essential for advancing our understanding of how ESG and CSR initiatives impact CFP. As the global business environment evolves, this new research will contribute to academic knowledge and provide valuable insights for practitioners seeking to navigate the complexities of sustainability and financial performance in diverse and dynamic contexts.

This study employs a combined methodology of bibliometric analysis and systematic review to address these research questions and provide a comprehensive overview of the current state of the literature. The bibliometric analysis starts by testing the bibliometric laws of Bradford and Lotka and follows techniques outlined by Ellili et al. (2022), including Trend Analysis, Bibliographic Coupling of Sources Analysis, Biblio- graphic Coupling of Articles Analysis, and Keyword Occurrence Analysis.

The systematic review follows the techniques of Junior and Godinho Filho (2010), Van Kampen et al. (2012), and Jabbour (2013), utilizing a codification system to identify gaps in the research field. The review systematically analyzes financial performance variables, ESG and CSR measures, assessment methods, theoretical foundations, and the role of database providers. It also explores the statistical modeling techniques employed and identifies key areas for future research, providing a roadmap for advancing knowledge in this domain.

The bibliometric analysis complements systematic review by mapping the intellectual structure of the field and identifying key au- thors, influential studies, and journals. This dual approach offers a comprehensive and up-to-date overview of the existing literature while uncovering the underlying patterns and connections that have shaped academic discourse on ESG, CSR, and financial performance under the influence of moderating variables.

2.4.1 Sample selection

The first step in this research involved carefully selecting articles, for which we utilized the Web of Science and Scopus databases. These databases are widely recognized in financial corporate research for their reliability and broad coverage. Scholars such as Aguinis et al. (2018) emphasize the importance of Web of Science and Scopus in conducting comprehensive literature reviews, particularly in the context of CSR.

The robustness of research findings is further reinforced by the use of Web of Science, as demonstrated by Orlitzky et al. (2003) in their meta-analysis of CSR and financial performance. Similarly, Peloza (2009) recognizes Scopus' extensive coverage, making it a key resource for collecting data on CSR and financial outcomes. Friede et al. (2015) also advocate

for the combined use of Web of Science and Scopus in examining ESG and CFP, acknowledging their value in sourcing reliable, peer-reviewed articles. These collective insights underline the essential role of Web of Science and Scopus in enhancing the rigor and depth of financial corporate research.

Beyond their extensive use in academic research, Web of Science and Scopus are valued for their comprehensive journal coverage, which includes a wide array of regional and non-English publications, especially within Scopus (Mongeon & Paul-Hus, 2016). Both databases maintain rigorous selection criteria to ensure that only high-quality, peer reviewed journals are included, which is vital for producing credible research outcomes (Harzing & Alakangas, 2016). Their advanced search capabilities allow for precise and thorough literature reviews, facilitating systematic analyses that are integral to corporate financial research (Bramer et al., 2017). Moreover, the citation analysis tools provided by Web of Science and Scopus enable researchers to track the influence of their work and assess academic impact, offering valuable insights into research trends and collaboration networks (Bornmann & Daniel, 2008).

The global reach and interdisciplinary scope of Web of Science and Scopus make them indispensable for cross-disciplinary research, further establishing their reliability and broad adoption in various academic fields (Archambault et al., 2009). The consistent use of these databases in research processes not only enhances the credibility of the studies but also broadens their scope and relevance, solidifying Web of Science and Scopus as essential resources for conducting comprehensive and impactful financial corporate research.

We searched the Web of Science and Scopus databases using a search string encompassing many articles. The search string utilized: (“ESG” or “CSR”) and (“financial performance” or “firm value”) AND (“moderating” or “moderator” or “moderate”). The last systematic search was completed on December 18, 2023.

The sample initially comprised 610 articles from the Web of Science and Scopus databases from the 2019 to 2023 period, excluding the more recent studies such as Ye et al. (2021) and Lee and Suh (2022). We applied native filtering criteria Scopus (Subject Area: Business, Management, and Accounting; Social Sciences; Economics) and Web of Science (Categories: Business; Management). Subsequently, we imported a sample of 610 articles and filtered it to Ryann Systematic Review for a content selection analysis. Rayyan is an increasingly popular tool for conducting systematic reviews, offering significant advantages in terms of efficiency, collabo- ration, and accuracy. Developed by Ouzzani et al. (2016), Rayyan is a web-based application specifically designed to streamline the screening and selection process in systematic reviews.

A key feature of Rayyan is its native PRISMA interface, which integrates seamlessly with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. PRISMA is widely recognized for enhancing the transparency, reproducibility, and rigor of systematic reviews (Moher et al., 2009). By incorporating PRISMA's framework, Rayyan ensures that researchers can easily document and visualize the flow of studies through the different phases of the review process, adhering to best practices in transparency and reporting (Haddaway et al., 2022).

We apply the native PRISMA protocol selection in the Rayyan soft- ware to guarantee the reliability of the selection process.

After selecting the articles, we start the analysis using the RStudio software with the Bibliometrics R package. This package allows us to use Biblioshiny, a web-based interface for bibliometric analysis. The sample includes 108 documents published between 2019 and 2023 by 320 authors from 76 academic sources. Figure 1 presents the PRISMA flow diagram, generated using Rayyan, which visually summarizes the selection process from initial identification through final inclusion.

2.4.2 Bibliometric analysis

Bibliometric Analysis is a quantitative research method that allows scholars to systematically explore the landscape of academic literature by analyzing patterns in publication data and citations. It is particularly effective for identifying key authors, influential studies, and significant trends within a given field. The method has grown in popularity due to its ability to provide a structured and data-driven approach to under- standing scholarly communication (Garfield, 1955; Price, 1963; Glanzel, 2003; Small, 2004; Van Raan, 2005; Aria & Cuccurullo, 2017; Bornmann & Daniel, 2008; Thelwall, 2008).

The origins of bibliometric analysis trace back to key figures such as Eugene Garfield, who pioneered the concept of citation indexing in the 1950s. His work, especially through the creation of the Science Citation Index, marked a breakthrough in the evaluation of academic impact and citation networks (Garfield, 1955).

Within this historical context, the application of foundational bibliometric laws — Bradford's Law and Lotka's Law — further refines our understanding of the structure and dynamics of academic literature.

Bradford's Law, introduced by Bradford (1934), identifies the core journals that publish the majority of significant articles, building on Garfield's concept of citation networks to reveal where research output is most concentrated (Aria & Cuccurullo, 2017).

Lotka's Law, formulated by Lotka (1926), examines author productivity, showing that a small number of authors produce most of the literature — a concept that aligns with Price's observations on the distribution of scientific productivity (Thelwall, 2008).

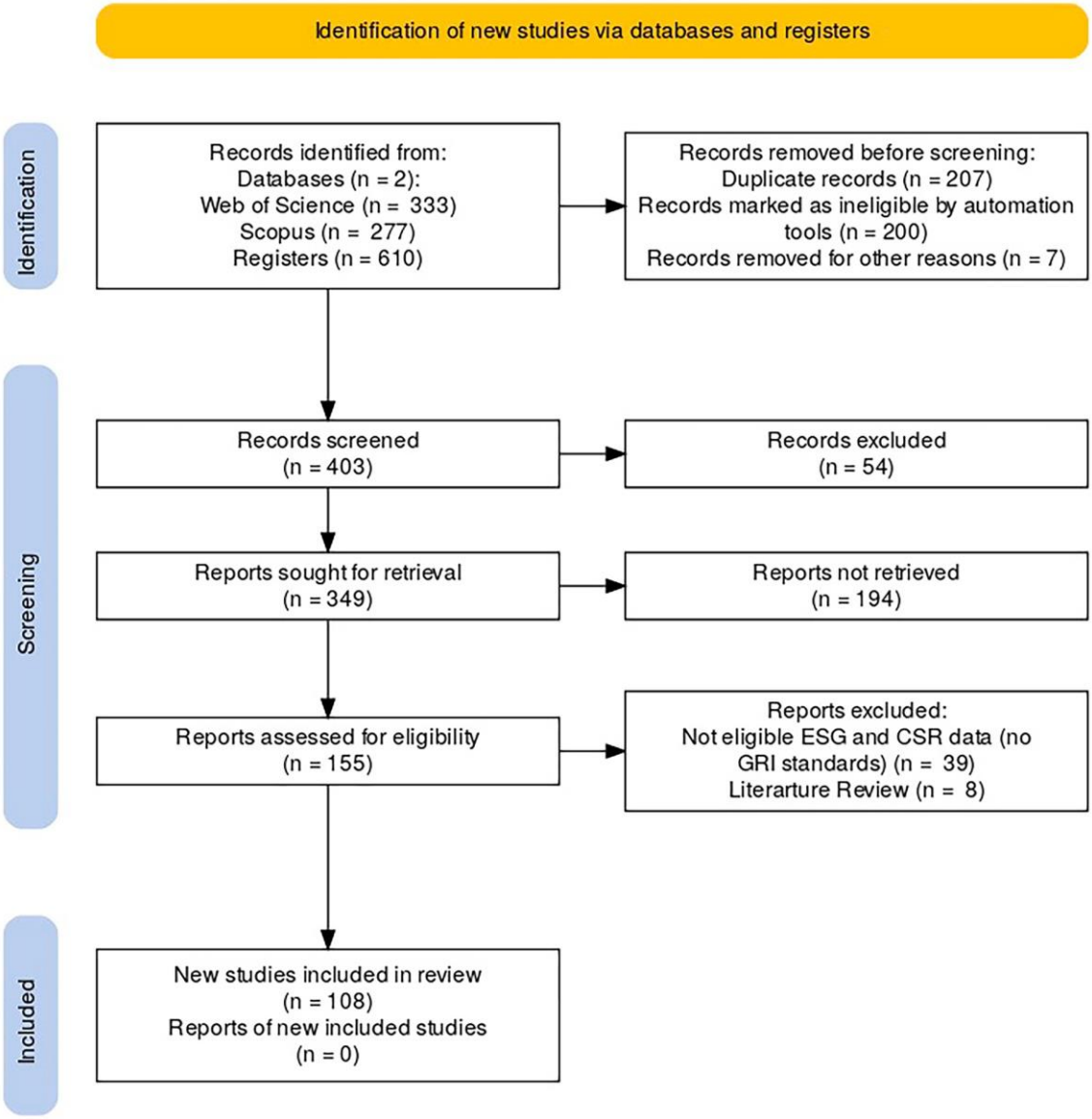


Figure 1 - PRISMA flow diagram for selection of studies.

Note: generated by Rayyan.

Researchers have applied these laws to identify key journals, authors, and themes within ESG and CSR research, particularly in business, demonstrating their utility in mapping the academic landscape (Chytis et al., 2024). Additionally, the comprehensive bibliometric methodologies developed by Maria Aria and Corrado Cuccurullo, including the use of the Bibliometrix package, integrate these laws to provide a structured and data-driven approach to understanding the impact of ESG and CSR on financial performance (Aria & Cuccurullo, 2017).

In modern research, tools such as VOSviewer and Biblioshiny have greatly enhanced bibliometric studies by offering sophisticated visualization capabilities for citation networks and large-scale data analysis. This has enabled a more in-depth examination of trends, research impact, and the evolution of academic fields (Donthu et al., 2021; Zupic & Cˇater, 2015). By leveraging techniques such as co-citation analysis, bibliographic coupling, and keyword mapping, researchers can explore the structure of academic disciplines and identify influential contributions over time.

2.4.2.1 Trend analysis

In conducting a trend analysis using Biblioshiny, linking the production of academic works over the years with the number of citations provides deeper insights into the impact and evolution of a research field. The software allows researchers to visualize the annual growth of publications, revealing how interest in topics such as ESG and CSR has expanded over time. This can be contrasted with citation data, highlighting which publications have had a lasting influence on the field. For example, a surge in publications might signal growing interest, but only highly cited works indicate significant contributions that shaped subsequent research.

By analyzing both production volume and citation trends, Biblioshiny enables a more nuanced understanding of how certain themes, such as sustainability integration or stakeholder engagement, have gained prominence and influenced CFP discussions. This dual approach helps researchers identify pivotal studies and periods of heightened scholarly attention, revealing key trends and gaps for further exploration.

Linking the article's production over the years with citation evolution is essential to understanding the relevance of the research field interest. While article production data reveal trends and periods of increased scholarly activity, citation analysis measures the lasting impact of these works, highlighting those that have significantly influenced subsequent research (Bornmann et al., 2008).

2.4.2.2 Bradford's law and bibliographic coupling analysis

In bibliometric research, Bradford's Law is a foundational principle that aids in understanding the distribution of scientific literature across journals. The law suggests that if journals are ranked according to the number of articles they publish on a specific topic, they can be categorized into a small core group followed by several broader groups, each containing a similar number of articles but spread over an increasing number of journals (Bradford, 1934). This distribution pattern highlights the concentration of crucial research outputs within a limited

number of journals, while the remaining contributions are scattered across a broader range of publications.

The primary advantage of applying Bradford's Law in bibliometric analysis is its ability to identify the most influential journals within a specific field. By isolating the core journals, researchers can more efficiently target their literature reviews, focus on the most impactful publications, and strategically decide where to submit their work for publication (Egghe & Rousseau, 1990). Moreover, the law provides a clear visual representation of how research output is distributed across journals, which can reveal trends in publication activity over time (Ding et al., 2000).

The application of Bradford Law is relatively simple. The articles are, then, sorted by the journals in which they were published and ranked in descending order according to the number of articles each journal has contributed to the topic. Once the journals are ranked, the next step is dividing them into zones as Bradford's Law prescribes them.

The total number of articles N is divided into three groups, each containing one-third of the articles. The first group, known as the core zone, will typically include the fewest journals but publish the largest share of significant research articles. The remaining two zones will contain the same number of articles as the core zone but distributed across more journals (Egghe & Rousseau, 1990).

The general formula for Bradford's Law, while not as mathematically strict as other bibliometric laws, is often expressed in terms of the geo- metric progression of the number of journals in each zone:

$$R: R^2: R^3 \qquad (1)$$

Where:

R represents the multiplier that reflects the increasing number of journals needed in each subsequent zone to account for the same number of articles as in the core zone.

The process can be outlined as follows:

Total Articles (N): Determine the total number of articles in the dataset.

Articles per Zone: Divide N by 3 to find the number of articles in each zone:

Articles per Zone = N3

Zone Division: Assign journals to the first zone until the cumulative number of articles reaches one-third of N. Continue this process for the subsequent zones.

The application of Bradford's Law using Biblioshiny, a web interface for the R package Bibliometrix, facilitates the identification of core journals within research fields by automating the ranking and categorization of journals based on their publication frequency. The software's capabilities ensure that even in fields with widely dispersed research outputs, Bradford's Law can be applied with accuracy, providing a clear visualization of the most influential journals (Aria & Cuccurullo, 2017). Furthermore, Bibliophagy's intuitive interface enhances the accessibility of bibliometric analysis, allowing researchers to maintain a focused and strategic approach to literature reviews and resource allocation, as recommended by Egghe and Rousseau (1990).

While Bradford's Law is a valuable tool for identifying core journals in specific fields, its application in broad and interdisciplinary areas such as ESG and CSR presents challenges due to the dispersed nature of research outputs across numerous journals (Egghe & Rousseau, 1990). This dispersion can blur the boundaries of core journals, making precise application difficult. Additionally, the dynamic nature of research fields means that core journals may shift over time, necessitating regular updates to maintain relevance. Furthermore, the overlap of journals across multiple disciplines complicates categorization, as journals might appear in more than one zone, further challenging the law's application (Li & Yin, 2017). To address these limitations, it is recommended that Bradford's Law be complemented with other bibliometric methods, such as Bibliographic Coupling, to achieve a more comprehensive understanding of the research landscape (Chiu & Ho, 2005).

Bibliographic coupling reveals connections between articles through shared references, highlighting clusters of related research that contribute to the same agenda. Together, these methods address the limitations of Bradford's Law by mapping the interconnected research streams and influential works that shape and advance the field.

By integrating Bradford's Law with bibliographic coupling, re- searchers can achieve a deeper, more structured view of the research landscape, enabling precise navigation of complex interdisciplinary fields such as ESG and CSR (Kessler, 1963; Small, 1973; Zupic & C˘ater, 2015).

This bibliographic coupling reveals the strength of connections between journals, with stronger links indicating a higher number of shared references. The software creates a visual map where journals are rep- resented as nodes, and their connections are depicted as lines, highlighting clusters of journals that share similar research areas (Van Eck & Waltman, 2010). These clusters indicate closely related journals, while the size and centrality of nodes reflect the influence of particular journals within the academic landscape.

2.4.2.3 Lotka's law

Lotka's Law, introduced by Alfred J. Lotka in 1926, is a critical bibliometric principle that describes the frequency of publication by authors within a given field. According to Lotka's Law, the number of authors producing n publications is inversely proportional to n^2; that is, meaning that a small number of authors contribute a large portion of the total publications, and in contrast, the majority of authors publish only a few papers (Lotka, 1926).

The application of Lotka's Law in bibliometric analysis involves several steps. First, a comprehensive dataset of publications in the ESG and CSR fields is collected. Authors are then identified, and their contributions (i.e., the number of publications) are counted. The distribution of these contributions is, then, analyzed to determine if it follows the inverse square law described by Lotka. Mathematically, Lotka's Law can be expressed as:

$$y = \frac{C}{x^n} \tag{2}$$

where,

y is the number of authors contributing x publications.

C is a constant representing the total number of authors.

n is typically equals 2, representing the inverse square law.

To apply this in practice, the distribution of authorship is plotted on a logarithmic scale. If the data follow a straight line with a slope of approximately -2, it confirms the applicability of Lotka's Law to the dataset (Pao, 1985; Egghe & Rousseau, 1990).

Lotka's Law is a valuable tool in bibliometric analysis for identifying the most influential authors within a research field, offering insights into the key figures and groups driving scholarly advancements (Aksnes & Sivertsen, 2004).

It also helps understand the structure of scientific output by highlighting the disparity between highly productive and less active authors (Pao, 1985). Additionally, the rise of multi-author publications com- plicates the attribution of individual productivity, further challenging the application of Lotka's Law (Aksnes & Sivertsen, 2004). To address these limitations, using Lotka's Law alongside other bibliometric mea- sures (e.g., Bibliographic Coupling Analysis) is recommended to comprehensively understand author influence (Pao, 1985).

Integrating Lotka's Law analysis with bibliographic coupling by article analysis provides a nuanced understanding of the research landscape in ESG/CSR and CFP under the effect of moderating variables. We perform bibliographic coupling by articles on VOSviewer.

Bibliographic coupling, which analyzes shared references between documents, is utilized in academic research to identify research trends, map intellectual structures, and trace the evolution of ideas within a field. This method helps to discover influential works, potential col- laborations, and the impact of research by clustering studies with overlapping citations (Kessler, 1963; Boyack & Klavans, 2010). It is particularly useful in systematic reviews and meta-analyses for efficiently identifying related studies (Zupic & C˘ater, 2015).

2.4.2.4 Keyword analysis

Similar to the coupling analysis, the co-occurrence analysis is very often used. A co-occurrence map based on keyword analysis is utilized in academic research to visualize relationships between key concepts and identify major themes within a research area. Co-occurrence maps provide a visual representation of how different concepts are inter- connected, which is valuable for understanding the intellectual structure of a field and guiding future research (Van Eck & Waltman, 2010; Donthu et al., 2021).

2.4.3 Systematic review

The systematic review in the article follows a five-step technique proposed by Jabbour (2013) and Lage Junior and Godinho Filho (2010), which has been successfully disseminated in various academic works. Several studies have utilized systematic literature reviews with categorization systems to advance research in different fields. Lee et al. (2021) employed a categorization system to review and classify sustainability-oriented applications of value stream mapping in manufacturing. Henrique et al. (2019) categorized machine learning techniques applied to financial market prediction, while Nazario et al. (2017) focused on categorizing studies related to technical analysis in stock markets. Pinto and Sobreiro (2022) provide a comprehensive literature review of anomaly detection approaches in digital business financial systems. It explores various methods used to identify irregularities within financial data, focusing on digital business environments, and evaluates their effectiveness in improving system security and performance.

In the field of environment and eco-innovation, Salim et al. (2019) systematically categorized internal capabilities for enhancing eco-innovation in manufacturing, and Masudin and Fernanda (2019) reviewed logistics performance using a structured approach to assess its impact on environmental outcomes. These studies illustrate the broad applicability of systematic reviews across diverse research fields, utilizing categorization to streamline complex data.

Systematic Search: The first step involved conducting a systematic search of available articles in a reliable database, specifically the Web of Science and Scopus databases. We improve the efficiency of the selection of the articles by using Rayyan and following the PRISMA's protocol.

Classification and Coding Framework Development: The second step involved developing a classification and coding framework to systematically categorize the identified articles. This framework was designed to capture various aspects of the studies, including the financial performance measures, ESG and CSR measures, moderating variable types, theoretical frameworks, statistical methods and common suggestions for research given by authors. The classification aimed to provide a comprehensive overview of the existing literature and identify research trends.

Application of Classification Framework: The third step entailed applying the classification framework to synthesize and present study trends on moderating variables and the ESG/CSR-CFP research field. Each article was classified according to the developed framework, allowing for a systematic analysis of the methodologies and findings.

Framework Outline: The fourth step involved outlining a framework that portrays the composition of the literature review, its main results, and what it represents according to the pre-established coding. This visual representation helped in understanding the results and trends in the current literature on detecting financial anomalies.

Analysis and Discussion of Results: The final step consisted of analyzing and discussing the results based on the synthesized data. This step also included proposing suggestions for future research, identifying gaps in the existing literature, and outlining potential areas for further investigation.

After selecting and reviewing the articles in this study, a classification framework was developed to categorize the papers in a manner relevant to the research theme. This framework consists of 7 primary categories, numbered from 1 to 7, each paired with a set of subcategories represented by letters (A, B, C, D, etc.). These codes facilitate a detailed analysis of research trends regarding methods, data sources, and variables. Each article was analyzed and assigned one or more codes from the framework, reflecting its alignment with the relevant categories and subcategories. Notably, a single article can receive multiple codes within a category for more than one subtopic. Table 1 presents this classification and coding framework, detailing the categories and subcategories used to organize the data systematically. The framework includes categories such as Financial Performance, ESG and CSR Measures, Moderating

Variables, Theoretical Framework, Suggestions for Future Research, Statistical Modeling, and Data Sources, each with specific subcategories and corresponding codes.

Category 1 is Financial Metrics. In evaluating the intersection between ESG/CSR measures and financial performance, categorizing financial variables into distinct groups — such as Valuation Metrics (1A), Profitability Metrics (1B), Risk Metrics (1C), Innovation Metrics (1D), and Market Performance Metrics (1D) offers a comprehensive framework for analysis. Valuation Metrics, grounded in Fama and French's (1992) seminal work on stock returns and further validated by Alareeni and Hamdan (2020) and Nguyen et al. (2021), help in understanding how market perceptions, influenced by ESG scores, affect firm valuation. Profitability Metrics, central to Friedman's (1970) emphasis on profit maximization, are reinforced by recent studies (e.g., Velte, 2023) that demonstrate how CSR initiatives can enhance a firm's profitability. Risk Metrics, originally conceptualized by Sharpe (1964) through Beta, are crucial for understanding financial risk and are shown in recent research by Broadstock et al. (2021) and Bae et al. (2021) to be mitigated by strong ESG practices. Innovation, as highlighted by Croitoru (2012) in Schumpeter's analysis of economic development, provides a comprehensive view of the dynamics of capital and innovation, which are further explored by Zhou et al. (2020), connecting ESG initiatives to increased innovation outputs. Lastly, Market Performance Metrics, pioneered by Jensen (1968) with the introduction of Alpha, are supported by Albuquerque et al. (2020) and Cheema-Fox et al. (2021), who reveal that firms with strong ESG performance often achieve superior market outcomes, particularly in times of market stress.

Table 1 - Classification and Coding Framework

Category	Subcategories	Code
Financial Performance	Valuation Metrics	1A
	Profitability Metrics	1B
	Risk Metrics	1C
	Innovation Metrics	1D
	Other	1E
	Market Performance Metrics	1F
ESG and CSR Measures		
	General ESG	2A
	Governance	2B
	Environmental	2C
	Expenditures	2D
	Other	2E
	Social	2F
	General CSR	2G

Moderating Variables		
	Firm Characteristics	3A
	Industry-Level Indicators	3B
	Institutional/Legal Environment	3C
	Corporate Governance	3D
	Social/Culture Indicators	3E
	CSR Engagement	3F
	Firm Strategy	3G
	Economic Indicators	3H
Theoretical Framework		
	Economic and Financial Theories	4A
	Strategic Management Theories	4B
	Social and Ethical Theories	4C
	Organizational Theories	4D
	Cultural and Behavioral Theories	4E
	Innovation and Technology Theories	4F
Suggestions for Future Research		
	Variables of Interest	5A
	Theoretical Framework	5B
	Data Sources	5C
	Type of Research Methodology	5D
	Sample Size/Population	5E
	Time Frame	5F
	Uncategorized	5G
	Geographical Focus	5H
	Research Focus Area	5I
	Industry/ Sector Focus	5J
	Suggested Contributions	5K
Statistical Modeling		
	Basic and Intermediate Models	6A
	Advanced Models	6B
Data Sources		
	Content Analysis	7A
	Reputational Indices	7B
	One-Dimensional Measures	7C

Note: elaborated by authors

Category 2 is ESG and CSR assessment. General ESG (2A) captures broad, aggregated metrics, supported by Eccles et al. (2014), who link comprehensive ESG scores to long-term financial success, with recent refinements by Berg et al. (2022). The Governance (2B) category, sustainability initiatives, while Industry-Level Indicators (3B) (e.g.,competition and regulatory pressures) contextualize the impact of these practices across sectors. The Institutional/Legal Environment (3C) reflect the regulatory landscape, and Corporate Governance (3D) is critical in aligning management actions with stakeholder expectations through effective governance

structures. Social/Culture Indicators (3E) capture the influence of societal norms on the success of CSR and ESG efforts, while the level of CSR Engagement (3F) highlights the importance of deeply integrating sustainability into business operations for enhanced financial returns. Firm Strategy (3G) includes strategic orientations such as innovation or cost leadership determining how these practices contribute to competitive advantage. Finally, Economic Indicators (3H), such as economic growth and market volatility, provide the broader macroeconomic context that influences how sustainability practices translate into financial performance. This categorization offers a comprehensive framework for understanding the complex dynamics between corporate sustainability and financial success.

Category 4 comprises the theoretical framework. We start with Economic and Financial Theories (4A), such as Fama's Efficient Market Hypothesis (1970) and its extensions by Fatemi et al. (2015), which emphasize the role of market efficiency in reflecting sustainability practices in stock prices. Strategic Management Theories (4B), including Barney's RBV (1991) and Porter's Competitive Advantage (1985), illustrate how CSR and ESG initiatives can serve as strategic assets, a view reinforced by Eccles et al. (2014). Social and Ethical Theories (4C), particularly Freeman's Stakeholder Theory (1984) and Carroll's CSR Pyramid (1979), provide a framework for understanding corporate responsibility, with Aguinis and Glavas (2012) confirming its positive impact on financial performance. Organizational Theories (4D), such as DiMaggio and Powell's Institutional Theory (1983) and Jensen and Meckling's Agency Theory (1976), along with recent works by Jamali and Karam (2018) and Ferrell et al. (2016), explore how internal structures and external pressures shape CSR effectiveness. Cultural and Behavioral Theories (4E), highlighted by Hofstede's Cultural Dimensions (1980) and furthered by Liang and Renneboog (2017), underscore the influence of cultural context on CSR practices. Finally, Innovation and Technology Theories (4F), from Rogers's Diffusion of Innovations (1962) to Christensen's Disruptive Innovation (1997) and Bocken et al. (2019), demonstrate how technological advancements drive sustainability and corporate transformation.

Category 5 resumes the future research suggestions collected in the sample articles. We have categorized future research suggestions into several key ones. These categories include Variables of Interest (5A) that focus on the specific constructs to be examined, such as ESG metrics or financial performance indicators, as highlighted by Kaplan and Norton (1996). The Theoretical Framework (5B) category emphasizes the importance of grounding research in robust theories such as Stakeholder Theory, as discussed by Whetten (1989), and further explored by Khan et al. (2016). Data Sources (5C) are critical for ensuring research validity and reliability, with Bryman and Bell (2015) underscoring the need for alignment between data

sources and research objectives. The Type of Research Methodology (5D) employed whether a quantitative, qualitative, or mixed methods approach is essential for methodological rigor, as advocated by Creswell and Plano Clark (2017) and exemplified in studies such as Flammer (2018). Considerations of Sample Size and Population (5E) ensure the generalizability of findings, with recent research by Fatemi et al. (2015) highlighting the importance of adequate sample sizes. The Time Frame (5F) of research, whether cross-sectional or longitudinal, is crucial for capturing trends and causality, as Menard (2002) noted. The Geographical Focus (5H) of a study, informed by the work of Hofstede (1980) and Liang and Renneboog (2017), reveals how cultural and institutional differences shape research outcomes. The Uncategorized (5G) includes miscellaneous suggestions that are difficult to categorize into other categories. The Research Focus Area (5I) defines the primary topics to be explored, with Whetten (1989) stressing the importance of clarity in this domain, supported by Eccles et al. (2014). Industry/Sector Focus (5J) is vital for understanding how different industries exhibit unique challenges and opportunities, as discussed by Porter (1985) and extended by Bocken et al. (2019). Finally, Suggested Contributions (5K) must be articulated, ensuring that research advances both academic knowledge and practical application, a principle highlighted by Whetten and Van de Ven (1989). By categorizing these areas, we provide a structured approach that not only identifies key areas for further investigation but also ensures that future research continues to build on foundational knowledge, contributing meaningfully to both theory and practice in corporate sustainability.

Category 6 shows the statistical methods categorized by complexity into Basic and Intermediate Models (6A) and Advanced Models (6B), providing a structured framework for selecting the appropriate tools for research. Ordinary Least Squares (OLS) regression is a foundational method classified under Basic Models (6A) due to its simplicity and widespread use in estimating linear relationships (Wooldridge, 2010). As we move to Intermediate Models (6B), techniques such as Propensity Score Matching and the Instrumental Variables (IV) approach address more complex issues (e.g., selection bias and endogeneity), offering more robust analysis options (Rosenbaum & Rubin, 1983; Greene, 2012).

Additionally, Fixed Effects Regression and Two-way Fixed Effects Models are essential in controlling unobserved heterogeneity in panel data, providing more accurate estimates (Baltagi, 2008). At the Advanced Models (6C), methods such as the Dynamic Panel Generalized Method of Moments (GMM) and System GMM tackle endogeneity in dynamic panel data with sophisticated instrument selection (Hansen, 1982). Moreover, techniques such as the Staggered Difference-in-Differences (DID) approach and robustness tests such as the Parallel Trend Test

and Goodman-Bacon Decomposition are critical for ensuring the validity of causal inference in complex datasets (Goodman-Bacon, 2021). Lastly, Two-stage Data Envelopment Analysis (DEA) and Directional Distance Function (DDF)-based models, often paired with bootstrapping, provide advanced tools for efficiency analysis, emphasizing the nuanced nature of these methods (Cooper et al., 2004). This categorization framework ensures that researchers select the most suitable methodologies, aligning with their research goals and the complexity of their data.

Finally, Category 7 is the data source regarding ESG and CSR information into three main categories: Content Analysis (7A), Reputation Indices (7B), and One-Dimensional Measures (7C). Questionnaire surveys gather direct insights from critical stakeholders but are vulnerable to biases and variations in response rates, a challenge common to CSR and ESG data acquisition (Rau & Yu, 2024). Acquiring and assessing CSR data is vital for understanding its impact on a company's financial performance, and various methods have been developed to capture these activities. Barauskaite and Streimikiene (2020) categorize these methods into reputation indices, content analysis, questionnaire surveys, and one-dimensional measures. Reputation indices (e.g., the Dow Jones Sustainability Index) offer a comprehensive view of a company's CSR efforts but are often limited to large, publicly traded firms and may introduce bias. Similarly, in ESG assessments, indices such as those provided by MSCI and Sustainalytics aggregate data from multiple sources, although they face similar limitations (Rau & Yu, 2024).

Content analysis allows for a detailed examination of company reports, providing flexibility in assessing specific CSR aspects, though it risks subjectivity and inconsistencies in reporting. This method is also applied in ESG assessments, which systematically review company dis- closures to evaluate performance across various dimensions (Barbosa, 2023). One-dimensional measures focus on particular CSR aspects, offering in-depth insights but potentially providing an incomplete picture of overall CSR performance. Each method presents unique strengths and limitations, requiring careful consideration to ensure an accurate and meaningful analysis of CSR's relationship with financial performance.

As illustrated in Figure 2, the graphical abstract provides an overview of the systematic literature review process, detailing the steps to categorize and analyze the articles. In the following paragraphs, we detail the categories' definitions and characteristics.

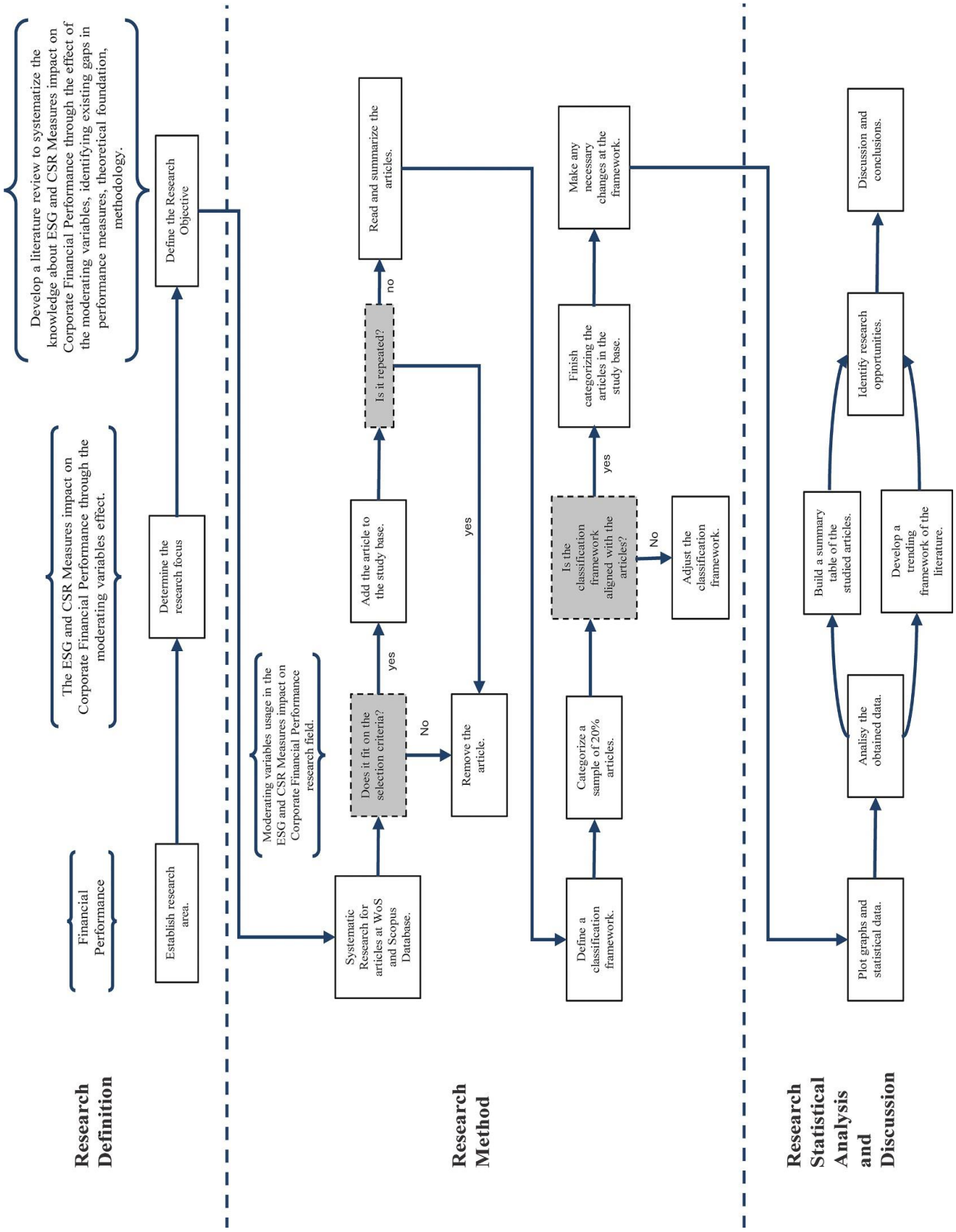


Figure 2 - Systematic literature review graphical abstract.

Note: elaborated by authors.

2.5 RESULTS AND DISCUSSION

2.5.1 Trend analysis

We analyzed the trends in the research field. With the annual growth rate of articles in the last five years, there has been an increase in the number of scholarly articles published on moderating variables in ESG/ CSR/CFP research. Figure 3 graphically depicts the annual scientific pro- duction alongside the average citations per year, highlighting trends over time. Figure 3 emphasizes the increased scholarly interest and sub- sequent citation impact, particularly notable in 2021 and 2022. The highest number of publications, with 38 articles, was in 2022, followed by 24 articles in 2023, 22 in 2021, 14 in 2020, and 10 in 2019. As shown in Table 2, the annual scientific production and average citations per year illustrate the fluctuation in scholarly output and impact within the field from 2019 to 2023. The data reveal that while the number of publications peaked in 2022, the most cited year was 2020, indicating a lag between publication and citation accumulation.

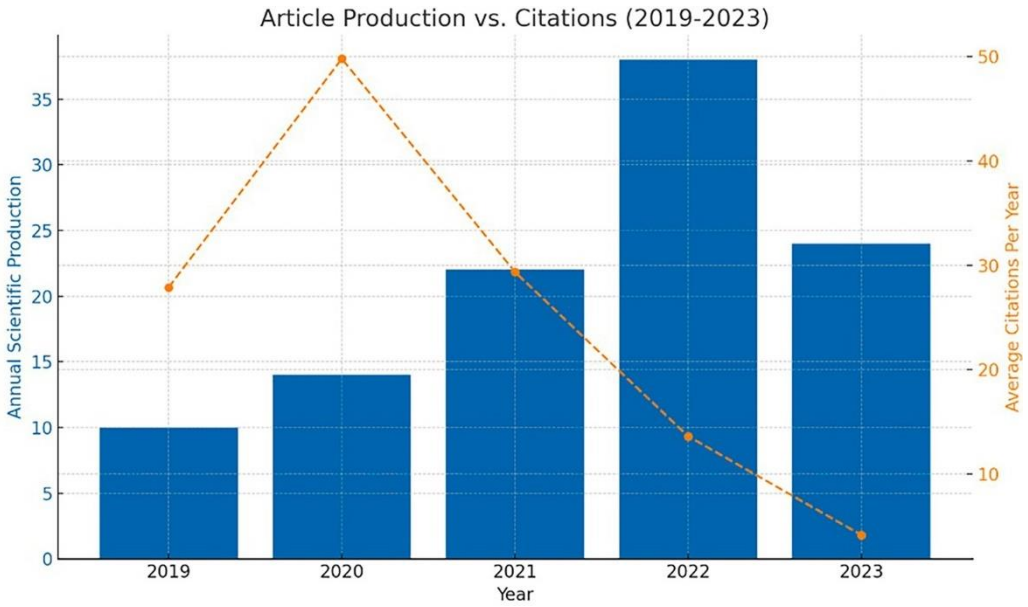


Figure 3 - Annual scientific production and average citation

Note: Elaborated by authors

Table 2 - Annual Scientific Production over the Years and Average Citations per Year

Year	Annual Scientific Production	Average Citations Per Year
2019	10	27,9
2020	14	49,79
2021	22	29,36
2022	38	13,63
2023	24	4,17

Source: Biblioshiny

It is possible to identify a high article production paired with low citations, which may indicate areas of research that are heavily explored but less impactful (Garfield, 2006). This trend may occur due to research fragmentation (Boyack & Klavans, 2010), increased competition (Merton, 1968), shifts toward newer topics (Kolk, 2016), citation fatigue (Tahamtan et al., 2016), reduced novelty (Kuhn, 1962), changing methodologies (Fiss, 2011), greater article availability but lower individual impact (Laakso et al., 2011), and shorter citation lifecycles (Redner, 1998). Three stages can be identified as follows:

Early Growth (2019–2020): In 2020, both article production and citations peaked, indicating that research on moderating variables in ESG/CSR-CFP was gaining significant attention. This may reflect an increased recognition of the importance of moderating factors (e.g., ownership structure, governance, or environmental regulations) in the ESG-CFP relationship. The heightened citations suggest that early publications on this topic were highly influential, potentially establishing key frameworks or models for analyzing moderating variables in ESG/CSR studies.

Declined Citations Despite Growth in Publications (2021–2023): Article production continued to grow, peaking in 2022, but citations have sharply declined. This suggests a possible saturation of the research field, where numerous studies are being published, but their impact (as measured by citations) has diminished. It could also indicate that subsequent studies may focus on more niche or specific aspects of moderating variables, which are not as widely cited as the foundational work from earlier years.

Dip in Article Production and Citations (2023): The drop in both production and citations in 2023 might suggest a leveling off of interest or innovation in the field. Researchers may be shifting focus or waiting for more substantial developments in moderating variables before further contributions. It could also mean that the research field is consolidating, with fewer new insights being introduced in comparison to the surge in prior years.

The sharp rise in early article production and citations likely points to the importance of exploring moderating variables in understanding the ESG-CFP relationship. These variables help explain why ESG performance may have varied impacts on financial outcomes depending on factors such as industry, ownership, and governance structures.

The decline in citations in recent years suggests that future studies need to explore new or underexplored moderating variables or perhaps take a more interdisciplinary approach to continue adding value to the ESG/CSR-CFP literature. This might involve integrating data from different regions, industries, or firm characteristics to uncover new patterns.

The short-term surge in article production (especially in 2022) may reflect a response to external pressures (e.g., the pandemic), leading to more studies investigating how these factors moderate the ESG-CFP link. However, the quick decline in citation suggests that a clear, dominant consensus or theoretical framework may still be lacking.

In conclusion, while the moderating variables in ESG/CSR-CFP research experienced a spike in scholarly interest, the declining citation trend indicates the need for more innovative approaches to sustain academic influence in this domain.

2.5.2 Bradford's law and bibliographic coupling analysis

Figure 4 visually summarizes the application of Bradford's Law, illustrating the concentration of core journals central to moderating variables and ESG/CSR-CFP research. Table 3 shows Bradford's Law zone distribution, where Zone 1 includes 7 core journals that account for 37 publications, representing the most concentrated sources of impactful research. Zones 2 and 3 consist of 26 and 35 journals, respectively, with decreasing frequency of publications, reflecting a broader, less concentrated distribution of research. This distribution, generated using Biblioshiny, illustrates the varying levels of journal influence within the academic landscape.

The core zone is composed of seven journals: Sustainability with 13 articles, Corporate Social Responsibility and Environmental Management with 9 articles, Cogent Business & Management with 3 articles, Finance Research Letters with 3 articles, Journal of Asian Finance Eco- nomics and Business with 3 articles, Journal of Sustainable Finance & Investment with 3 articles and Management Decision with 3 articles. Table 4 further breaks down the annual scientific production and citations by source, allowing for a more detailed analysis of the most influential journals in the field. Figure 5 displays the bibliographic coupling map generated by VOS-viewer. It reveals the clustering of journals that share similar research agendas and citation patterns. The color of the nodes corresponds to the groups formed by a set of journals, with three clusters identified.

The Red Cluster focuses on the broad, cross-industry effects of CSR and ESG practices, highlighting how comprehensive ESG scores and detailed CSR disclosures influence financial metrics such as Tobin's Q, Return on Assets (ROA), and misvaluation measures. Bofinger et al. (2022) demonstrate that higher CSR scores are associated with reduced misvaluation and enhanced firm value, indicating that the market better values firms with robust CSR practices. Additionally, Rashid et al. (2020) explore the moderating effect of CEO power on the relationship between CSR disclosures and firm performance, finding that stronger CEO power

can negatively impact the transparency of CSR disclosures, which, in turn, affects firm valuation. Sreepriya et al. (2023) further investigate the role of Global Reporting Initiative (GRI) compliance as a moderating variable, revealing that adherence to GRI standards significantly enhances the impact of sustainability disclosures on firm value. These studies collectively underscore the importance of integrating ESG factors into business strategies and maintaining transparency in sustainability efforts to drive financial success and market confidence.

The Blue Cluster examines sector-specific impacts of CSR practices, particularly within industries such as tourism and energy. This cluster explores how CSR activities, especially those related to environmental sustainability and social governance, affect financial performance metrics such as Net Interest Margin (NIM), ROA, and Tobin's Q. Javeed and Lefen (2019) highlight the positive impact of CSR on financial performance in the energy sector, particularly through environmental sustainability initiatives that enhance firm value and profitability. Similarly, Nagalingam et al. (2022) analyzed the tourism sector and found that companies with strong CSR practices in environmental and community engagement experienced higher financial returns. Alani and Makhlouf (2023) add to this perspective by examining how ownership structures, such as state-owned versus private firms, moderate the relationship between CSR and financial performance, with state-owned firms benefiting more from CSR activities in emerging markets. These findings suggest that tailored CSR strategies that align with sector-specific dynamics and regulatory environments can maximize financial returns, particularly in industries that face high public and regulatory scrutiny.

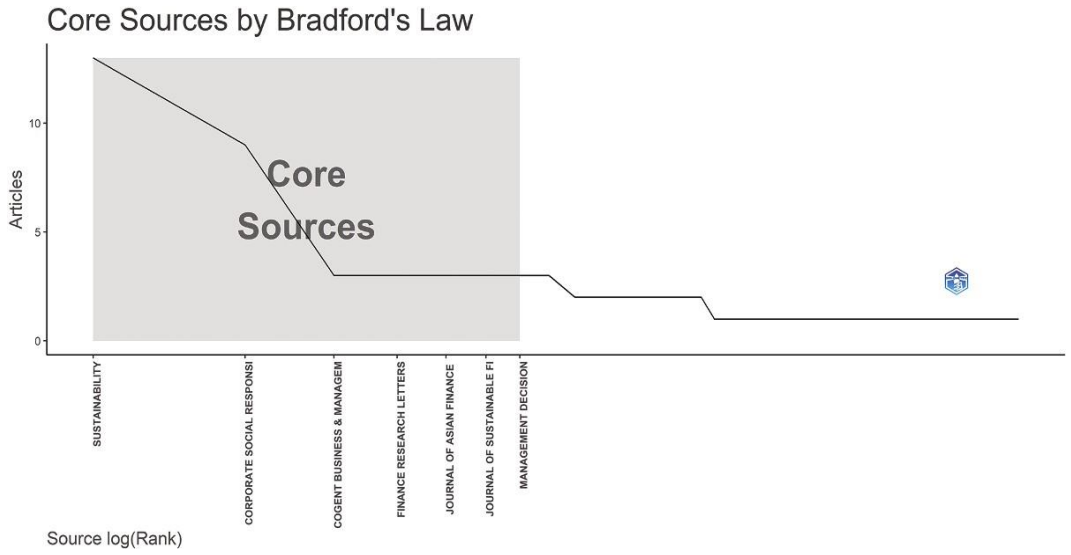


Figure 4 - Graphic resume of Bradford's Law

Note: Generated by Biblioshiny

Table 3 - Bradford's Law Zone Distribution

Zone	Total of Journals	Total_Frequency	Cumulative Frequency
Zone 1	7	37	37
Zone 2	26	36	73
Zone 3	35	35	108

Note: Extracted from Biblioshiny software.

Table 4 - Annual Scientific Production and Average Citations Per Year

Source	Documents	Citations
Sustainability	13	155
Corporate Social Responsibility and Environmental Management	9	259
Cogent Business & Management	3	56
Finance Research Letters	3	87
Journal of Asian Finance Economics and Business	3	40
Journal of Sustainable Finance & Investment	3	11
Management Decision	3	57
Review of Managerial Science	3	68

Note: Extracted from Biblioshiny software.

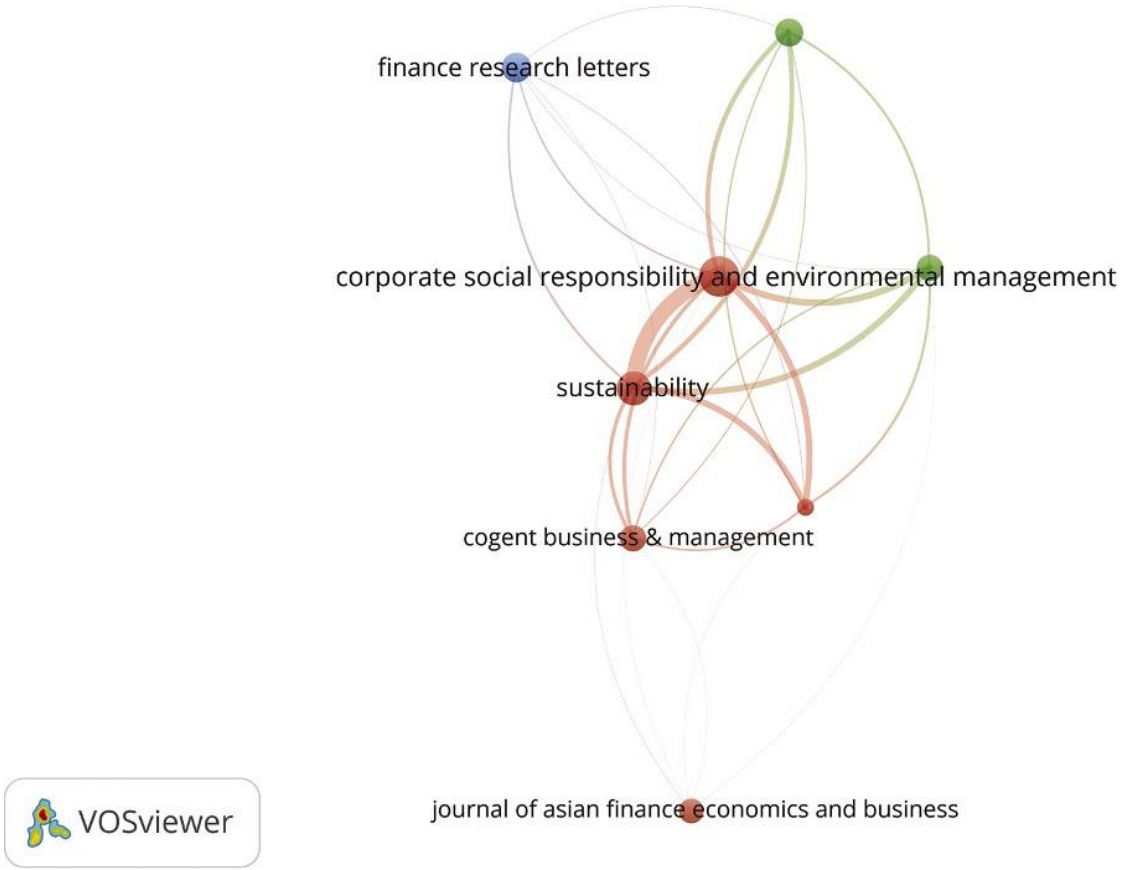


Figure 5 - Bibliographic coupling maps by sources (Journals)

Note: Generated by VOSviewer.

At last, in the Green Cluster, the focus shifts to a more nuanced exploration of CSR and ESG initiatives within specific contexts, such as food safety and finance, emphasizing the strategic alignment of these initiatives with corporate goals. Jihadi et al. (2021) examine the food industry and find that financial ratios (e.g., liquidity and leverage) moderate the impact of CSR activities on firm value, with firms possessing higher liquidity and lower leverage better positioned to leverage CSR for financial gains. Brooks et al. (2020) investigate the finance sector and reveal that firms adopting differentiation strategies benefit more from CSR initiatives in terms of profitability and firm value, highlighting the importance of aligning CSR efforts with a firm's strategic objectives. Naseem et al. (2019) further explore the moderating role of financial ratios in the CSR-financial performance relationship across various industries in emerging markets, finding that high leverage can dampen the positive effects of CSR on firm value. This cluster emphasizes the need for firms to carefully consider their strategic positioning, financial health, and market environment when designing and implementing CSR and ESG strategies to ensure maximum financial and competitive gains.

Table 5 applies Lotka's Law to analyze author productivity, showing he relationship between the number of authors and publications in moderating variables and the ESG/CSR-CFP research field. It illustrates a typical distribution pattern of authorship in academic literature, where a few authors are highly prolific, while the majority contribute only occasionally.

In this dataset, 296 authors wrote only 1 document each — accounting for 95.2 % of the total — reflecting a large pool of infrequent contributors. A smaller group of 14 authors — representing 4.5% — wrote 2 documents each, showing moderate productivity. Finally, a single author wrote three documents, showcasing the high productivity that Bradford's Law predicts for a "core" group of authors who contribute most significantly to literature. This pattern confirms the law's principle that a few authors generate most publications in any field, highlighting the uneven distribution of scientific output. Figure 6 illustrates author productivity by Lotka's Law, highlighting the disproportionate contribution of a few highly prolific authors.

Table 5 - Author´s Productivity through Lotka's Law

Documents Written	Nº of Authors	Proportion of Authors
1	296	95,2%
2	14	4,5%
3	1	0,3%

Source: Biblioshiny

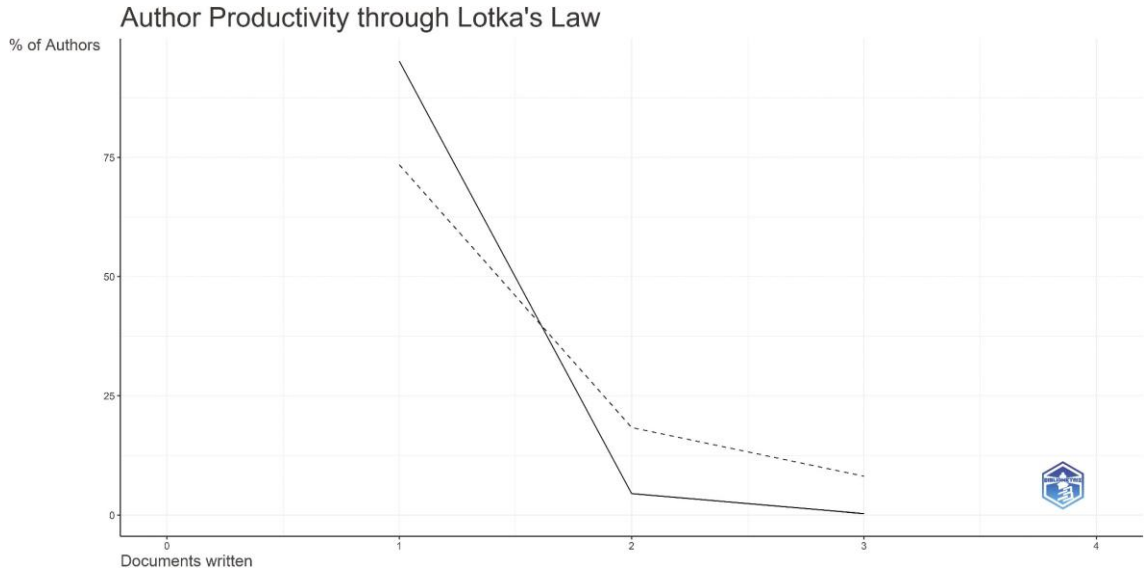


Figure 6 - Author productivity through Lotka's Law

Note: Generated by Biblioshiny.

The VOSviewer software permits the creation of maps based on bibliographic data. We applied the bibliographic coupling analysis based on the shared references. Figure 7 shows the bibliographic coupling map of articles, identifying key clusters and the intellectual structure of the field as determined by shared references. We identified five research clusters: yellow, green, blue, red, and purple. Each one is based on the shared bibliographic references.

In examining the dynamics of CSR and its impact on financial performance, the Yellow Cluster primarily focuses on how CSR disclosures influence market perceptions and firm value. Research in this cluster (e. g., Bofinger et al., 2022) highlights that transparency in CSR practices significantly enhances investor confidence and reduces information asymmetry, leading to improved financial performance. Bofinger et al. (2022) found that firms with robust CSR reporting often enjoy higher market valuations, especially in regions with stringent regulatory frameworks and high stakeholder expectations. Moderating variables such as firm size, industry type, and CEO characteristics is also critical. For example, Rashid et al. (2020) show that larger firms benefit more from CSR disclosures due to their greater visibility and scrutiny from investors and regulators. In contrast, Butt et al. (2020) demonstrate that firms in highly regulated or consumer-facing industries, where public scrutiny is intense, see enhanced financial outcomes from engaging in visible CSR activities. Moreover, the power dynamics within firms (e.g., the influence of a strong CEO) can direct the strategic focus of CSR efforts, aligning them more closely with market expectations and thereby amplifying their positive

impact on firm value. These findings underscore the importance of considering both internal and external factors when evaluating the financial benefits of CSR activities.

The Green Cluster delves into the integration of ESG criteria into business strategies and their subsequent impact on firm value and financial stability. Studies within this cluster, such as those by Sreepriya et al. (2023), demonstrate that firms effectively integrating ESG practices into their operations tend to perform better in terms of market valuation and financial resilience. Sreepriya et al. (2023) found that adherence to recognized sustainability standards (e.g., GRI) increases a firm's credibility and trust among investors, which can enhance financial outcomes. Similarly, Nisar Ahmad et al. (2021) emphasize that the financial benefits of ESG practices are moderated by economic conditions and market maturity. In developed markets, where governance structures are robust and investor sentiment favors sustainability, firms with high ESG scores are more likely to attract institutional investors and achieve higher valuations. Conversely, Jihadi et al. (2021) point out that in emerging markets, where ESG practices are still evolving, the impact on firm value may be less pronounced but is gaining traction as awareness and regulatory support increase. These studies collectively highlight that while ESG integration generally boosts firm value, the extent of its impact is influenced by market-specific factors and the broader economic context.

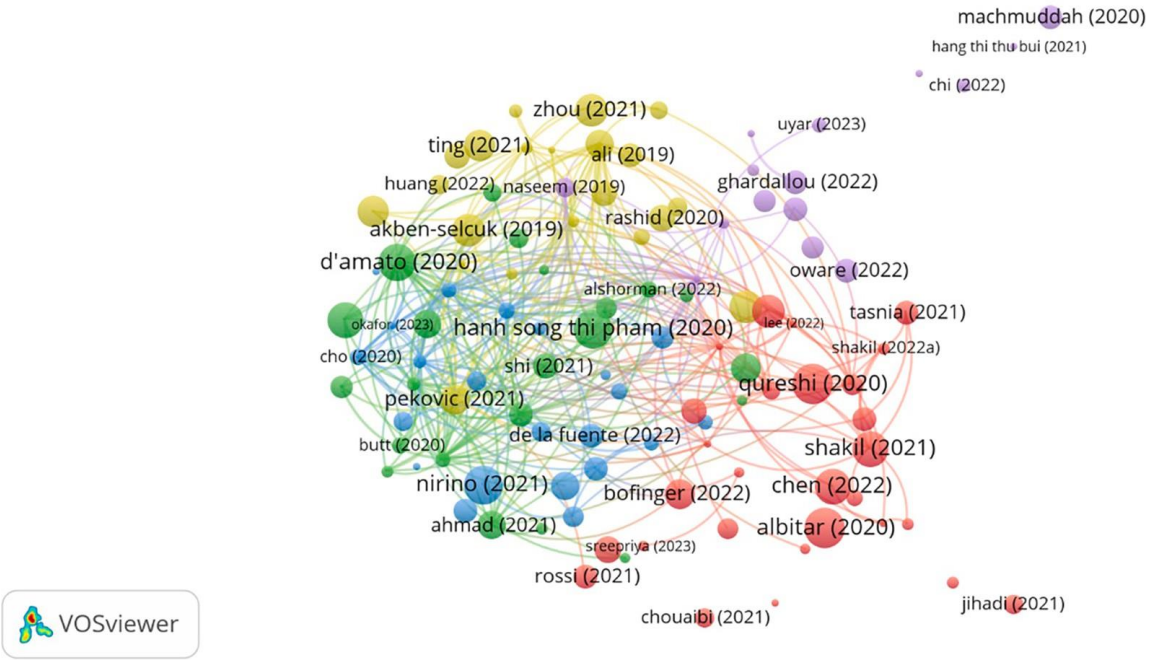


Figure 7 - Bibliographic coupling articles map

Note: Generated by VOSviewer.

The Blue Cluster focuses on identifying and analyzing the moderating variables that influence the outcomes of ESG and CSR initiatives on financial performance. This cluster emphasizes that the effectiveness of these initiatives is not uniform and depends heavily on a range of firm- specific characteristics and external factors. For example, Brooks et al. (2022) show that larger firms often see greater benefits from their ESG and CSR activities due to their higher visibility and the greater scrutiny they receive from investors and the public. Similarly, Vuong (2022) highlights the crucial role of audit firm tenure; firms with longer audit relationships may present more credible ESG disclosures, which enhances their market valuation. Chang et al. (2019) add another layer by examining the impact of media freedom, finding that in markets with higher media freedom, CSR disclosures lead to improved firm value due to increased public scrutiny. These studies underscore the importance of understanding the specific contexts in which ESG and CSR activities are deployed, as the interplay of these moderating variables can significantly alter their impact on financial performance.

The Red Cluster explores the integration of ESG criteria into financial performance metrics and its effect on overall firm performance, particularly during periods of economic volatility. Wang (2023) suggests that firms that incorporate ESG considerations into their strategic management and financial planning are better positioned to navigate market fluctuations and maintain stable performance. For instance, integrating ESG criteria into financial planning can enhance a firm's resilience and adaptability in volatile markets, leading to more sustainable long-term growth (Wang, 2023). This integration is particularly beneficial in industries facing frequent disruptions, where ESG-driven strategies help firms manage risks and seize opportunities more effectively. E-Vahdati (2023) further explores this theme by examining how ESG integration fosters financial innovation, focusing on the development of new metrics for measuring sustainability impacts. The role of economic cycles as a moderating variable is also crucial; during economic downturns, Albuquerque (2020) found that firms with strong ESG practices tend to perform better because their focus on sustainability and ethical practices fosters greater trust and confidence among stakeholders, thereby, stabilizing their financial performance.

Finally, the Purple Cluster comprises niche studies focusing on the unique impacts of ESG considerations within specific sectors or under unique market conditions. This cluster emphasizes that the financial impact of ESG activities can vary significantly depending on sector- specific dynamics and market maturity. For instance, Zhou et al. (2021) investigated the banking sector and find that firms engaging in responsible lending and promoting financial

inclusion see enhanced reputational benefits and reduced risk, leading to improved financial performance. Similarly, in the real estate sector, Sebastiano Cupertino et al. (2021) show that firms prioritizing sustainable building practices and energy efficiency attract environmentally conscious investors, thereby, boosting their market valuation. Ali Uyar et al. (2022) explored the tourism sector, demonstrating that strategic orientations (cost leadership vs. differentiation) significantly affect CSR performance. These studies highlight the importance of tailoring ESG strategies to fit the specific needs and conditions of different sectors, as well as the varying levels of market maturity, to optimize their impact on financial performance.

2.5.3 Keyword analysis (Co-Occurrence analysis)

Keyword analysis using VOSviewer is a valuable bibliometric approach that identifies key research themes and trends by examining the co-occurrence of keywords within academic publications. By con- structing co-occurrence networks, VOSviewer visually maps how frequently specific keywords appear together in the same documents, revealing clusters of related concepts. This helps researchers identify core topics, emerging trends, and potential research gaps within a field. Additionally, the strength of the connections between keywords offers insights into the relationships between different research areas, guiding future research directions and supporting interdisciplinary exploration. VOSviewer is widely used in systematic reviews and research planning due to its ability to quantify and visualize the intellectual structure of a domain effectively.

Figure 8 presents a VOSviewer-generated keyword co-occurrence map depicting the major themes and conceptual linkages within the literature. We analyzed the five clusters that were automatically identified based on the keywords shared by articles.

The Red Cluster focuses on CSR's impact on management strategies and financial outcomes, grounded in Stakeholder Theory and Shareholder Value Maximization. Bofinger et al. (2022) explore how CSR initiatives can enhance market efficiency and firm value, using financial metrics such as Tobin's Q. Their findings align with Stakeholder Theory, suggesting that CSR activities cater to a broader range of stakeholders, thereby, improving market perceptions and, subsequently, financial performance. Similarly, Cho and Tsang (2020) investigate how aligning CSR with product strategies, such as cost leadership or differentiation, enhances a firm's financial performance. The research supports the idea that CSR, when integrated into strategic business operations, can significantly boost firm value, especially when external market sentiment toward sustainability is favorable.

The Blue Cluster delves into the interplay between financial performance, corporate governance, and CSR, with a foundation in Agency Theory and Stakeholder Theory. Rashid et al. (2020) examined how CEO power moderates the relationship between CSR and financial performance, revealing that stronger governance mechanisms can enhance the positive effects of CSR on the firm value measured by Tobin's Q. This finding aligns with Agency Theory, which posits that effective governance can mitigate agency problems and ensure that CSR activities are aligned with shareholder interests. Guo et al. (2020) further explore this dynamic, finding that governance structures, such as board size and independence, significantly influence how CSR initiatives translate into firm value, underscoring the importance of robust governance frame- works in maximizing the financial benefits of CSR.

The Green Cluster emphasizes sustainability strategies and their impact on performance metrics, drawing on Signaling Theory and Legitimacy Theory. Vuong (2022) examines how investor sentiment toward CSR affects financial performance, highlighting that positive investor sentiment can significantly boost financial outcomes. This supports Signaling Theory, which suggests that companies engaging in CSR signal their commitment to ethical practices, thereby, attracting positive investor attention. Chang et al. (2020) investigated the moderating role of media freedom on the CSR-financial performance relationship, demonstrating that greater media freedom amplifies the positive impact of CSR on financial outcomes. This aligns with Legitimacy Theory, which posits that organizations strive to operate within the norms and values of their social environment to maintain legitimacy. The Yellow Cluster focuses on risk management, shareholder value, and their determinants, underpinned by Risk Management Theory and Agency Theory. Leopizzi (2023) examines how CSR initiatives affect firm value in the context of various risk management strategies, finding that effective risk management enhances the positive effects of CSR on firm value. Sreepriya et al. (2023) discusses the role of GRI compliance as a moderating factor in the CSR-financial performance link, emphasizing the importance of transparency and risk mitigation. These findings suggest that companies that integrate robust risk management practices and adhere to sustainability reporting standards are better positioned to maximize the financial benefits of CSR.

The Purple Cluster explores governance issues, ownership structures, and legitimacy concerning CSR and financial performance, drawing from Agency Theory and Stakeholder Theory. Brooks et al. (2022) analyzed the relationship between audit firm tenure and CSR, demonstrating that longer audit firm tenure, combined with robust CSR practices, enhances firm value. Javeed and Lefen (2019) examine how different ownership structures impact the CSR-financial performance relationship, revealing that firms with diverse ownership structures

are better able to capitalize on CSR initiatives. These studies underscore the importance of governance mechanisms in moderating the impact of CSR activities on firm value, suggesting that internal governance factors play a crucial role in shaping CSR effectiveness.

Finally, the Cyan Cluster examines ESG criteria, particularly environmental performance, and their financial implications, guided by the RBV and Institutional Theory. Ahmad et al. (2021) revisit the impact of ESG on financial performance, highlighting that environmental performance has a particularly strong impact on financial metrics such as ROA and stock returns. Naseem et al. (2019) discuss how traditional financial metrics such as liquidity and leverage moderate the impact of ESG ef- forts on overall financial performance, indicating that companies with healthier financial positions are better positioned to benefit from ESG investments. These findings suggest that firms with solid financial foundations can more effectively leverage their ESG initiatives to improve financial outcomes.

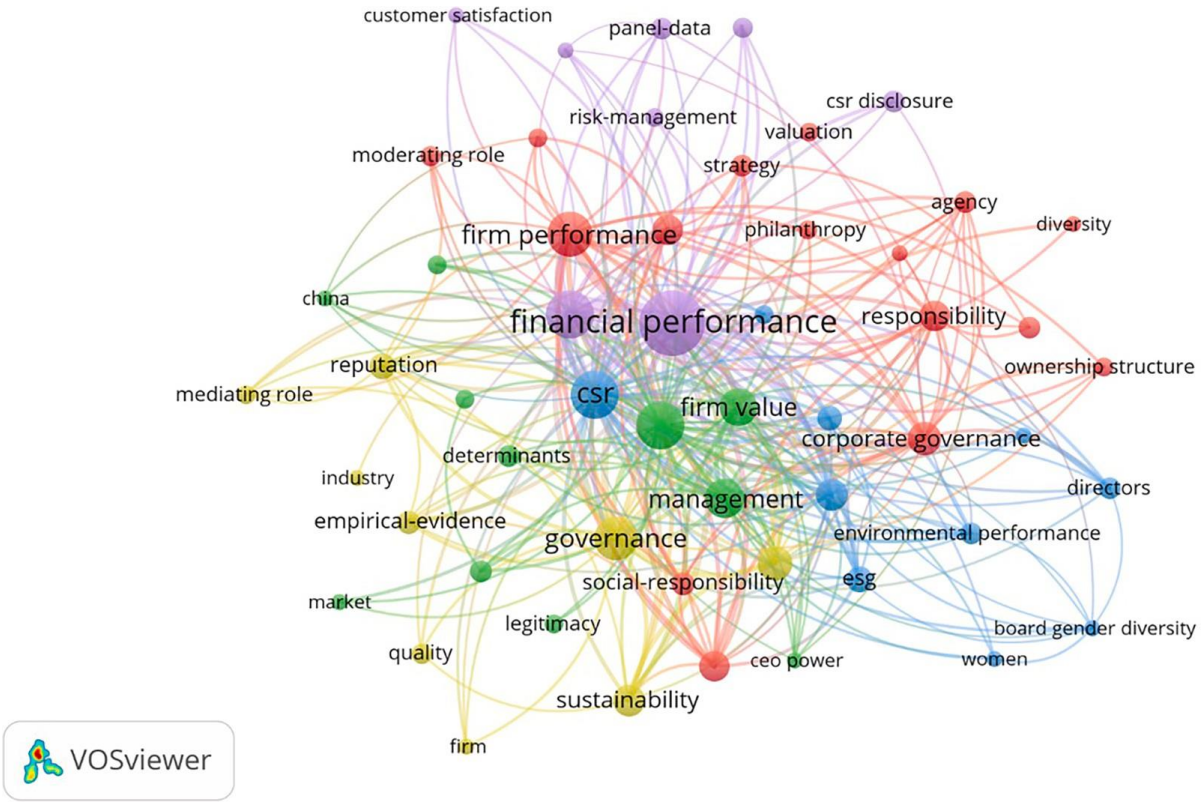


Figure 8 - Keyword Map

Note: Generated by VOSviewer.

2.5.4 Moderating variables analysis

Based on the frequent count analysis of the 108 articles, we identified the most common moderating variables at play in the ESG/CSR-CFP research. Table 6 provides a detailed

classification of the articles based on the primary moderating variables identified. This categorization highlights critical factors such as audit firm tenure, audit quality, board characteristics, and CEO characteristics, frequently examined to assess their moderating effects on the ESG/CSR-CFP relationship.

Figure 9 presents a frequency count of these moderating variables, highlighting which factors have been most examined in recent studies. This visualization helps to underscore the importance of specific moderating variables, such as board and firm characteristics, in shaping the ESG/CSR-CFP dynamic. The five most frequent moderating variables influencing the relationship between ESG/CSR measures and CFP are Board Characteristics, Firm Characteristics, Ownership Structure, Social Influence, Industry Characteristics, and CEO Characteristics. These categories collectively represent the most studied moderators shaping the impact of sustainability practices on financial outcomes.

Board Characteristics are the most frequent moderating variables, accounting for 19.3 % of the total occurrences. This category includes elements such as board size, diversity, independence, and the presence of sustainability committees. The board's composition and structure significantly influence a company's strategic direction, especially concerning ESG and CSR initiatives. A well-composed board, characterized by diversity and independence, is better equipped to oversee and sup- port sustainability practices that align with long-term shareholder value. Strong governance frameworks provided by the board can enhance the credibility and effectiveness of ESG/CSR strategies, thus, positively impacting financial performance.

Similarly, Firm Characteristics make up 17.4 % of the moderating variables, reflecting their substantial influence. This category includes factors such as firm size, age, profitability, leverage, and operational efficiency, which affect a firm's ability to implement and benefit from ESG and CSR initiatives. For example, larger firms often have more re- sources and capabilities to invest in comprehensive sustainability pro- grams, while firms with high leverage may face financial constraints that limit their ability to engage in CSR activities. The prominence of firm characteristics as a moderator underscores the importance of under- standing company-specific traits when assessing the financial impacts of sustainability efforts.

Ownership Structure also plays a critical role, accounting for 7.3 % of the total moderating variables. This category includes aspects such as ownership concentration, institutional ownership, family ownership, and shareholder activism. Different ownership structures can influence a firm's strategic priorities and openness to adopting ESG and CSR measures. For instance, firms with concentrated ownership, such as those dominated by a few

large shareholders, might prioritize long-term sustainability if it aligns with the owners' preferences and objectives. Conversely, firms with dispersed ownership might experience challenges in achieving consensus on sustainability strategies, potentially impacting their financial benefits from ESG and CSR initiatives. The high frequency of ownership structure as a moderator highlights its role in shaping corporate governance and strategic decision-making.

Table 6 - Moderating Variable Classification

Moderator	**Articles**
Audit Firm Tenure	Brooks, L. Z., et al. (2020)., Brooks, L. Z., et al. (2022).
Audit Quality	Dakhli, A. (2022)., Fuadah, L. L., et al. (2022)., Byrne, J. A., et al. (2022).
Board Characteristics	Rossi, M., et al. (2021)., Albitar, K., et al. (2020)., Chijoke-Mgbame, A. M., et al. (2020)., Karim, S., et al. (2020)., Lee, S.-P. (2021)., Li, Y., et al. (2022)., Sampong, F., et al. (2021)., Ahmad, N., et al. (2023)., Pekovic, S., & Vogt, S. (2021)., Shakil, M. H., et al. (2022)., Schoenmaker, D., & Schramade, W. (2022)., Kahloula, I., et al. (2022)., Al-Shammari, M. A., et al. (2023)., Nirino, N., et al. (2022)., Nekhili, M., et al. (2021)., Brinette, S., et al. (2023)., Shakil, M. H. (2021)., E-Vahdati, S., et al. (2023)., Ooi, C.-A., et al. (2022)., Yeon, J., et al. (2021)., Butt, A. A., et al. (2020).
CEO Characteristics	Ghardallou, W. (2022)., Pham, H. S. T., & Tran, H. T. (2020)., Almulhim, A. A., & Aljughaiman, A. A. (2023)., Velte, P. (2020)., Okafor, C. E., et al. (2023).
Competitive Environment	Rasheed, B., & Ahmad, M. (2022)., Bashir, M. (2022).
Cultural Factors	Shi, W., & Veenstra, K. (2021)., Le, M.-H., et al. (2022)., DasGupta, R., & Roy, A. (2023)., Liu, S., et al. (2023).
Sustainability Indicators	Chen, Z., & Xie, G. (2022)., Grassmann, M. (2021)., J., S., et al. (2023)., Zhou, G., et al. (2021).
Economic Indicators	Alfalih, A. A. (2022).
Financial Aspects	Saadaoui, A., & Ben Salah, O. (2021)., Coleman, M., & Wu, M. (2021)., Ryu, D. (2019).
Financial Flexibility	Guo, Z., et al. (2020)., Guo, Z., et al. (2020).
Firm Characteristics	Javed, A., et al. (2020)., Dkhili, H. (2023)., Akben-Selcuk, E. (2019)., Machmuddah, Z., et al. (2020)., Ting, P.-H. (2021)., Handayati, P., et al. (2022)., Shakil, M. H. (2022)., Jiang, G. F., et al. (2020)., Wirawan, A. W., et al. (2020)., Wang, X., et al. (2023)., Alshorman, S., et al. (2022)., Aqabna, S. M., et al. (2023)., Espinosa-Méndez, C., & Maquieira, C. P. (2023)., D'Amato, A., & Falivena, C. (2020)., Abdi, Y., et al. (2022)., Sánchez-Infante Hernández, J. P., et al. (2020)., Al-Dah, B. (2019)., Naseem, M. A., et al. (2019)., Ahmad, N., et al. (2021).
Firm Strategy	Uyar, A., et al. (2022).
Growth and Market-Based Assets	de la Fuente, G., et al. (2022)., Lin, W. L., et al. (2020).

Industry Characteristics	Jeong, N. (2021)., Wang, K., & Qiao, Y. (2022)., Jia, X. (2020)., Qureshi, M. A., et al. (2020)., Kaupke, K., & Knyphausen-Aufseß, D. (2023).
Internationalization	Sang, M., et al. (2022).
Market Sentiment	Bofinger, Y., Heyden, K. J., & Rock, B. (2022)., Vuong, N. B. (2022).
Marketing and Media	Sun, W., et al. (2019)., Fu, L., et al. (2022)., Chang, K., et al. (2019).
Other	Wen, H., et al. (2022)., Asante-Appiah, B., & Lambert, T. A. (2023)., Boulhaga, M., et al. (2023)., Cupertino, S., et al. (2021).
Ownership Structure	Ang, R., et al. (2022)., Chi, L. H. D., et al. (2022)., Muda, I., et al. (2019)., Ali, S., et al. (2019)., Bui, H. T. T. (2021)., Bai, Y. (2022)., Rastogi, S., et al. (2023)., Tarighi, H., et al. (2022).
Political and Governance Factors	Huang, J. (2022)., Lee, E. T., & Li, X. (2022).
Product and Strategic Management	Cho, E., & Tsang, A. (2020).
R&D Investment	Al-Shammari, M. A., et al. (2022)., Duan, Y., et al. (2023).
Regulatory Environment	Forgione, A. F., et al. (2020).
Social Influence	Khan, A., et al. (2023)., Garel, A., et al. (2022)., Oware, K. M., & Mallikarjunappa, T. (2022)., Zhao, W., et al. (2022)., Gallego-Álvarez, I., & Pucheta-Martínez, M. C. (2022)., Bifulco, G. M., et al. (2023).
Social Ties	Jang, S. S., et al. (2019).
Stakeholder Influence	Rashid, A., et al. (2020)., Tsang, A., et al. (2022).
Tax and Regulation	Tasnia, M., et al. (2021).
Technology and Innovation	Chouaibi, S., & Chouaibi, J. (2021).

Note: Created by Authors

In addition to governance-related factors, Social Influence represents 5.5 % of the moderating variables. This category captures the impact of societal expectations, cultural norms, and stakeholder pressures on a company's ESG and CSR practices. Factors such as community engagement, social responsibility pressures from customers and investors, and media attention significantly influence corporate behavior. Companies that effectively respond to social pressures often experience enhanced reputation, customer loyalty, and, consequently, improved financial performance. The presence of social influence as a frequent moderator emphasizes its importance in driving corporate sustainability strategies and outcomes.

Industry Characteristics are also prominent, identified as moderating variables in 4.6 % of the cases. This category includes sector-specific dynamics, regulatory requirements, competitive intensity, and market maturity. Certain industries, such as energy, mining, and manufacturing, face higher regulatory scrutiny and societal expectations due to their

environmental impact. Firms in these sectors may need to adopt more robust ESG strategies to comply with regulations and maintain their social license to operate. Conversely, firms in less-regulated industries might not experience the same level of pressure, leading to varied financial impacts of their sustainability initiatives. The frequency of this moderator underscores the significant role of industry context in determining the financial implications of ESG and CSR activities.

Finally, CEO Characteristics also account for 4.6 % of the moderating variables. This category includes factors such as CEO power, tenure, ethical orientation, and leadership style. The CEO's vision and commitment to sustainability can significantly influence a firm's strategic direction and the effectiveness of its ESG/CSR initiatives. A CEO who prioritizes ethical governance and sustainability is likely to foster a corporate culture that values long-term success over short-term gains, which can enhance a firm's reputation and financial stability. The frequent mention of CEO characteristics as a moderating variable underscores the critical influence of executive leadership on the success of sustainability initiatives.

In conclusion, the aforementioned moderating variables collectively account for a significant portion of the dataset. These variables highlight the complex interplay between governance, leadership, social pressures, industry-specific contexts, and ownership dynamics in shaping the financial outcomes of ESG and CSR initiatives. Understanding these moderators is crucial for businesses looking to optimize their sustain- ability strategies and enhance financial performance.

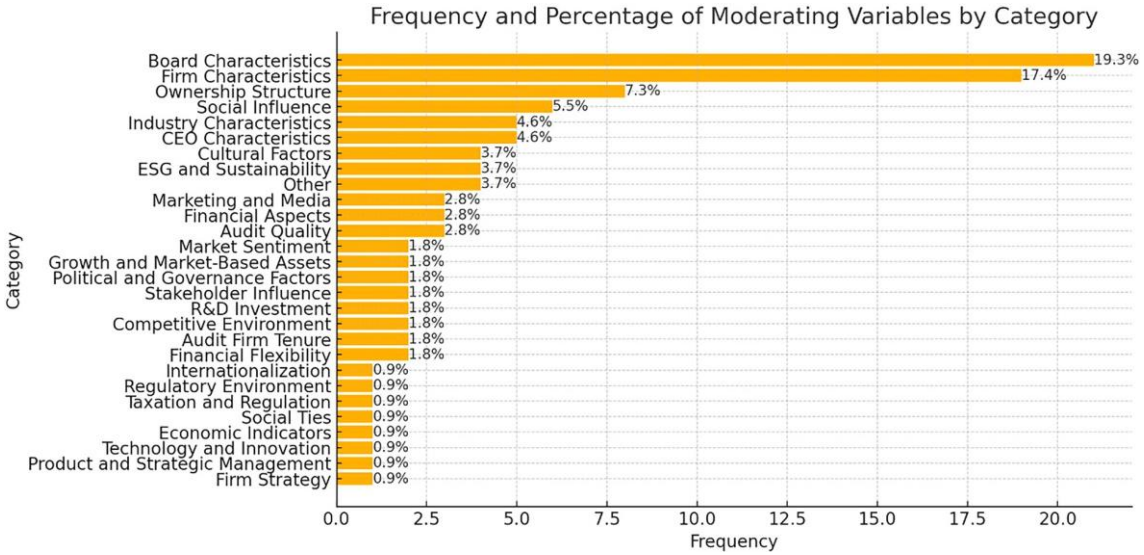


Figure 9 - Moderating variables frequency count

Note: Elaborated by authros

2.6 SYSTEMATIC REVIEW

Following the proposed classification framework by Junior and Godinho Filho (2010), Van Kampen et al. (2012), and Jabbour (2013), we applied a codification system and a frequency count analysis to identify research gaps. We conducted a comprehensive review and analysis of 108 selected articles, subsequently categorizing them based on their specifications. Table 7 provides a complete categorization matrix of articles based on the classification and coding framework. This classification endeavor facilitates a coherent comprehension and analysis of the knowledge and methodologies in the selected papers.

Table 7 - Categorized Articles based on the Classification and Coding Framework

References	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Abdi, Y., et al. (2022).	1A	2C/ 2F/ 2B/ 2A	3A	4C/ 4A	5I/ 5A	6B	7A/ 7C
Ahmad, N., et al. (2021).	1A/ 1B	2C/ 2F/ 2B/ 2A	3A	4C/ 4D	5I/ 5E	6B	7B/ 7C
Akben-Selcuk, E. (2019).	1B	2G	3D	4C/ 4D	5I	6B	7B/ 7C
Albitar, K., et al. (2020).	1A	2A	3D	4C/ 4D	5I/ 5A/ 5C	6B	7C
Al-Dah, B. (2019).	1A	2G	3D	4D/ 4F	5I	6B	7B/ 7C
Alfalih, A. A. (2022).	1A/ 1B	2C/ 2F/ 2B/ 2A	3H	4C/ 4D	5I/ 5A	6C	7C
Ali, S., et al. (2019).	1A	2F/ 2G	3C	4C/ 4D	5I/ 5A	6B	7C
Almulhim, A. A., & Aljughaiman, A. A. (2023).	1B	2C/ 2F/ 2B/ 2A	3D	4D/ 4C	5I	6C	7B/ 7A/ 7C
Al-Shammari, M. A., et al. (2022).	1A	2G	3A	4C/ 4D	5I	6C	7B/ 7C
Al-Shammari, M. A., et al. (2023).	1B	2G	3D/ 3A	4D	5I	6B	7B/ 7C
Alshorman, S., et al. (2022).	1C	2C/ 2F/ 2B/ 2A/ 2G	3H	4C	5I	6B	7B/ 7C
Ang, R., et al. (2022).	2E	2C/ 2F/ 2B/ 2A	3E	4C	5G	6C	7B/ 7C
Aqabna, S. M., et al. (2023).	1A/ 1B	2C/ 2F/ 2B/ 2A/ 2G	3D	4C/ 4D/ 4A	5I/ 5H/ 5C	6C	7B/ 7A/ 7C
Asante-Appiah, B., & Lambert, T. A. (2023).	1A/ 1B	2A	3C	4A/ 4C	5I	6B	7B/ 7C
Bai, Y. (2022).	1D	2G	3A	4C/ 4D	5I	6B	7C
Bashir, M. (2022).	1A/ 1B	2G/ 2D	3B	4C/ 4D	5D/ 5F	6C	7B/ 7A/ 7C

Bifulco, G. M., et al. (2023).	1F	2A	3F	4C/ 4D	5I/ 5A/ 5J	6B	7B/ 7C
Bofinger, Y., Heyden, K. J., & Rock, B. (2022).	1A	2A	3G/ 3F	4C	5I/ 5J	6C	7B/ 7C
Boulhaga, M., et al. (2023).	1A	2A	3D	4C/ 4D	5I/ 5A	6C	7B/ 7C
Brinette, S., et al. (2023).	1A	2A	3D	4C/ 4D	5A/ 5E	6B	7B/ 7C
Brooks, L. Z., et al. (2020).	1A	2F/ 2A/ 2G	3C	4C	5I/ 5J	6A	7B/ 7C
Brooks, L. Z., et al. (2022).	1A	2A/ 2G	3C	4C/ 4D	5J/ 5F	6A	7B/ 7C
Bui, H. T. T. (2021).	1B	2F/ 2G/ 2D	3D	4C	5A/ 5J	6C	7A/ 7C
Butt, A. A., et al. (2020).	1A/ 1B	2F/ 2G	3D/ 3A	4C/ 4D	5A	6B	7A/ 7C
Byrne, J. A., et al. (2022).	1A/ 1B	2G	3C	4C/ 4D	5I	6B	7A/ 7C
Chang, K., et al. (2019).	1A	2C/ 2F/ 2G	3E	4C	5I/ 5A	6B	7B/ 7C
Chen, Z., & Xie, G. (2022).	1A/ 1B/ 2E	2A	3F	4C/ 4D	5I	6C	7B/ 7C
Chi, L. H. D., et al. (2022).	1B	2G	3D	4C/ 4E/ 4B	5G	6A	7B/ 7C
Chijoke-Mgbame, A. M., et al. (2020).	1B	2F/ 2G	3D/ 3A	4D/ 4C	5I	6B	7A/ 7C
Cho, E., & Tsang, A. (2020).	1A	2G	3G	4B	5I/ 5A	6B	7C
Chouaibi, S., & Chouaibi, J. (2021).	1A	2F	3G/ 3F	4C/ 4F	5I/ 5A/ 5H	6C	7B/ 7C
Coleman, M., & Wu, M. (2021).	1A/ 1B	2B	3H	4D/ 4C	5G	6B	7A/ 7C
Cupertino, S., et al. (2021).	1B	2C/ 2F/ 2B/ 2A	3A	4B/ 4C	5A/ 5J	6A	7B/ 7C
Dakhli, A. (2022).	1A/ 1B	2G	3C	4D/ 4C	5A/ 5J	6B	7B/ 7C
D'Amato, A., & Falivena, C. (2020).	1A/ 1F	2G	3A	4C/ 4D	5I/ 5A/ 5H	6B	7B/ 7A/ 7C
DasGupta, R., & Roy, A. (2023).	1A/ 1B	2A	3E	4E	5I	6B	7B/ 7C
de la Fuente, G., et al. (2022).	2E/ 1A	2F/ 2A	3G	4A/ 4F/ 4C/ 4D	5H/ 5J/ 5F	6B	7B/ 7C
Dkhili, H. (2023).	1A	2A	3B	4C/ 4D	5I/ 5A/ 5H/ 5J	6C	7B/ 7C
Duan, Y., et al. (2023).	1A	2A	3G	4C	5I/ 5A	6A	7B/ 7C
Espinosa-Méndez, C., & Maquieira, C. P. (2023).	1A	2C/ 2F/ 2B/ 2A	3H	4D	5C	6C	7B/ 7C

E-Vahdati, S., et al. (2023).	1F	2C/ 2F/ 2B/ 2A	3D	4C	5I/ 5A	6C	7B/ 7C
Forgione, A. F., et al. (2020).	2E	2C/ 2F/ 2B/ 2G	3C	4D/ 4C	5J/ 5F	6B	7B/ 7C
Fu, L., et al. (2022).	1C	2F/ 2A	3F	4C	5I	6B	7B/ 7C
Fuadah, L. L., et al. (2022).	1A/ 1B	2A	3C	4C/ 4D	5A/ 5C	6C	7C
Gallego-Álvarez, I., & Pucheta-Martínez, M. C. (2022).	1A/ 1B	2E	3D/ 3A	4C/ 4D	5I/ 5H/ 5J	6B	7B/ 7C
Garel, A., et al. (2022).	1A	2C/ 2F/ 2G	3F	4C	5I/ 5J	6B	7B/ 7C
Ghardallou, W. (2022).	1A/ 1B	2A/ 2G	3D	4C/ 4D/ 4B	5I/ 5E	6C	7B/ 7A/ 7C
Grassmann, M. (2021).	1A	2C/ 2F/ 2G/ 2D	3F/ 3C	4D/ 4C	5I	6A	7B/ 7C
Guo, Z., et al. (2020).	1A	2F/ 2G	3H/ 3G	4D	5I/ 5A	6B	7A/ 7C
Guo, Z., et al. (2020).	1A/ 1C	2G	3H/ 3G	4C/ 4D/ 4B	5A	6B	7B/ 7C
Handayati, P., et al. (2022).	1A	2F/ 2G	3A/ 3A	4C	5A/ 5J	6A	7A/ 7C
Huang, J. (2022).	1B	2G	3C	4C/ 4D	5I/ 5A/ 5J	6B	7B/ 7C
J., S., et al. (2023).	1A/ 1B	2A	3F	4C	5G	6C	7B/ 7C
Jang, S. S., et al. (2019).	1B	2G	3E	4C/ 4D	5I/ 5A	6B	7B/ 7C
Javed, A., et al. (2020).	1A/ 1B	2G	3D	4D/ 4C	5A	6A	7A/ 7C
Jeong, N. (2021).	1B	2F	3B	4F	5I/ 5J/ 5C	6A	7B/ 7C
Jia, X. (2020).	1A	2G	3B	4C	5I/ 5J/ 5C	6C	7B/ 7A/ 7C
Jiang, G. F., et al. (2020).	2E	2C/ 2F/ 2G	3B	4C	5A/ 5J/ 5E	6C	7B/ 7C
Kahloula, I., et al. (2022).	1A/ 1B	2A/ 2G	3D	4C/ 4D/ 4E	5I/ 5A	6C	7B/ 7A/ 7C
Karim, S., et al. (2020).	1A/ 1B	2B/ 2G	3D	4D/ 4C	5I/ 5J	6C	7B/ 7A/ 7C
Kaupke, K., & Knyphausen-Aufseß, D. (2023).	1A	2A	3B	4C	5I/ 5F	6B	7B/ 7C
Khan, A., et al. (2023).	1B	2C/ 2F/ 2G	3F	4C/ 4D	5I	6A	7B/ 7C
Le, M.-H., et al. (2022).	1A	2G	3D	4D/ 4C	5I/ 5J	6B	7B/ 7A/ 7C
Lee, E. T., & Li, X. (2022).	1B	2A	3E	4C/ 4A	5I/ 5E/ 5K	6C	7B/ 7C
Lee, S.-P. (2021).	1B	2C/ 2G	3D	4C/ 4D	5A/ 5J/ 5E	6B	7B/ 7C

Li, Y., et al. (2022).	1A/ 1B	2G	3D	4C	5I/ 5J	6C	7C
Lin, W. L., et al. (2020).	1A	2G	3C	4C	5J/ 5F	6C	7B/ 7C
Liu, S., et al. (2023).	2E	2B	3E	4D	5I	6B	7B
Machmuddah, Z., et al. (2020).	1A	2G	3A	4C	5B/ 5A	6B	7C
Muda, I., et al. (2019).	1A	2F/ 2G	3C	4C/ 4D	5A	6A	7A/ 7C
Naseem, M. A., et al. (2019).	1A	2G	3A	4C/ 4D/ 4E	5I/ 5D	6A	7C
Nekhili, M., et al. (2021).	1A	2C/ 2F/ 2B/ 2A	3D	4C/ 4D	5I/ 5A/ 5J	6C	7B/ 7A/ 7C
Nirino, N., et al. (2022).	1C	2C/ 2F	3D	4C/ 4D	5I/ 5A/ 5C	6B	7C
Okafor, C. E., et al. (2023).	1A	2F/ 2G	3D	4D	5G	6B	7B/ 7C
Ooi, C.-A., et al. (2022).	1A	2F	3D	4C	5I	6A	7B/ 7C
Oware, K. M., & Mallikarjunappa, T. (2022).	1A/ 1B/ 1F	2G/ 2D	3F	4C/ 4D	5I	6B	7C
Pekovic, S., & Vogt, S. (2021).	1A/ 1B	2F/ 2G	3H	4D/ 4C	5I	6C	7B/ 7C
Pham, H. S. T., & Tran, H. T. (2020).	1A/ 1B	2A/ 2G	3D	4C/ 4E	5I/ 5A/ 5J	6C	7B/ 7C
Qureshi, M. A., et al. (2020).	1F	2C/2F/ 2B	3B	4C	5I/ 5J	6B	7B/ 7C
Rasheed, B., & Ahmad, M. (2022).	1B/ 1F	2G/ 2D	3B	4C/ 4D	5I/ 5J	6C	7A/ 7C
Rashid, A., et al. (2020).	1A	2B	3E	4C/ 4D	5I/ 5J	6B	7B/ 7A/ 7C
Rastogi, S., et al. (2023).	1A	2A	3D	4D/ 4C	5E	6B	7A/ 7C
Rossi, M., et al. (2021).	1B	2G	3D/ 3A	4C/ 4D	5I/ 5A/ 5C	6B	7B/ 7A/ 7C
Ryu, D. (2019).	1A	2G	3H	4C	5I/ 5A/ 5J	6B	7B/ 7C
Saadaoui, A., & Ben Salah, O. (2021).	1B	2C/ 2F/ 2B/ 2G	3H	4C/ 4D	5A	6B	7B/ 7C
Sampong, F., et al. (2021).	1A	2F/ 2G	3D	4D/ 4F	5I/ 5A/ 5H/ 5C	6B	7B/ 7C
Sánchez-Infante Hernández, J. P., et al. (2020).	2E	2C/ 2F/ 2G	3A	4C	5A/ 5J	6B	7C
Sang, M., et al. (2022).	1B	2F	3G	4C/ 4D	5H/ 5J	6A	7B/ 7C
Schoenmaker, D., & Schramade, W. (2022).	1A/ 1B	2A/ 2G	3D	4C/ 4D	5I	6B	7B/ 7C

Shakil, M. H. (2021).	1C	2A	3D	4C	5I/ 5A	6B	7B/ 7C
Shakil, M. H. (2022).	1C	2A	3A	4C	5I/ 5A/ 5J	6B	7B/ 7C
Shakil, M. H., et al. (2022).	1A	2C/ 2F/ 2G	3G	4C	5I/ 5J/ 5E	6A	7B/ 7C
Shi, W., & Veenstra, K. (2021).	1A/ 1B	2C/ 2F/ 2A	3E	4C/ 4E	5J/ 5F	6B	7B/ 7C
Sun, W., et al. (2019).	1A	2F/ 2G	3H	4D/ 4B/ 4F/ 4C	5I/ 5H/ 5J	6B	7A/ 7C
Tarighi, H., et al. (2022).	1C	2F/ 2G	3C	4D/ 4C	5I/ 5J	6C	7C
Tasnia, M., et al. (2021).	1A/ 1B	2C/ 2F	3D	4C	5D/ 5J	6B	7B/ 7C
Ting, P.-H. (2021).	1A/ 1B	2G	3A	4C	5I/ 5J	6B	7B/ 7A/ 7C
Tsang, A., et al. (2022).	1A	2G	3H/ 3E/ 3C	4C/ 4D	5I	6A	7B/ 7C
Uyar, A., et al. (2022).	1A	2C/ 2F/ 2B/ 2G	3G	4D	5I/ 5A	6B	7B/ 7C
Velte, P. (2020).	1B	2C/ 2F/ 2B/ 2A	3D	4C	5I/ 5A	6B	7B/ 7C
Vuong, N. B. (2022).	1A/ 1B	2C/ 2F/ 2B/ 2A	3H	4C/ 4E	5I/ 5A	6B	7B/ 7C
Wang, K., & Qiao, Y. (2022).	1A	2G	3A	4D/ 4C	5G	6B	7B/ 7C
Wang, X., et al. (2023).	1B	2C	3D	4C	5J	6C	7A/ 7C
Wen, H., et al. (2022).	1A	2F/ 2G	3A/ 3A	4C/ 4D	5I/ 5J	6B	7A/ 7C
Wirawan, A. W., et al. (2020).	1A	2F/ 2G	3D	4D	5G	6B	7B/ 7C
Yeon, J., et al. (2021).	1A	2G	3D	4C/ 4D	5G	6B	7B/ 7A/ 7C
Zhao, W., et al. (2022).	1B	2G	3E	4C/ 4D	5I/ 5E	6A	7B/ 7A/ 7C

Note: Produced by Authors.

In the following subsections, we discuss the results of the seven categories provided by the categorization matrices, identifying the existing knowledge base and the gaps in it that require further studies. We graphically show the variable's distribution of each category, providing an easy view of how the knowledge has been explored in the sample articles. We identified the 20 most representative gaps. Each gap is identified by the letter "G" and the numbers 1 to 20 in highlighted text format for easy identification.

2.6.1 Financial performance research gaps

This classification category was structured to list the selected articles according to their Financial Performance Measure. Figure 10 depicts the percentage distribution of these financial performance categories, indicating that Category 1A, which includes valuation metrics such as Tobin's Q and market-to-book ratio, accounts for the highest proportion at 51.06 %. This suggests a predominant focus on market-based performance measures in the literature that is particularly relevant for assessing the impact of ESG and CSR initiatives, which are generally designed to enhance long-term value rather than immediate financial returns (Verma & Mukhtaruddin, 2023; Wagner, 2011).

Profitability metrics (1B), including ROA, return on equity (ROE), and net profit margin, represent another significant area of focus, making up 34 % of the studies. For example, Wagner (2011) shows that socially responsible practices can enhance profitability by driving operational efficiencies, reducing costs, and fostering customer loyalty. However, the impact of moderating variables such as corporate governance structures and regulatory environments in this relationship is less explored, indicating an area for further research. Understanding these moderating effects could provide deeper insights into optimizing financial outcomes through ESG practices.

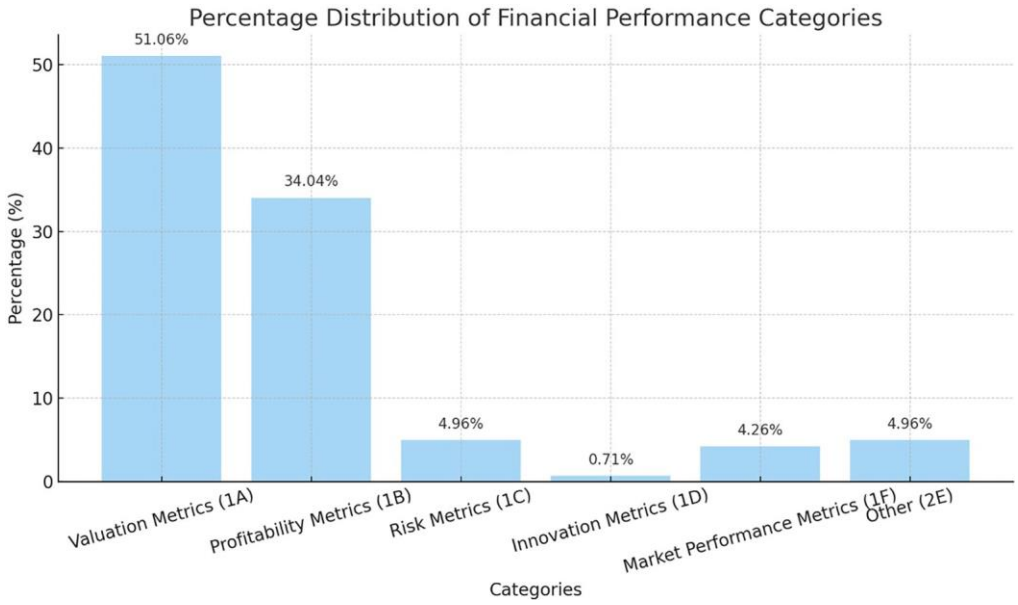


Figure 10 - Percentage distribution of financial performance categories

Note: elaborated by authors

Risk management metrics (1C) are another critical yet underrepresented area in ESG research, constituting less than 5 % of the studies. These metrics, which include credit risk,

volatility, and beta, are essential for assessing how ESG and CSR initiatives contribute to financial risk mitigation. The systematic review by Verma and Mukh- taruddin (2023) underscores the limited exploration of these metrics, particularly in how different moderating variables (e.g., geographic location and industry-specific risks), influence the effectiveness of ESG practices in reducing financial risks. This gap suggests a need for more focused research on the role of ESG in enhancing financial resilience.

G1: Explore risk metrics such as credit risk, volatility, and beta systematic risk in upcoming articles

Innovation metrics (1D), including R&D expenditure and patent counts, are also notably underrepresented in the literature, representing only 0.71 % of the studies. These metrics are crucial for assessing the role of ESG and CSR initiatives in driving innovation within firms. The limited focus on innovation suggests a gap in exploring how sustain- ability practices contribute to long-term financial performance through innovation, particularly when moderated by factors such as industry innovation rates and access to sustainable technologies. The work by Bartolacci et al. (2020) highlights the need for more research to understand the broader implications of ESG on innovation.

G2: Explore innovative metrics in future articles, such as R&D expenditures and patent counts

Emerging financial performance indicators (1E), such as those related to stakeholder value creation and intangible assets, are gaining importance as integrated reporting frameworks such as the GRI become more widely adopted. Common variables in this category include brand value, employee satisfaction, and customer loyalty. The absence of extensive research on Total Shareholder Return (TSR) in the context of ESG and CSR initiatives represents a significant gap in the current literature. TSR is a key indicator of overall shareholder value, combining both capital gains and dividends, and remains underexamined. This gap is noteworthy because TSR directly reflects the financial benefits delivered to shareholders over time, making it a crucial measure of how ESG and CSR efforts translate into tangible returns for investors.

Market performance metrics (1F), such as stock price, market share, and sales growth, account for 4.26 % of the studies. These metrics are vital for understanding the competitive advantages conferred by sustainability practices. The study by Hang et al. (2019) indicates that

stock price directly reflects market perceptions and investor sentiment, and moderating factors such as market volatility and investor behavior can significantly influence its relationship with ESG activities.

The research on valuation and profitability metrics in the context of ESG and CSR initiatives is well-developed and supported by robust evidence from recent literature. However, significant gaps remain in the study of risk management, market performance, innovation, and emerging financial performance indicators, particularly the TSR when considering the effects of moderating variables.

G3: Explore more financial indicators, such as brand value, employee satisfaction, customer loyalty, and TSR

G4: Explore stock prices in future articles, for example, in different time frames ESG and CSR measures research gaps

2.6.2 ESG and CSR Measures

Notable disparities exist in how different ESG and CSR categories are covered. These disparities reveal areas that have been extensively explored and others that remain underdeveloped, presenting valuable opportunities for future research. Figure 11 illustrates the percentage distribution of these categories, revealing that “General CSR” (2G) and “General ESG” (2A) are the most frequently studied, accounting for the highest percentages at approximately 35 % and 20 %, respectively. This suggests a broad interest in overarching ESG and CSR themes. In contrast, categories such as “Expenditures” (2D) have significantly lower representation, indicating less emphasis on these specific aspects in the existing research.

General CSR is the most extensively studied category, with numerous investigations exploring how CSR activities can enhance corporate reputation, stakeholder engagement, and financial outcomes. Early research established a positive correlation between CSR and financial performance, suggesting that CSR engagement often leads to improved risk management and corporate image enhancement (Khan et al., 2016; Jo & Harjoto, 2011; Brammer & Millington, 2008). Some studies have reinforced these findings, with Lim and Tsutsui (2012) demonstrating that CSR engagement yields long-term financial benefits, particularly when it aligns with stakeholder expectations and corporate strategy.

Similarly, the social aspect of ESG, which includes labor practices, community involvement, and human rights, has also garnered substantial attention. Waddock and Graves

(1997) were among the early proponents of the view that socially responsible firms performed better financially due to enhanced employee morale and customer loyalty. This perspective has been further validated by more recent research from Ye et al. (2021), who highlight that social performance is a critical driver of financial success, especially in industries where consumer and employee relations are crucial.

In addition to these well-explored areas, the integrated approach to ESG, which considers environmental, social, and governance factors collectively, has gained considerable focus. Friede et al. (2015) conducted a comprehensive meta-analysis, confirming that a holistic ESG strategy is associated with superior financial performance. Recent contributions (e.g., Pellegrini, 2022) support this view, suggesting that companies with high ESG scores are better positioned to navigate market uncertainties and achieve long-term financial stability.

Despite these advances, significant gaps remain, particularly in the governance and environmental categories. Governance, a crucial component of ESG, is less explored than social and CSR factors. While Gompers et al. (2003) linked strong governance to reduced agency costs, more recent studies (e.g., Freeman & Evan, 1990) indicate that the impact of specific governance practices on financial performance is context-dependent, requiring further exploration across different sectors and regions.

G5: Explore disaggregated governance scores in future articles combined with moderating variables

Environmental factors, although increasingly recognized as critical to financial performance, are still underrepresented in the literature relative to social and CSR issues. Research by Clarkson et al. (2011) suggests that proactive environmental strategies can result in cost savings and revenue growth. Moreover, studies such as McCright and Dunlap (2011) emphasize the growing importance of environmental sustainability in corporate strategy, particularly in response to climate change and regulatory pressures. Nevertheless, more empirical research is necessary to assess the financial impact of specific environmental initiatives across diverse industries.

G6: Explore disaggregated environmental scores in future articles combined with moderating variables

Furthermore, the literature on expenditures associated with ESG and CSR initiatives is notably sparse. Understanding the financial trade-offs and Return of Interest of these expenditures is essential for informed decision-making. Recent studies (e.g., Jorion, 2007) underscore the need for a more rigorous analysis of the costs and benefits of ESG and CSR investments, particularly concerning risk management and long-term value creation.

G7: Explore expenditures associated with ESG and CSR initiatives in future articles combined with moderating variables

Finally, the "Other" category, representing emerging or niche areas within ESG and CSR, remains virtually unexplored. This category includes new technological advancements in sustainability reporting, the role of artificial intelligence (AI) in monitoring ESG performance, and the impact of emerging social issues such as mental health on financial outcomes. The lack of research in this area highlights a substantial opportunity for pioneering studies that could significantly influence the future direction of ESG and CSR practices.

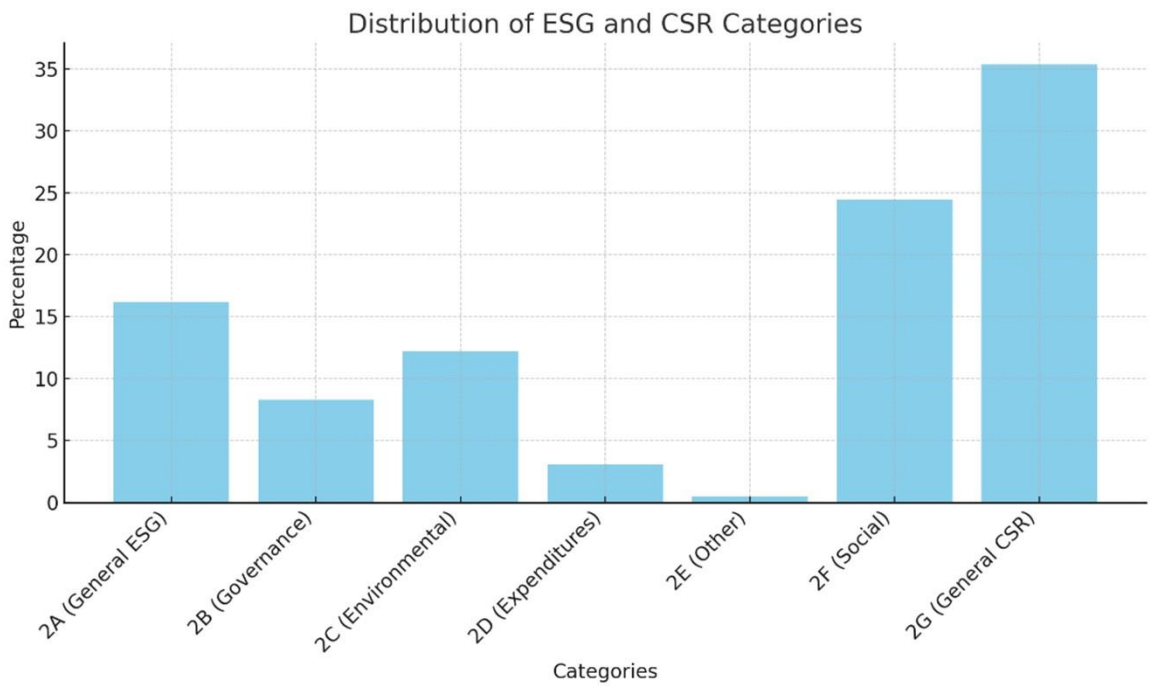


Figure 11 - ESG and CSR measures gaps

Note: Elaborated by authors

G8: Explore new technological advancements in sustainability reporting and the role of AI in future articles combined with moderating variables

2.6.3 Moderating variables category analysis

The analysis of moderating variables in the relationship between ESG/CSR practices and CFP highlights key factors influencing these outcomes. Figure 12 illustrates the percentage of occurrences by moderating variable categories, with “Corporate Governance” (3D) being the most frequently studied at approximately 30 %. This underscores the importance of governance structures in shaping the outcomes of ESG and CSR initiatives. Meanwhile, categories such as “Firm Characteristics” (3A) and “Economic Indicators” (3H) are also significantly represented, suggesting these factors are critical considerations in the research. In contrast, less attention has been given to “Industry-Level Indicators” (3B) and “Institutional/Legal Environment” (3C), indicating areas where future research could provide additional insights.

Corporate governance (3D) is the most frequently explored moderating variable category in the sample articles analyzed regarding ESG and CSR impacts on financial performance. This focus is substantiated by numerous studies, including Shen et al. (2019), who emphasize that robust governance structures (e.g., board independence and shareholder engagement) are pivotal in aligning corporate strategies with ESG goals, thereby, enhancing financial outcomes. Moreover, Bartolacci et al. (2020) illustrate that good corporate governance mitigates risks and strengthens the legitimacy of ESG initiatives in the eyes of investors, which can lead to improved financial performance. The importance of corporate governance (3D) is further underscored by Yu (2022), who systematically reviews the relationship between CEO duality and firm performance, highlighting how this aspect of governance can influence ESG and CSR outcomes depending on various contextual factors, such as ownership structures and national governance quality.

In addition to corporate governance (3D), firm characteristics (3A), such as size, market position, and industry sector, also receive substantial attention in the literature. Barauskaite and Streimikiene (2020) argue that larger firms typically have more resources to invest in ESG and CSR initiatives, which can result in better financial performance. This is consistent with findings by Shen et al. (2019), who show that firm size and market share significantly influence the effectiveness of CSR strategies, particularly in industries where consumer expectations for corporate responsibility are high. However, while these firm characteristics (3A) are widely recognized, there remains a need to examine in-depth how specific attributes, such as innovation capacity or leadership style, further moderate the ESG-financial performance relationship.

The extent of a firm's CSR engagement (3F) plays a crucial role in determining its financial outcomes. Barauskaite and Streimikiene (2020) suggest that firms with proactive CSR strategies, which go beyond regulatory compliance and actively seek to address social and environmental issues, are more likely to see positive financial returns. This is echoed by Shen et al. (2019), who observed that the depth of CSR engagement (3F), measured by integrating CSR into core business strategies, significantly enhances financial performance, particularly in consumer-facing industries where brand reputation is vital.

Economic conditions (3H), including macroeconomic factors such as GDP growth and inflation, are also recognized as significant moderators in the relationship between ESG/CSR and financial performance. Bartolacci et al. (2020) show that firms in more stable economic environments can better capitalize on their ESG investments, while during economic downturns, the financial benefits of ESG and CSR initiatives may be less pronounced. This highlights the importance of economic context (3H) in evaluating the effectiveness of these strategies.

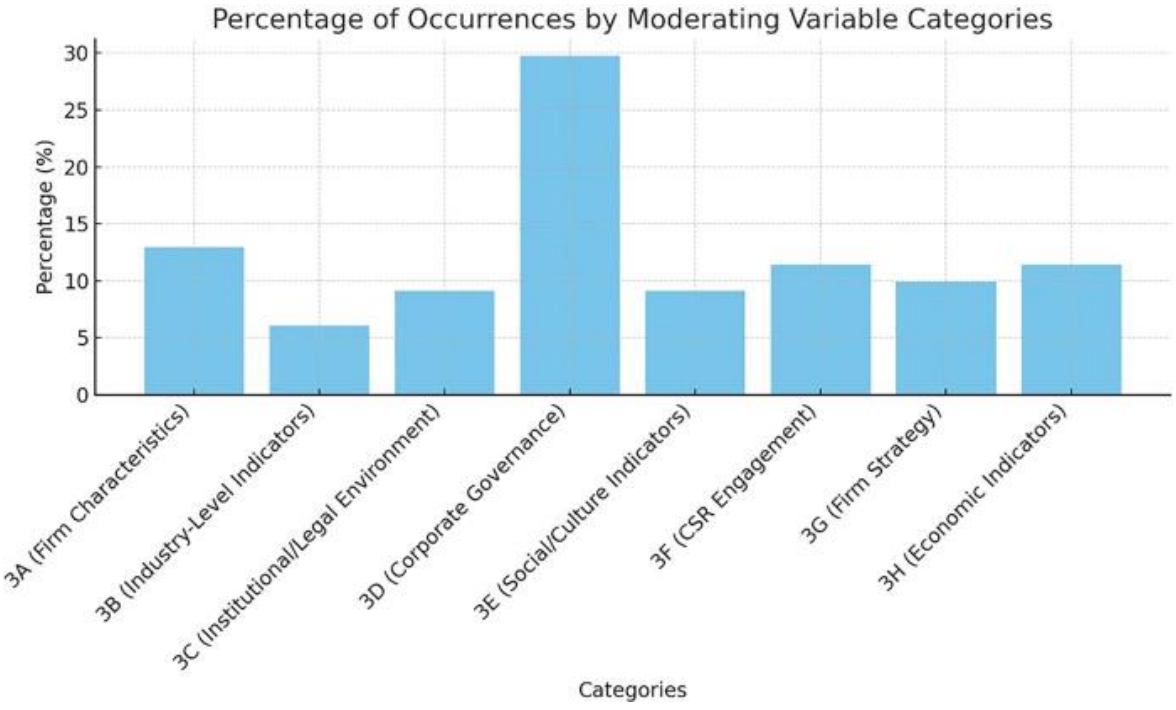


Figure 12 - Percentage distribution of moderating variables categories

Note: Elaborated by authors

Despite the industry-specific nature of many ESG and CSR initiatives, industry-level indicators (3B) remain underexplored. Shen et al. (2019) point out that industry characteristics, such as regulatory intensity and competitive dynamics, can significantly influence the success

of ESG strategies. For instance, industries with stringent environmental regulations may derive more financial benefits from ESG initiatives than less-regulated sectors. However, the relatively low focus on this area suggests a gap in understanding how industry-specific factors (3B) interact with ESG and CSR strategies.

G9: Explore industry-level indicators in future articles, such as regulatory intensity and competitive dynamics as moderating variables

The role of institutional and legal environments (3C) in moderating the ESG-financial performance relationship is another area with room for further exploration. Bartolacci et al. (2020) indicate that differences in regulatory frameworks across countries can lead to varying financial outcomes from ESG initiatives. For example, firms operating in regions with solid environmental regulations may see higher financial returns from ESG investments than those in less-regulated environments. This highlights a need for more cross-country comparative studies to understand better the impact of institutional and legal environments (3C) on ESG and CSR outcomes.

G10: Explore institutional and legal environments, such as regulatory frameworks across countries, as moderating variables

Social and cultural factors (3E), despite their potential to significantly influence corporate behavior and consumer expectations, are underrepresented in current research. Barauskaite and Streimikiene (2020) emphasize that cultural differences, such as varying levels of consumer activism or social norms around sustainability, can affect how ESG and CSR initiatives are perceived and rewarded in different markets. This suggests that more research is needed to explore how social and cultural contexts (3E) influence the financial impact of ESG and CSR activities, particularly in multinational or culturally diverse settings.

G11: Explore social and cultural factors, such as cultural dif- ferences across countries, as moderating variables

Finally, firm strategy (3G) is another underexplored moderating variable. A firm's strategic orientation, whether it focuses on cost leadership, differentiation, or innovation, can significantly impact the implementation and financial success of ESG and CSR initiatives.

Bartolacci et al. (2020) highlight that firms with strategies closely aligned with ESG principles, such as those prioritizing sustainability or ethical business practices, tend to achieve better financial outcomes. However, more research is needed to understand how different strategic approaches (3G) affect the relationship between ESG/CSR and financial performance, particularly in dynamic or competitive markets.

G12: Explore firm strategies, such as strategic orientation, as moderating variables

2.6.4 Theoretical framework research gaps

The analysis of the theoretical framework within ESG and CSR research reveals both areas of extensive exploration and those with significant gaps. Figure 13 presents the percentage distribution of theoretical framework categories, with “Social and Ethical Theories” (4C) being the most commonly applied, representing 51.00 % of the total. This is followed by “Organizational Theories” (4D) at 35.00 %, indicating a strong reliance on these perspectives to explain the relationship between ESG/CSR practices and financial outcomes. Theories such as “Innovation and Technology” (4F), “Strategic Management” (4B), “Cultural and Behavioral” (4E), and “Economic and Financial” (4A) are less represented, suggesting opportunities for expanding the theoretical bases in future research.

The dominance of Social and Ethical Theories (4C), which comprise over half of the theoretical frameworks used in the literature, highlights the critical role these theories play in understanding corporate responses to social expectations and ethical challenges. This is particularly evident in how businesses are increasingly held accountable for their social impact, a trend extensively documented in the literature. For example, the work of McCright and Dunlap (2011) underscores the societal pressures that compel firms to adopt socially responsible practices. Additionally, Carroll’s (1979) classic model of CSR remains a corner-stone in analyzing ethical business behavior.

Organizational Theories (4D) also feature prominently, reflecting their importance in examining how companies integrate ESG and CSR into their internal structures and cultures. This theoretical perspective is crucial for understanding the practical implementation of sustainability initiatives within firms. Lim and Tsutsui (2012) illustrate how organizational culture and leadership can either facilitate or hinder the adoption of sustainable practices. These insights are vital for businesses seeking to align their internal processes with broader sustainability goals, as outlined in the strategic management frameworks discussed by Freeman and Evan (1990).

In contrast, Economic and Financial Theories (4A) are significantly underrepresented despite their potential to offer valuable insights into the financial implications of ESG and CSR activities. This category's scant 2.67 % occurrence suggests a gap in understanding the economic rationale behind sustainability efforts. This gap is critical because integrating financial models into ESG and CSR research could illuminate how these practices impact firm value and risk, as suggested by the financial performance categories proposed by Orlitzky et al. (2003). Moreover, Jorion's (2007) work on risk management could provide a robust framework for analyzing the financial risks associated with sustainability initiatives, an underexplored area.

G13: Explore economic and financial theories, such as Fama's efficient market hypothesis or behavioral finance

Similarly, Innovation and Technology Theories (4F), which account for only 3.21 % of the theoretical frameworks, represent another area ripe for further investigation. In an era marked by rapid technological advancements, understanding how innovation drives or is driven by ESG and CSR practices is crucial. The limited exploration of this category may overlook how technological innovations can support sustainability goals – an issue that is increasingly relevant as firms leverage new technologies to enhance their ESG performance. Studies that focus on the intersection of technology and sustainability, such as those examining the role of digital transformation in CSR, could significantly enrich this area of research.

G14: Explore innovation and technology theories, such as open innovation theory or absorptive capacity theory

Cultural and Behavioral Theories (4E) also appear to be underutilized, with a representation of just 3.74 %. These theories are essential for understanding how cultural contexts and behavioral factors influence the effectiveness of ESG and CSR initiatives. Given the global reach of many corporations, cultural considerations play a crucial role in shaping stakeholder expectations and responses to corporate actions. Expanding research in this area could provide deeper insights into how cultural values impact the implementation and success of sustainability strategies, as highlighted by Pellegrini's (2022) exploration of cultural influences on corporate behavior.

Strategic Management Theories (4B) similarly occupy a small portion of the theoretical landscape, despite their importance in understanding how firms strategically manage their ESG

and CSR initiatives. The integration of these theories could shed light on how companies align their sustainability efforts with broader business objectives, ultimately enhancing their competitive advantage. The strategic perspective is crucial for identifying the drivers behind successful ESG and CSR integration, a topic that could benefit from further exploration, particularly in relation to the frameworks proposed by Davis (1960) and Freeman and Evan (1990).

G15: Explore strategic management and behavioral theories, such as dynamic capabilities and prospect theory

In summary, while Social and Ethical Theories and Organizational Theories are well-represented in the literature, significant gaps exist in the application of Economic and Financial Theories, Innovation and Technology Theories, Cultural and Behavioral Theories, and Strategic Management Theories. Addressing these gaps could provide a more comprehensive understanding of the complex relationship between ESG/CSR practices and CFP. Integrating these underexplored theoretical perspectives will not only enrich academic discourse but also offer practical insights for businesses striving to achieve sustainable success in an increasingly complex and demanding global environment.

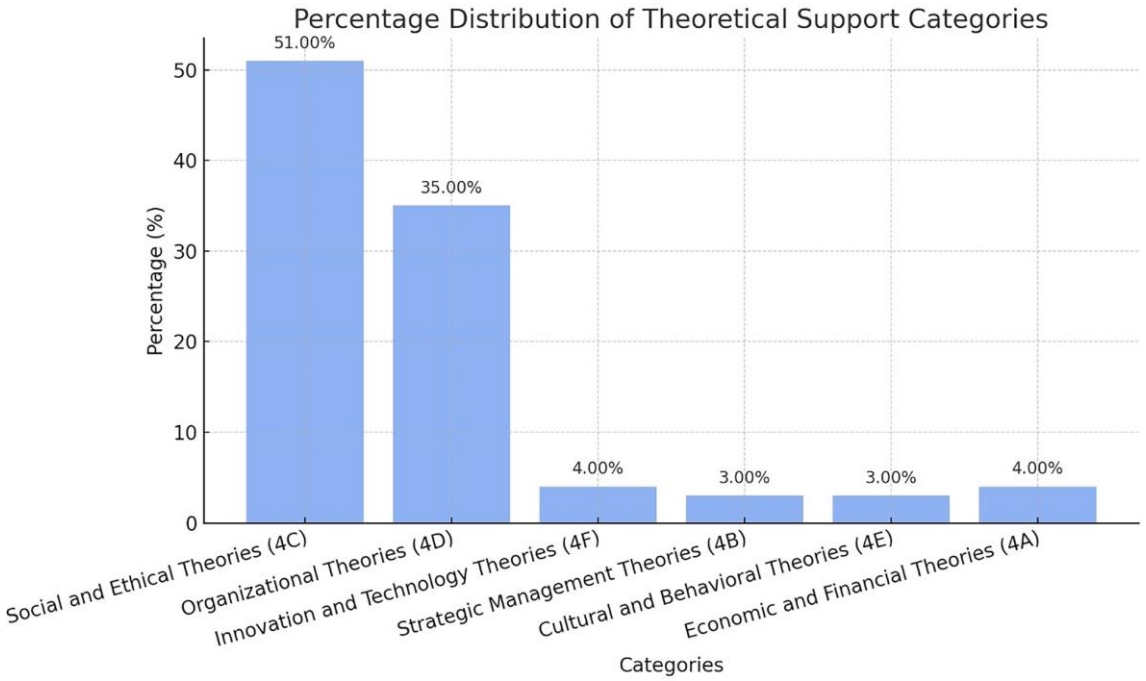


Figure 13 - Percentage distribution of theoretical framework categories

Note: elaborated by authors

2.6.5 Significant gaps to be explored in the ESG/CSR-CFP research

Analyzing the Suggestions for Future Research categories reveals exciting trends and significant gaps in the current ESG and CSR research landscape. Figure 14 presents the percentage distribution of suggestions for future research categories. Notably, “Industry/Sector Focus” (5J) emerges as the most frequently suggested area, accounting for 36.27 % of the total, indicating a strong need to explore ESG/CSR impacts across different sectors. This is followed by “Variables of Interest” (5A) and “Research Focus Area” (5I), which highlight the need for more specific studies on particular variables and broader research domains. Other categories, such as “Theoretical Framework” (5B) and “Suggested Contributions” (5K), appear less frequently, suggesting they are less of a priority but still relevant for building a comprehensive research agenda.

A prominent focus has been placed on the Research Focus Area (5I), which suggests that many scholars prioritize exploring specific sectors and themes within ESG and CSR. This emphasis aligns with these fields’ increasing complexity and specificity, where sector-specific studies can yield more actionable insights. Chang et al. (2020) and Nagalingam et al. (2022) highlight the need for tailored approaches, as the impact of ESG practices can vary greatly between industries due to differing regulatory pressures and stakeholder expectations.

Similarly, the frequent mention of Variables of Interest (5A) in the suggestions indicates a robust interest in uncovering new dimensions or factors that might influence the relationship between ESG and financial performance. This reflects the dynamic nature of the field, where scholars are continuously seeking to refine and expand the set of variables under consideration. For instance, Li and Zheng (2023) introduce novel variables such as corporate digitalization, which could potentially mediate the ESG-financial performance relationship in ways that have not been fully explored.

However, the Industry/Sector Focus (5J) category also receives considerable attention, underscoring the importance of understanding ESG impacts within specific contexts. This trend is supported by research from Clark et al. (2015), who have found that sector-specific factors significantly moderate the effectiveness of ESG initiatives, particularly in industries with high environmental risks. The insights gained from such sectoral studies are crucial, as they provide more precise guidelines for companies operating in different environments.

However, the analysis also uncovers notable gaps in literature. Theoretical Frameworks (5B) are surprisingly underexplored, suggesting that many studies may lack a solid theoretical foundation. This could potentially limit the generalizability of the findings. The importance of

grounding empirical research in robust theories is well-documented, with scholars like Whetten (1989) arguing that theory-building is essential for advancing any field of study. The relative scarcity of sug- gestions related to theoretical frameworks may indicate a need for more rigorous theoretical contributions in future ESG and CSR research.

G16: Explore a more comprehensive theoretical foundation

Another significant gap is the minimal focus on Suggested Contributions (5K). This could imply that many studies fail to adequately articulate their broader impact or relevance to ongoing academic and practical debates. For example, Aguinis and Glavas (2012) have noted that many studies lack a clear articulation of their contributions, particularly in terms of how they advance theoretical debates or inform practical applications. This often leaves a gap in understanding the broader implications of research findings.

Methodological rigor also appears somewhat overlooked, as evidenced by the relatively few mentions of Research Methodology (5D). This is concerning because the robustness of research findings depends heavily on the methodologies employed. Podsakoff et al. (2012) emphasize that innovative and rigorous methodologies are crucial for producing reliable and valid results. The current trend suggests that there may be room for improvement in this area, with a need for more diverse and sophisticated research methods in ESG and CSR studies.

G17: Utilize robust research methodology

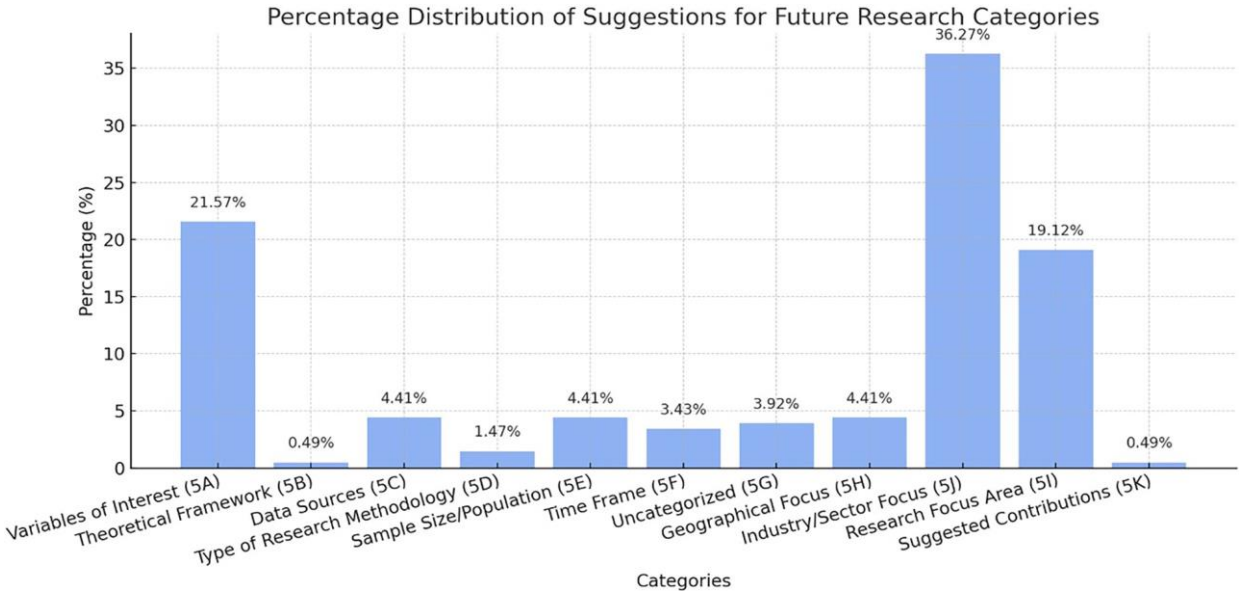


Figure 14 - Percentage distribution of suggestions for future research categories

Note: Elaborated by authors

The moderate emphasis on Data Sources (5C) and Geographical Focus (5H) suggests that they are not as prioritized as others. Expanding the diversity of data sources and exploring

ESG impacts across different geographical contexts could enrich the field significantly. For example, Enciso-Alfaro and García-Sánchez (2024) demonstrate how diverse data sources can uncover regional variations in ESG effectiveness, which might otherwise be overlooked.

G18: Utilize different sources of ESG and CSR databases, acquisition techniques, and broad geographic samples

In summary, while the research focus on specific areas, variables, and industries within ESG and CSR is well-established, there are critical gaps in theoretical grounding, methodological rigor, and the articulation of contributions. Addressing these gaps through more theoretically informed, methodologically robust, and globally diverse research will be essential for advancing the field and providing more actionable insights for practitioners.

2.6.6 Statistical modeling research gaps

The choice of statistical models plays a crucial role in shaping the conclusions that researchers can draw in the study of ESG/CSR and CFP. Figure 15 shows the percentage distribution of statistical modeling categories, revealing that “Basic and Intermediate Models” are the most frequently utilized, accounting for 72.22 % of the total usage. This predominance suggests a reliance on simpler models, possibly due to their ease of interpretation and application. In contrast, “Advanced Models” comprise only 27.78 % of the total, indicating that more sophisticated techniques are less commonly employed, potentially reflecting the need for specialized knowledge and computational re- sources. This distribution underscores the importance of encouraging advanced modeling techniques to capture the complexities of ESG/CSR impacts more accurately.

Basic models, including simple linear regressions, are often used in exploratory phases of research or studies with limited data complexity. They are particularly useful when the goal is to establish foundational relationships or when the research context does not require the handling of multiple interacting variables. However, these models have limitations, especially in their assumptions of linearity and the potential for oversimplification of complex relationships. Intermediate models (e.g., multivariate regressions and panel data techniques) build on this foundation by incorporating more variables and allowing for the control of certain types of data structure, such as time series or cross-sectional data. These models are well-suited for studies that require a more nuanced analysis of how ESG/CSR factors influence financial performance across different contexts or over time.

Despite their widespread use, basic and intermediate models are often critiqued for their inability to fully capture the complexities inherent in the CSR-CFP relationship. They may not adequately account for non-linear interactions or the simultaneous effects of multiple moderating variables. Consequently, while these models are valuable for their simplicity and clarity, they may lead to oversimplified conclusions that overlook important nuances.

In contrast, advanced models, such as GMM, Structural Equation Modeling (SEM), and machine learning techniques, offer a more sophisticated approach to analyzing ESG/CSR impacts on financial performance. These models are designed to handle the complexities that basic and intermediate models often struggle with. For instance, GMM is particularly effective in addressing issues of endogeneity, which is a common challenge in ESG/CSR research where the direction of causality between CSR activities and financial outcomes can be ambiguous. SEM allows researchers to model complex relationships between variables, including latent variables, and to test for direct and indirect effects within a single framework. This is particularly useful when examining how multiple ESG/CSR dimensions interact to influence financial performance.

Advanced models are also better equipped to handle non-linear relationships and model the interactions between multiple moderating variables. This capability is crucial for capturing the full complexity of the CSR-CFP relationship, where the effects of CSR initiatives may not be uniform across different contexts or periods. Additionally, machine learning techniques provide a powerful tool for uncovering patterns in large datasets that might not be apparent through traditional statistical methods, offering new insights into the impact of ESG/CSR activities on financial performance.

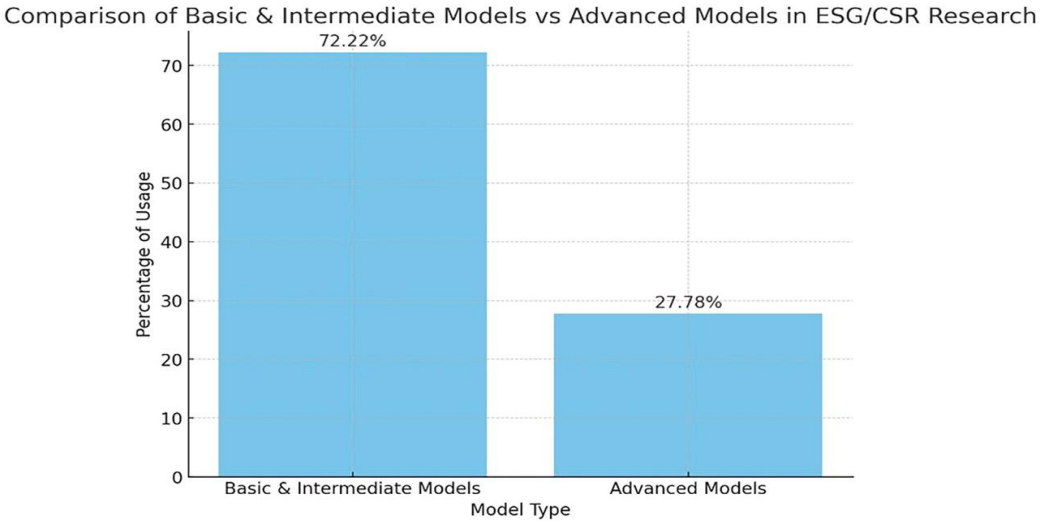


Figure 15 - Percentage distribution of statistical modeling categories

Note: Elaborated by authors

However, the use of advanced models comes with its own set of challenges. These models require a higher level of technical expertise and a deeper understanding of the underlying assumptions and limitations. They also demand more sophisticated data collection and preparation processes, as the quality of the inputs significantly impacts the validity of the results. Moreover, the complexity of these models can sometimes make their results more difficult to interpret, particularly for practitioners who may not have a strong background in econometrics or statistics.

G19: Utilize Advanced Statistical and Econometric Modeling.

2.6.7 Data Sources research gaps

The research landscape on ESG and CSR reveals a significant focus on methodologies that are quantifiable and easily comparable across con- texts, such as one-dimensional measures and reputational indices. Figure 16 shows the percentage distribution of data source categories, highlighting that “One-Dimensional Measures” (7C) are the most frequently used, accounting for 49.32 % of the occurrences. This suggests a reliance on single-factor data points, such as individual ESG scores or specific financial indicators. “Reputational Indices” (7B) are also commonly utilized, representing 36.53 % of the data sources, reflecting the importance of firm reputation in CSR/ESG studies. In contrast, “Content Analysis” (7A) is the least frequently used data source at 14.16 %, indicating a lesser focus on qualitative assessments of ESG and CSR practices. This distribution highlights the need for more diverse data collection methods to capture a comprehensive picture of ESG and CSR impacts.

One-dimensional measures (7A), often concentrated on specific aspects such as environmental scores or social responsibility indices, are particularly prominent. These measures are favored for their ability to distill complex CSR activities into easily comparable figures, making them a staple in the analysis of ESG impacts. Similarly, reputational indices, such as the Dow Jones Sustainability Index, serve as proxies for broader CSR engagement and are widely accepted as reliable indicators of corporate success. The connection between corporate reputation and financial stability, as highlighted by Barauskaite and Streimikiene (2020), reinforces the importance of these indices in assessing long-term value creation and risk management.

However, the preference for quantifiable metrics may lead to underutilizing more nuanced approaches, such as content analysis. This method, which involves a detailed examination of CSR activities through corporate reports, press releases, and other qualitative data, is notably less represented in literature. Despite its potential to provide deeper insights

into the authenticity of CSR practices and their strategic motivations, content analysis is often overlooked. This gap in research indicates a broader trend where qualitative methods are under-emphasized, potentially overlooking the rich context that they can provide.

The limited use of content analysis may stem from a broader inclination within the research community to prioritize methods that yield clear, comparable results. However, as highlighted by Wang and Lu (2021), qualitative methods such as content analysis are crucial for understanding the subtleties of CSR practices. These methods can reveal discrepancies between what companies report and what they actually do, offering valuable insights into issues such as greenwashing or the genuine integration of sustainability into business strategies. Barauskaite and Streimikiene (2020) further support this view by emphasizing that content analysis can uncover deeper layers of corporate behavior that quantitative measures might miss.

The emphasis on one-dimensional measures and reputational indices is not without academic support. For instance, Gillan et al. (2021) argue that these tools are effective in linking CSR to financial performance, particularly in capturing the market's perception of a company's social responsibility. Similarly, Orlitzky et al. (2003) demonstrate through a meta-analysis that the use of these measures is widespread and often correlates with positive financial outcomes. This is supported by findings from Jorion (2007), who discusses the role of reputational indices in risk management, highlighting their importance in maintaining long-term financial stability.

However, the gap in the use of content analysis reflects a missed opportunity to explore CSR practices more comprehensively. Future research could address this gap by integrating content analysis with quantitative measures to provide a fuller picture of how ESG activities are reported versus their actual impact. For instance, longitudinal studies that track the evolution of CSR reporting and its effects on financial performance could provide insights into the long-term efficacy of CSR strategies. Furthermore, applying content analysis in cross-cultural studies could reveal important differences in how CSR is practiced and communicated across different regions, especially in emerging markets where CSR is still developing.

The current emphasis on one-dimensional measures and reputational indices, while well-supported by the existing literature, represents only part of the potential analytical landscape. To advance the field, re- searchers must also embrace the richness of qualitative methods like content analysis, as highlighted by Melo and Galan (2011) and Han et al. (2019). Consequently, they can uncover new insights and contribute to a more nuanced understanding of the complex relationship between CSR and financial success. This balanced approach could lead to more authentic and effective CSR practices, ultimately benefiting both companies and society at large.

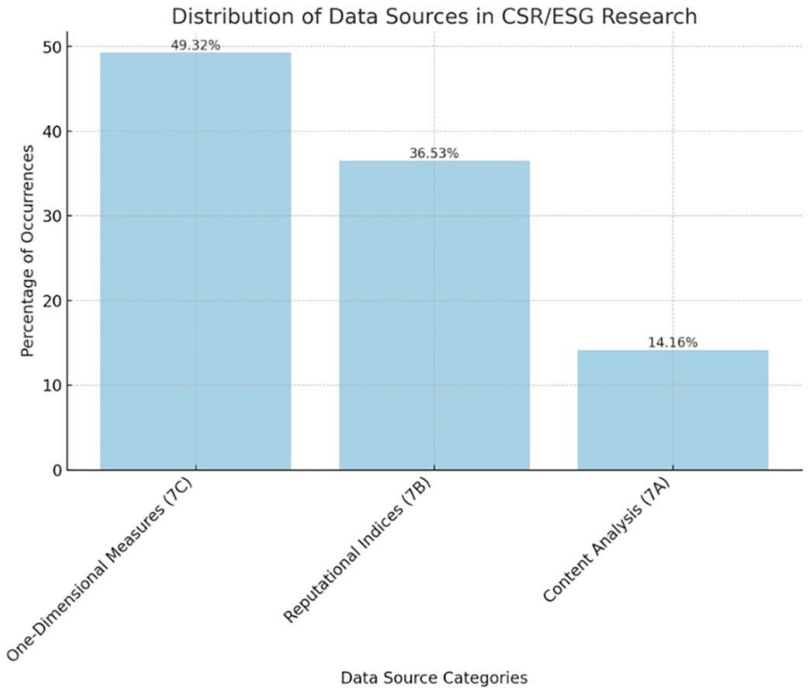


Figure 16 - Percentage distribution of data sources of ESG and CSR categories

Note: elaborated by authors

G20: Utilize content analysis to measure ESG and CSR

This approach aligns with the findings of Barauskaite and Streimikiene (2020), who emphasize the importance of diverse methodologies in capturing the full scope of CSR's impact on financial performance. By broadening the scope of research to include both quantitative and qualitative methods, scholars can provide a more comprehensive understanding of how CSR practices influence corporate success, paving the way for more informed and effective business strategies.

In the dynamic and evolving field of ESG/CSR-CFP, under the effect of moderating variables research, it is imperative to systematically map out the existing body of knowledge to identify the strengths and weaknesses within literature. This comprehensive mapping provides a clear understanding of what has already been explored and established and reveals research gaps or inconsistencies. By pinpointing these gaps, researchers can better comprehend the complexities and multifaceted nature of ESG and CSR impacts on CFP and other outcomes. Table 8 summarizes all research gaps identified, providing a future research agenda. Figure 17 provides a graphical summary of these findings, categorizing them into seven key areas: Financial Performance, ESG and CSR Measures, Moderating Variables, Theoretical Framework, Suggestions for Future Research, Statistical Modeling, and Data Sources. The figure illustrates the current state of knowledge and the areas where future research is needed,

offering a structured pathway for advancing the field. This visual representation helps researchers identify critical gaps
and prioritize topics for further investigation.

Table 8 - Resume of the Gaps identified

Gaps	Description
G1	Explore risk metrics such as credit risk, volatility, and beta systematic risk in upcoming articles
G2	Explore innovative metrics in future articles, such as R&D expenditures and patent counts
G3	Explore more emerging financial indicators, such as brand value, employee satisfaction, customer loyalty and TSR
G4	Explore stock prices in future articles, for example, in different time frames
G5	Explore disaggregated governance scores in future articles combined with moderating variables
G6	Explore disaggregated environmental scores in future articles combined with moderating variables
G7	Explore expenditures associated with ESG and CSR initiatives in future articles combined with moderating variables
G8	Explore new technological advancements in sustainability reporting and the role of AI in future articles combined with moderating variables
G9	Explore industry-level indicators in future articles, such as regulatory intensity and competitive dynamics as moderating variables
G10	Explore institutional and legal environments, such as regulatory frameworks across countries, as moderating variables
G11	Explore social and cultural factors, such as cultural differences across countries, as moderating variables
G12	Explore firm strategies, such as strategic orientation, as moderating variables
G13	Explore economic and financial theories, such as Fama's efficient market hypothesis or behavioral finance
G14	Explore innovation and technology theories, such as open innovation theory or absorptive capacity theory
G15	Explore strategic management and behavioral theories, such as dynamic capabilities theory and prospect theory
G16	Explore a more comprehensive theoretical foundation
G17	Utilize robust research methodology
G18	Utilize different sources of ESG and CSR databases, acquisition techniques, and broad geographic samples
G19	Utilize advanced statistical and econometric modeling
G20	Utilize content analysis to measure ESG and CSR

This strategic approach to identifying knowledge gaps is essential for developing well-informed and targeted future research agendas. It encourages scholars to delve deeper into underexplored areas, address methodological weaknesses, and build upon the foundations laid by previous studies. Furthermore, understanding these gaps allows for formulating new

research questions and hypotheses to drive innovation and theoretical advancement in the field. This methodical process enriches the academic discourse and has practical implications, guiding practitioners and policymakers in understanding the nuanced impacts of ESG and CSR initiatives.

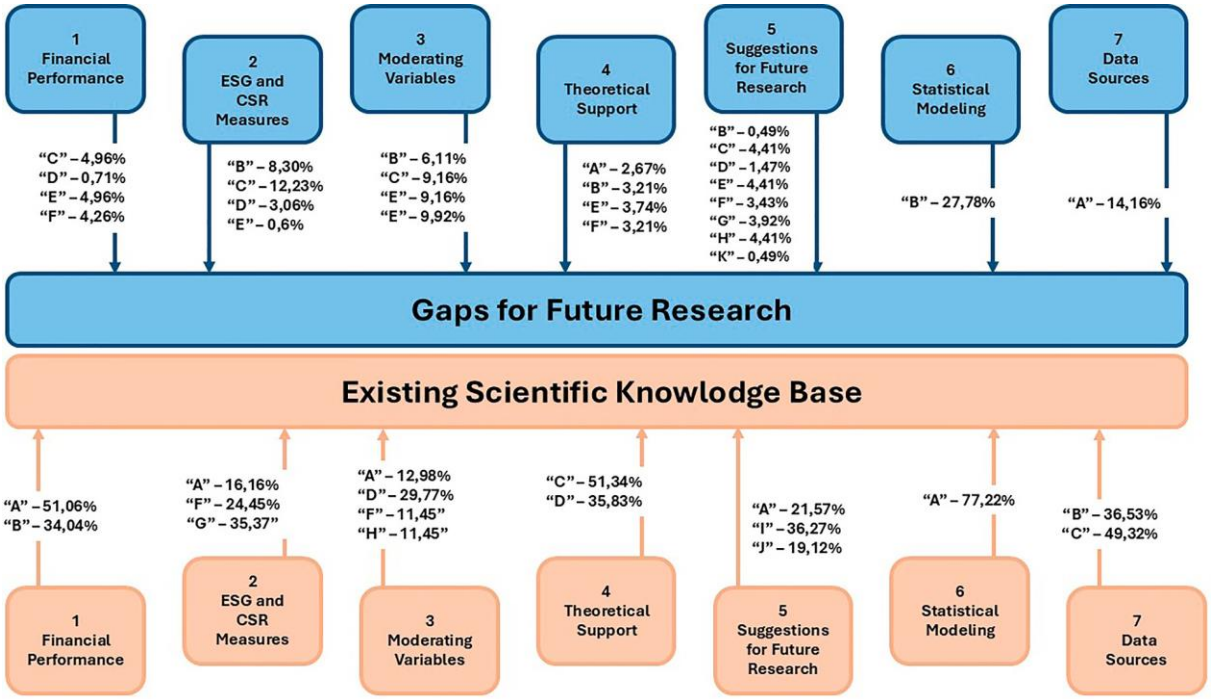


Figure 17 - Graphic resume of the suggestions for a research agenda

Note: Elaborated by authors.

2.7 LIMITATIONS, IMPLICATIONS, AND CONCLUSION

2.7.1 Research limitations

This study has several limitations. It specifically focuses on the direct and unidirectional relationship between ESG/CSR measures and financial performance while acknowledging that this relationship can also be bidirectional and influenced by various contexts and processes. The deliberate choice to emphasize this direct relationship was made to understand better how moderating variables impact it. Nonetheless, the study is constrained to articles published within a five-year period, and findings may vary over a longer timeframe. Furthermore, the analysis is limited to firms with publicly available ESG/CSR data, thereby, excluding studies that use internal measures from questionnaire-based surveys. To keep the research manageable, studies employing conceptual frameworks with mediating or mediating-moderating models were not included. Additionally, many articles view ESG/CSR measures as moderators or

mediators of other variables; although these perspectives have not been addressed in this review, they present opportunities for future research.

Moreover, the study is restricted to articles published between 2019 and 2023, which may omit earlier foundational works that could offer additional insights into the ESG/CSR and CFP relationship. The analysis relied on two highly regarded databases — Web of Science and Scopus—, which may not capture all relevant literature, particularly in regional or niche journals not indexed by these databases.

Another limitation is the diversity of ESG/CSR metrics and financial performance indicators used in the reviewed studies. The lack of standardization across different research designs and variable definitions introduces challenges in synthesizing results and forming generalized conclusions. While the study focuses on moderating variables such as governance, market maturity, and firm strategy, it does not exhaustively cover all possible influencing factors, such as cultural, geographic, or regulatory variations, which may also impact the relationship between ESG/CSR and financial outcomes.

Finally, while this study identifies trends and gaps in literature, the analysis is predominantly quantitative, relying on bibliometric and systematic review techniques. The absence of qualitative insights from case studies or interviews limits the depth of understanding regarding the nuances and contextual factors influencing the ESG/CSR-CFP nexus. Expanding future research to include qualitative methodologies could provide a more comprehensive perspective on the complexities of this relationship.

2.7.2 Theoretical implications

This research provides significant contributions to the theoretical landscape surrounding the ESG/CSR-CFP relationship by emphasizing the role of moderating variables. It highlights the importance of governance structures, industry characteristics, and firm strategy in shaping the impact of ESG and CSR initiatives on financial performance. The study reinforces and extends the relevance of existing theoretical frameworks, such as Stakeholder Theory, Agency Theory, and the RBV, by demonstrating how these theories interact with moderating factors to influence financial outcomes.

Moreover, this research uncovers evolving trends in academic focus, offering new directions for future theoretical exploration. For instance, the study suggests that the underexplored role of social and cultural variables as moderators presents a valuable avenue for expanding current theoretical models. Additionally, this work highlights the need for more

standardized frameworks for measuring ESG and CSR outcomes, which could strengthen the integration of these concepts into established financial and strategic management theories.

This study also contributes to the academic debate by clarifying the complex relationship between ESG, CSR, and financial performance through a deeper focus on the moderating variables. The research calls for multi-theoretical approaches to capture the nuances of the ESG-CSR- CFP relationship, thus, advancing the theoretical understanding of corporate sustainability, for example, in digital transformation and technological readiness as emerging factors.

2.7.3 Practical implications

From a practical standpoint, this research offers actionable insights for corporate leaders and policymakers. Identifying key moderating variables, such as firm governance and industry characteristics, can help organizations tailor their ESG and CSR initiatives to align with their specific operational contexts. Companies can use this information to improve decision-making processes, enhance stakeholder engagement, and optimize financial performance by integrating sustainability into core business strategies more effectively.

Additionally, the study highlights the potential for standardizing ESG/CSR metrics, which could aid businesses in benchmarking their sustainability efforts against industry standards and improving transparency in reporting. Policymakers may also find the results useful in shaping regulations that encourage the adoption of ESG principles, particularly by acknowledging the diverse ways in which these initiatives influence financial outcomes across different sectors and regions.

2.7.4 Social implications

On a broader societal level, this study contributes to the ongoing discourse on sustainability and corporate responsibility. By revealing the conditions under which ESG and CSR initiatives are most likely to lead to positive financial outcomes, the research supports the integration of social and environmental considerations into corporate strategies. This alignment between profitability and sustainability goals not only benefits shareholders but also has a positive impact on broader society by promoting ethical business practices and addressing pressing global challenges like climate change and social inequality.

Furthermore, the study’s call for standardized ESG metrics aligns with societal demands for greater corporate accountability and transparency. As more firms adopt robust ESG and CSR

practices, the cumulative impact can drive meaningful progress toward achieving global sustainability goals, benefiting both the business community and society at large.

2.8 CONCLUSION

This study provides a comprehensive examination of the complex relationship between ESG measures, CSR activities, and CFP, highlighting the critical role that moderating variables play in shaping this relationship. The bibliometric analysis and systematic review revealed that key moderating variables, such as governance structures, cultural norms, economic conditions, and industry characteristics, profoundly influence the ESG-CSR-CFP dynamic.

Governance structures, such as board diversity and executive compensation aligned with sustainability goals, enhance the positive financial impact of ESG and CSR initiatives. These findings show that corporate governance can amplify or weaken the link between sustain- ability practices and financial outcomes, underscoring that the compo- sition and incentivization of leadership are crucial to successful ESG implementation. Similarly, cultural norms emerge as significant moderators, affecting how stakeholders perceive and value CSR activities. In regions where environmental and social responsibility is highly prioritized, ESG initiatives are more likely to lead to positive financial outcomes. Conversely, in markets where such initiatives are undervalued, their financial impact may be muted. This highlights the importance of tailoring ESG strategies to the cultural context in which firms operate to maximize their financial benefits.

Economic conditions, such as market stability and growth, also moderate the effectiveness of ESG initiatives. In stable economic environments, ESG practices can enhance financial returns by improving a firm's reputation and reducing operational risks. However, during eco- nomic downturns, the financial benefits of ESG efforts may be diminished, as firms struggle to balance short-term survival with long-term sustainability goals. This underscores the need for context-specific strategies when implementing ESG and CSR initiatives, especially in volatile economic climates.

Industry characteristics play a significant moderating role, as factors such as competition and regulatory pressures shape the financial returns of ESG investments. In highly regulated industries, such as energy or finance, companies may achieve greater financial stability from ESG practices due to heightened scrutiny and potential penalties for non-compliance. In contrast, firms in less-regulated sectors may experience different financial outcomes from similar initiatives, given the varying regulatory and competitive dynamics across industries.

Innovation plays a crucial role in determining the effectiveness of ESG strategies. R&D expenditures and patent generation, for example, influence how much value firms can generate from their sustainability initiatives. Future research should explore how these innovation metrics moderate the relationship between ESG efforts and financial performance, particularly in industries that heavily rely on technological advancements. By understanding the link between innovation and ESG, firms can better leverage their sustainability efforts to foster competitive advantage and long-term growth.

Emerging financial performance indicators (e.g., those related to stakeholder value creation and intangible assets) are gaining importance as frameworks such as the Global Reporting Initiative (GRI) become widely adopted. Indicators like brand value, employee satisfaction, and customer loyalty reflect how sustainability practices impact intangible assets and stakeholder relationships. However, there is a notable gap in research on Total Shareholder Return (TSR), a critical measure that combines capital gains and dividends to reflect the financial benefits delivered to shareholders over time. Given TSR's direct link to share- holder value, future research should examine how ESG and CSR efforts translate into tangible returns for investors through this key metric.

Another important area for future exploration is stock price movements in response to ESG initiatives. While short-term stock price re- actions have been studied, less is known about how stock prices evolve over different time horizons due to sustained ESG efforts. A more detailed examination of stock price movements can provide insights into how market perceptions of sustainability influence both short-term and long-term financial outcomes.

This study also identifies gaps in the understanding of disaggregated governance and environmental scores as moderating variables. Governance and environmental practices are often evaluated in aggregate, but disaggregating these scores could provide more precise insights into which specific aspects drive financial outcomes. For example, components such as board diversity or shareholder rights may have different effects on financial performance depending on other moderating factors, such as industry and market conditions. Similarly, focusing on disaggregated environmental factors, such as carbon emissions or energy efficiency, could reveal new insights into how specific environmental practices contribute to financial success.

Additionally, expenditures as an ESG and CSR measure remain underexplored. Firms that allocate substantial resources to sustainability initiatives may experience varying financial outcomes based on how effectively they manage these expenditures, and how these efforts are influenced by various moderating variables. Factors such as organizational context, market

conditions, or governance structures can all impact the relationship between expenditure levels and financial performance. Understanding the financial impact of different levels of spending on ESG and CSR, while considering the role of these moderating variables, is essential for firms aiming to optimize resource allocation. By analyzing these dynamics, companies can better align their sustainability investments with improved financial outcomes.

Technological advancements in sustainability reporting, particularly AI integration, represent another important yet underexplored moderating factor. AI has the potential to improve the accuracy and efficiency of ESG reporting. Future research should investigate how technological innovations such as AI moderate the effectiveness of ESG practices in driving financial performance, as AI-driven ESG reporting could enhance transparency and provide stakeholders with more reliable data, leading to better-informed decision-making and stronger financial outcomes.

At the industry level, indicators such as regulatory intensity and competitive dynamics further influence how ESG initiatives translate into financial performance. Highly regulated industries tend to derive greater financial returns from ESG investments due to increased focus on compliance and risk management. In contrast, firms in less-regulated industries may need to rely more on competitive dynamics, such as differentiation through sustainability, to achieve financial success.

Finally, institutional and legal environments, including varying regulatory frameworks across countries, play a significant moderating role in the ESG-CFP relationship. Firms operating in jurisdictions with supportive legal environments for sustainability are more likely to achieve financial success through ESG initiatives. In regions with less stringent regulations, however, the financial impact of ESG practices may be weaker, highlighting the need for region-specific ESG strategies. In summary, this study highlights the crucial role of moderating variables in shaping the ESG-CSR-CFP relationship. However, to fully grasp the mechanisms driving the financial impact of ESG and CSR initiatives, it is essential to address the gaps identified in this research. Future studies should delve deeper into these moderating factors, utilizing advanced methods such as longitudinal analysis and sophisticated econometric techniques to capture their dynamic influence. By bridging these gaps, both scholars and practitioners will be able to unlock richer insights into the strategic value of ESG and CSR, paving the way for more effective corporate sustainability practices that are closely aligned with long-term financial performance.

2.9 COMBINING RESEARCH GAPS FOR EMPIRICAL EXPLORATION

The second article of this thesis was designed as a direct empirical response to the conceptual and methodological gaps identified in the systematic literature review presented in the first article. The review revealed that, despite increasing academic interest in the ESG–CFP relationship, significant inconsistencies remain in how ESG is operationalized, measured, and linked to financial performance. Fragmented constructs, narrow financial indicators, methodological limitations, and insufficient treatment of disclosure–performance dynamics continue to hinder theoretical advancement and practical applicability. Against this backdrop, the second article addresses four core research gaps by developing a multidimensional and empirically rigorous approach.

1. Expansion of Financial Metrics (G1, G3, G4): The ESG–CFP literature has long relied on traditional profitability indicators—such as Return on Assets (ROA) or Return on Equity (ROE)—as proxies for corporate financial performance. However, these metrics capture only one side of the financial reality, leaving aside broader market-based and investor-oriented measures. Responding to this limitation, the second article incorporates Total Shareholder Return (TSR) and Tobin's Q, in addition to ROA and ROE, to provide a more comprehensive and diversified representation of financial outcomes. TSR reflects shareholder value creation through both capital gains and dividends, while Tobin's Q measures market valuation relative to asset replacement costs, offering insights into investor expectations and firm valuation dynamics. This multidimensional approach allows the study to identify how ESG variables influence both internal efficiency and external market perceptions, thus expanding the theoretical and empirical scope of ESG-related financial effects. By aligning accounting-based and market-based metrics, the article addresses the call for broader and more representative performance indicators and tests the hypothesis that ESG strategies have heterogeneous impacts depending on the type of financial outcome considered.

2. Disaggregation of ESG Dimensions (G5, G6) versus Aggregated Measures: A significant gap in ESG–CFP research lies in the frequent use of aggregated ESG scores that obscure the distinct contributions and financial effects of environmental, social, and governance dimensions. The second article directly responds to this limitation by adopting a disaggregated approach to both ESG performance and disclosure, allowing for separate and combined analyses of each pillar. Building on the typological frameworks of Hahn & Kühnen (2013) and Lueg & Pesheva (2021), the article classifies ESG strategies into three configurations: isolated (focus on a single dimension), selective (engagement in two dimensions), and holistic (balanced

engagement across all three). This typology enables a nuanced analysis of the scope, coherence, and depth of ESG engagement, which are critical yet underexplored variables in determining financial outcomes.

3. Combining ESG Performance and Disclosure Measures in the ESG–CFP Nexus: Another central innovation of the second article is the combined treatment of ESG performance and disclosure, which are often examined in isolation despite being deeply interrelated. While ESG performance reflects substantive actions, disclosure represents the firm's ability to communicate its sustainability commitments to stakeholders. The article models the interaction between performance and disclosure, both in aggregated and disaggregated forms, to examine the diferente impact on financial outcomes. In addition to exploring the interaction between performance and disclosure, the article analises a quadratic specification of ESG performace-discloure variables to test for potential nonlinearities in the ESG–CFP relationship.

4. Adoption of Robust Econometric Techniques and Global Databases (G17, G18, G19): The literature review also highlighted methodological limitations in ESG–CFP research, particularly with respect to endogeneity control and generalizability across diverse contexts. In response, the second article adopts a two-stage least squares (2SLS) approach using instrumental variables, which helps address potential reverse causality and omitted variable bias. This is complemented by the use of a panel structure, allowing the model to control for unobserved heterogeneity across firms and over time. The data sample includes 2,987 firm-year observations from 453 listed companies across BRICS nations, extracted from the Bloomberg database. This choice enhances both the geographic scope and institutional relevance of the analysis, situating the results in the complex realities of emerging economies where ESG frameworks are less institutionalized, disclosure standards are evolving, and stakeholder expectations are highly variable. The article's methodological rigor ensures that its findings are both robust and policy-relevant, contributing to a more causally credible literature and expanding the empirical frontiers of ESG–CFP research beyond developed markets.

3 ARTICLE 2 - THE ESG DISCLOSURE–PERFORMANCE PARADOX IN BRICS: EVIDENCE FROM SHAREHOLDER RETURNS AND FIRM VALUE

3.1 ABSTRACT

This study examines the financial implications of corporate sustainability disclosure and performance across emerging economies, using a panel of 2,987 firm-year observations from 2016 to 2023. A Two-Stage Least Squares approach is employed to address endogeneity and assess both integrated and selective sustainability strategies. The results indicate that integrated

sustainability disclosure—encompassing environmental, social, and governance (ESG) dimensions—positively influences profitability and investor confidence, whereas selective disclosure and performance erode shareholder returns and overall profitability. The study reveals a key paradox in emerging markets: although the financial market tends to reward transparent and well-structured sustainability communication, there is no evidence that sustainability performance alone enhances financial returns. Conversely, even strategically aligned ESG practices may diminish firm value when they exceed an optimal threshold, as illustrated by the Too-Much-of-a-Good-Thing (TMGT) effect. Findings suggest that moderate ESG performance, when paired with high-quality disclosure, yields significant financial benefits by leveraging reputational capital and signaling advantages—especially in institutional contexts marked by regulatory asymmetries and information gaps. In contrast, firms positioned in the "gray zone" of intermediate ESG performance and disclosure tend to produce the least favorable financial outcomes, lacking both credibility and operational excellence. Drawing on stakeholder, agency, legitimacy, signaling, and institutional theories, the study suggests that in emerging economies, financial returns from sustainability are shaped not solely by the extent of ESG engagement, but by the strategic alignment and perceived credibility of both communication and performance.

Keywords: Corporate Sustainability, ESG disclosure, ESG performance, Total shareholder returns, Sustainability report, BRICS, Emerging economies, Greenwashing, TMGT effect.

3.2 INTRODUCTION

Corporate sustainability has evolved from a voluntary initiative into a strategic imperative for firms in both developed and emerging economies. The integration of environmental, social, and governance (ESG) practices into corporate decision-making is now essential for long-term financial resilience, regulatory compliance, and competitive advantage (Friede, Busch, & Bassen, 2015). Beyond regulatory pressure, firms pursue sustainability to strengthen risk management, improve operational efficiency, and meet growing stakeholder expectations (Eccles, Ioannou, & Serafeim, 2014).

In emerging markets, however, the adoption of sustainability practices is hindered by institutional weaknesses, limited enforcement capacity, and heterogeneous corporate governance standards (Marquis & Qian, 2014). As these economies deepen their integration into global capital markets, firms' ability to demonstrate sustainability commitment—through both disclosure and actual performance—has become a critical factor influencing investor

confidence and access to capital (Amel-Zadeh & Serafeim, 2018). Recent studies show that, in such contexts, ESG disclosure often plays a symbolic signaling role that can decouple reporting from actual practices, increasing the risk of greenwashing in weakly enforced institutional environments (Khatib, 2024a; 2024b; Shahab et al., 2024; Khatib, 2025). Similar dynamics have been observed in other emerging economies such as Turkey, Indonesia, and Jordan, where ESG adoption is also shaped by institutional voids and legitimacy pressures (Aliyev & Alishov, 2023; Syarkani, Subu, & Waluyo, 2024; Al Amosh & Khatib, 2023a, 2023b).

A key distinction in the sustainability literature lies between disclosure and performance. Sustainability disclosure refers to the extent and quality of publicly reported ESG information, providing transparency on impacts and practices. From an agency theory perspective, such disclosure helps reduce information asymmetry and allows investors to better price sustainability-related risks (Jensen & Meckling, 1976; Kotsantonis, Pinney, & Serafeim, 2016). However, disclosure alone does not guarantee substantive engagement, as firms may strategically manipulate information to enhance legitimacy without implementing meaningful initiatives (Mahoney, Thorne, Cecil, & LaGore, 2013).

In contrast, sustainability performance reflects the tangible integration of responsible practices into business operations—such as emissions reduction, ethical labor policies, or board diversity (Clark, Feiner, & Viehs, 2015). From a stakeholder theory perspective (Freeman, 1984), such practices enhance trust and relational capital. Strong performance is often associated with operational efficiency, reputational gains, and reduced regulatory and social risks (Fatemi, Glaum, & Kaiser, 2018). However, under the shareholder theory view (Friedman, 1970), sustainability efforts must generate measurable returns; otherwise, they risk being perceived as a misallocation of resources.

Unlike firms in developed markets, companies in BRICS economies—Brazil, Russia, India, China, and South Africa—operate in fragmented regulatory systems with varying investor expectations (Li, Gong, Zhang, & Koh, 2018). In these contexts, disclosure often serves symbolic purposes, allowing firms to signal legitimacy without substantive sustainability transformation—a phenomenon commonly associated with greenwashing (Mahoney et al., 2013; Ioannou & Serafeim, 2017).

The BRICS economies offer a compelling context for investigating these dynamics, given their economic relevance, rapid industrialization, and evolving institutions. While Brazil and South Africa have developed relatively advanced frameworks, enforcement remains inconsistent (de Villiers & Marques, 2016). In China, state-owned enterprises often adopt sustainability disclosures aligned with national policy, suggesting a political legitimacy

function—consistent with legitimacy theory (Suchman, 1995). Similarly, in Russia and India, weak governance and low transparency encourage firms to use disclosure as a signaling mechanism to attract international investors. According to signaling theory (Spence, 1973), firms use disclosure to convey credibility, even when underlying performance is limited.

Although studies in developed markets link strong sustainability performance to lower capital costs, enhanced valuation, and improved risk-adjusted returns (Giese, Lee, Melas, Nagy, & Nishikawa, 2019), emerging markets face institutional voids and political risks that complicate these relationships (Godfrey, Merrill, & Hansen, 2009). Global investors increasingly demand standardized sustainability metrics, yet these often fail to reflect local business realities, creating tension between global expectations and national contexts (Jamali & Karam, 2018).

Nonetheless, sustainability has become a key determinant of firm value in BRICS countries. Firms that embed responsible practices into their strategies are more likely to attract long-term capital and enhance financial resilience. However, excessive disclosure without commensurate performance may lead to diminishing returns and reputational risk (Boiral, 2016).

This study investigates how sustainability disclosure and performance—individually and jointly—affect firm-level financial outcomes in BRICS economies. First, it adopts a multi-theoretical foundation, integrating stakeholder, agency, legitimacy, signaling, and institutional theories to explain the heterogeneous effects of ESG strategies under varying institutional conditions. Second, it expands prior research by adopting a multidimensional perspective on financial performance, incorporating profitability indicators, market valuation metrics, and total shareholder returns to offer a more comprehensive view of value creation. Third, it examines the financial implications of holistic versus selective ESG strategies, distinguishing between integrated and fragmented sustainability efforts. Fourth, it compares BRICS with developed markets to explore the contextual influence of institutional voids, greenwashing risks, and regulatory asymmetries on ESG effectiveness.

Finally, it evaluates the curvilinear joint interaction between ESG disclosure and performance, revealing a Too-Much-of-a-Good-Thing (TMGT) effect, whereby the financial benefits of ESG engagement diminish—or even reverse—beyond an optimal threshold. The analysis shows that firms with modest ESG performance, when paired with high-quality, transparent disclosure, can still achieve substantial financial returns by leveraging reputational and signaling advantages. In contrast, firms that exhibit intermediate levels of both ESG performance and disclosure fall into a “gray zone” where neither credibility nor operational

excellence is sufficiently strong to generate market rewards—resulting in the least favorable financial outcomes. By addressing these complex, non-linear dynamics, the study offers both theoretical advancement and actionable insights for firms navigating sustainability strategies in emerging markets.

The remainder of this article is structured as follows. Section 2 presents the theoretical framework and hypothesis development, drawing on agency, stakeholder, legitimacy, and signaling theories. Section 3 details the methodology, including sample construction and a Two-Stage Least Squares (2SLS) strategy to address endogeneity. Section 4 presents the empirical results; Section 5 discusses the findings and concludes with theoretical and practical contributions, as well as recommendations for future research in sustainability management within emerging markets. Complementary pre- and post-estimation tests and robustness checks are provided in the Appendix.

3.3 THEORETICAL BACKGROUND AND HYPOTHESES DEVELOPMENT

3.3.1 Theoretical Perspectives on Sustainability, Disclosure, and Performance

The theoretical foundations of corporate sustainability reflect an enduring tension between shareholder and stakeholder perspectives. Stakeholder theory (Freeman, 1984) posits that firms create long-term value by addressing the interests of multiple stakeholders. Sustainability engagement—through environmental, social, and governance (ESG) practices—enhances relational capital, mitigates reputational and regulatory risks, and supports financial performance (Freeman et al., 2018). In contrast, shareholder theory (Friedman, 1970) holds that firms should prioritize profit maximization, viewing sustainability expenditures that do not yield measurable returns as potential erosions of shareholder value (Jensen, 2002). This tension is particularly salient in emerging markets, such as BRICS economies, where regulatory ambiguity and political influence often shape corporate sustainability strategies (Carvalho Ferreira et al., 2016).

Similar patterns are observed in non-BRICS emerging economies, including Mexico and several Middle Eastern countries, where sustainability strategies are frequently adopted for symbolic or investor-driven purposes under weak enforcement conditions (Jamali, Sidani, & El-Asmar, 2009; Dempere, Alamash, & Mattos, 2024).

Complementing these views, agency theory underscores the role of sustainability disclosure in reducing information asymmetry between managers and investors (Jensen & Meckling, 1976; Dhaliwal et al., 2011). While credible disclosure may improve governance and reduce capital costs, disclosure that lacks substantive performance raises concerns about

opportunism and greenwashing, especially in weak institutional environments (Mahoney et al., 2013; Marquis & Qian, 2014).

Legitimacy theory interprets sustainability reporting as a mechanism for securing societal and institutional approval, particularly when external pressures intensify (Suchman, 1995; Deegan, 2002). It also addresses the phenomenon of decoupling, wherein firms adopt ESG reporting to signal compliance with social expectations while disconnecting disclosure from actual practices—particularly under fragmented institutional oversight. In parallel, signaling theory suggests that firms use disclosure to communicate quality and reduce uncertainty in capital markets (Spence, 1973; Connelly et al., 2011). However, excessive or purely symbolic reporting can generate investor skepticism and diminish perceived authenticity (Giese et al., 2019; Friede et al., 2015).

Institutional theory further complements these perspectives by emphasizing how corporate behavior is shaped by formal and informal institutional environments (DiMaggio & Powell, 1983; North, 1990). In emerging markets, institutional voids—such as weak legal enforcement, limited regulatory capacity, and underdeveloped capital markets—significantly influence ESG adoption strategies (Khanna & Palepu, 2010; Peng, Wang, & Jiang, 2008).

Recent studies also highlight the role of sovereign-level ESG policies in shaping corporate behavior, particularly in the financial sector. For example, Athari et al. (2024) demonstrate that sovereign ESG practices exert curvilinear effects on firm profitability in Gulf Cooperation Council countries. Similarly, Athari (2024) finds that national ESG initiatives influence banking stability in Arab economies. These findings reinforce the view that country-level policy environments condition the effectiveness of corporate ESG initiatives, especially in regions marked by regulatory asymmetries and political intervention.

This sector-specific influence is especially evident in financial institutions, where sustainability strategies often rely on green banking practices to meet ESG objectives. Gulzar et al. (2024) argue that banks can substantially improve their environmental performance by integrating ESG principles into credit policies, investment screening, and operational frameworks. These findings underscore the importance of industry-specific dynamics and institutional contexts in shaping both the implementation and perception of ESG efforts.

Mounting evidence confirms that, in fragile institutional environments, firms frequently adopt ESG disclosure as a form of symbolic compliance rather than a commitment to substantive transformation. This is particularly evident when sustainability reporting is used to enhance external legitimacy without addressing underlying operational deficiencies. Khatib (2024) shows that firms involved in corrupt practices tend to engage in superficial ESG

reporting, particularly during the shake-out stage of the corporate lifecycle, using disclosure to signal compliance rather than commitment. Similarly, Khatib (2025) finds that in contexts of low formal enforcement, firms often adopt ESG reporting as a remedial measure after regulatory violations, relying on social trust and reputational repair to regain legitimacy.

Consequently, firms may engage in symbolic compliance—adopting ESG disclosure as a legitimacy-seeking strategy without corresponding sustainability performance (Marquis & Raynard, 2015; Dieleman & Sachs, 2008). These dynamics help explain the heterogeneity of ESG strategies across emerging economies and support the expectation that the interaction between disclosure and performance is shaped by institutional strength.

Literature increasingly emphasizes the conceptual distinction between sustainability disclosure and performance. While disclosure functions as a strategic communication tool that enhances transparency and stakeholder confidence (Clark, Feiner, & Viehs, 2015), it does not inherently reflect substantive action. Firms may use disclosure opportunistically to signal commitment without implementing meaningful ESG practices (Mahoney et al., 2013). In contrast, sustainability performance refers to tangible outcomes—such as emissions reduction, workforce inclusion, or board reform—that enhance operational efficiency, mitigate risks, and improve long-term resilience (Gibson et al., 2022).

The interaction between disclosure and performance is particularly salient in emerging markets, where inconsistent regulatory oversight and low investor trust demand careful alignment of signals and actions. Firms that combine strong ESG performance with credible disclosure are more likely to enhance firm valuation and financial stability (Fatemi, Glaum, & Kaiser, 2018; Li, Gong, Zhang, & Koh, 2018). Conversely, misalignment—such as high disclosure paired with weak performance—can lead to reputational backlash and reduced investor confidence (Hummel & Schlick, 2016). Thus, understanding the independent and joint effects of ESG disclosure and performance is critical to assessing the financial relevance of corporate sustainability strategies.

3.3.2 Corporate Sustainability Framework

Corporate sustainability can be conceptualized along a continuum ranging from fully integrated, holistic strategies to fragmented or isolated initiatives (Hahn & Kühnen, 2013). Holistic sustainability refers to the strategic alignment and integration of environmental, social, and governance (ESG) dimensions across all levels of the organization.

Firms that adopt this comprehensive approach tend to disclose sustainability information in a unified and connected format, linking financial, environmental, and social data

into a cohesive narrative. Such integration not only enhances transparency but also strengthens stakeholder trust and supports long-term value creation through improved financial performance and risk management. In contrast, firms that pursue isolated or compliance-based ESG activities—such as stand-alone environmental reports or selective initiatives—often fail to generate strategic synergy. These practices may signal commitment but lack operational depth, thereby weakening stakeholder engagement and diminishing financial outcomes (Lueg & Pesheva, 2021).

Figure 1 presents the conceptual framework adopted in this study to distinguish between holistic, selective, and isolated ESG strategies. Drawing on the integrative perspectives of Hahn and Kühnen (2013) and Lueg and Pesheva (2023), the framework classifies ESG approaches across six analytical dimensions: strategic orientation, ESG coverage, motivation, disclosure coherence, institutional context, and expected impact.

Conceptual Distinction between Holistic and Selective ESG Strategies

Dimension	Holistic ESG Strategy	Selective or Isolated ESG Strategy
Strategic Orientation	Integrated into core business strategy	Reactive or peripheral to strategy
ESG Coverage	Balanced engagement with **ESG** dimension	Focus on one or two **E, S, and G** dimensions
Motivation	Driven by long-term value creation and stakeholder alignment	Often driven by compliance, visibility, or specific stakeholder pressure
Disclosure Coherence	Consistent with actual ESG practices and performance	May emphasize disclosure over substance (risk of greenwashing)
Institutional Context	More common in **developed or strongly regulated** institutional settings	More common in **emerging markets with institutional voids**
Expected Impact	Greater stakeholder trust, reputational capital, and financial resilience	Limited or inconsistent impact on performance

Figure 18 - Conceptual Distinction between Holistic and Selective ESG Strategies

Note. This framework is based on conceptual distinctions adapted from Hahn and Kühnen (2013) and Lueg and Pesheva (2023). Created by the authors.

Holistic ESG strategies are characterized by their integration into the core business strategy, balanced engagement with all ESG dimensions, and a long-term value orientation aligned with stakeholder interests. These strategies are typically observed in developed or strongly regulated institutional contexts, where firms tend to exhibit consistency between

disclosure and actual ESG performance. Consequently, holistic strategies are associated with greater stakeholder trust, reputational capital, and enhanced financial resilience.

In contrast, selective ESG strategies reflect a fragmented or reactive approach, with emphasis on one or two ESG pillars and motivations often driven by regulatory compliance or external visibility. These strategies are more prevalent in emerging markets with institutional voids, where weak enforcement enables firms to prioritize symbolic disclosure over substantive implementation. As a result, the financial impact of selective strategies tends to be limited or inconsistent.

This framework guides our empirical investigation by offering a structured lens through which to interpret the performance implications of different ESG configurations in emerging economies.

3.3.3 The Financial Effects of Holistic versus Isolated ESG Strategies

Corporate sustainability strategies differ in scope and level of integration. Holistic approaches embed environmental, social, and governance (ESG) dimensions across core business operations, aligning them with long-term strategic objectives. In contrast, isolated ESG efforts are often fragmented and ad hoc, lacking strategic coherence and integration. This distinction is critical for understanding the divergent financial outcomes associated with different ESG practices.

Empirical research underscores the superior financial performance associated with holistic ESG integration. Eccles et al. (2014) demonstrate that firms embedding ESG principles into their core strategies consistently outperform those adopting superficial or compliance-oriented practices. Similarly, Lee and Li (2022) find that selective or unbalanced ESG initiatives frequently lead to diminishing returns, driven by inefficiencies and stakeholder skepticism. Khan et al. (2016) further argue that ESG activities yield financial benefits only when they address financially material issues.

Collectively, these findings emphasize the importance of strategic alignment, consistency, and material relevance in ESG engagement. When ESG performance and disclosure are integrated and coherent, they enhance transparency, strengthen stakeholder trust, and contribute to improved firm valuation.

Accordingly, we propose the following hypotheses:

H1a. Holistic sustainability performance influences financial performance.

H1b. Isolated environmental, social, and governance performance influences financial performance.

H2a. Holistic sustainability disclosure influences financial performance.

H2b. Isolated environmental, social, and governance disclosure influences financial performance.

3.3.4 The Financial Impact of Selective ESG Strategies

While holistic ESG strategies integrate environmental, social, and governance dimensions in a cohesive manner, many firms—particularly in emerging markets—adopt selective approaches. These strategies emphasize one or two ESG pillars, often in response to sector-specific demands or external stakeholder pressure (Kotsantonis & Serafeim, 2019). For example, high-emission industries may prioritize environmental reporting, whereas financial institutions might focus on governance reforms. Although such targeted initiatives may yield localized improvements in stakeholder relations or regulatory compliance, they risk undermining the synergistic benefits associated with fully integrated ESG adoption (Hahn & Kühnen, 2013; Lee & Li, 2022).

Selective strategies may also signal opportunism or superficial engagement, especially when firms emphasize more "visible" ESG components to appeal to investors while neglecting more complex or less tangible dimensions. In these cases, the coherence and depth of ESG integration are compromised, which can limit long-term financial benefits (Marquis & Raynard, 2015). This issue is particularly relevant in emerging markets, where institutional voids enable firms to leverage symbolic disclosure as a tool for gaining legitimacy, without necessarily improving underlying practices (Peng et al., 2008; Dieleman & Sachs, 2008).

Accordingly, we hypothesize:

H3a. Selective environmental, social, and governance performance—when combined—influences financial performance.

H3b. Selective environmental, social, and governance disclosure—when combined—influences financial performance.

3.3.5 Curvilinear Effects of ESG Engagement

Recent research increasingly suggests that the relationship between ESG engagement and financial performance may follow a non-linear trajectory. While moderate levels of ESG performance and disclosure are generally associated with improved firm value, excessively high levels can lead to diminishing returns due to rising operational costs, stakeholder skepticism, and perceived inefficiencies (Clark et al., 2015; Giese et al., 2019; Margolis et al., 2009). This phenomenon—commonly referred to as the Too-Much-of-a-Good-Thing (TMGT)

effect—implies decreasing marginal benefits and even potential reversals beyond an optimal threshold of ESG commitment (Zhang et al., 2022).

Firms that strike a balance between substantive ESG actions and coherent disclosure tend to benefit from reputational gains, lower capital costs, and stronger stakeholder engagement (Fatemi et al., 2018). Conversely, firms that overemphasize disclosure without matching performance risk being perceived as engaging in greenwashing, which can erode trust and depress firm valuation (Mahoney et al., 2013; Boiral, 2016). Similarly, firms that exhibit high ESG performance but fail to disclose it adequately may be unable to capitalize on their achievements, thereby limiting access to ESG-sensitive capital (Lys et al., 2015).

These curvilinear dynamics are particularly salient in BRICS economies, where ESG regulatory frameworks remain underdeveloped and investor expectations are evolving rapidly. In such environments, the key challenge lies in calibrating ESG efforts to avoid both under- and over-engagement, while maintaining strategic alignment and credibility.

Accordingly, we propose the following hypotheses:

H4a. Holistic sustainability performance has a curvilinear (quadratic) relationship with financial performance.

H4b. Holistic sustainability disclosure has a curvilinear (quadratic) relationship with financial performance.

3.3.6 ESG Disclosure–Performance Interaction

While ESG disclosure and ESG performance are often examined as distinct constructs, growing evidence suggests that their interplay is critical to financial outcomes. Firms that not only achieve substantive ESG results but also communicate them transparently tend to foster greater stakeholder trust, reduce perceived risks, and secure preferential access to capital (Ioannou & Serafeim, 2017; Giese et al., 2019). This alignment enhances credibility and strengthens the signaling effect of ESG disclosures, particularly in markets characterized by high information asymmetry.

Conversely, when firms engage in extensive ESG disclosure without corresponding operational substance, they risk being perceived as symbolic or disingenuous—a phenomenon widely recognized as greenwashing (Boiral, 2016; Mahoney et al., 2013). Such misalignment can trigger investor skepticism, damage reputational capital, and result in weaker financial performance. Empirical evidence from emerging markets such as Jordan and Indonesia confirms that in weakly regulated environments, ESG disclosure often precedes—or even substitutes—actual sustainability practices (Al Amosh & Khatib, 2023a; Syarkani et al., 2024).

At the same time, firms that demonstrate strong ESG performance but under-communicate their achievements may fail to capture associated reputational or valuation benefits. This underscores the importance of a balanced ESG signaling strategy (Lys et al., 2015). Ultimately, the financial value of ESG engagement depends not only on what firms do, but also on how effectively they communicate it.

In line with signaling and legitimacy theories, we hypothesize:

H5a. Holistic sustainability performance and holistic disclosure jointly influence financial performance.

H5b. This joint effect follows a curvilinear relationship with financial performance.

3.4 METHODOLOGY

3.4.1 Sample and data sources

The dataset is derived from the Bloomberg database, encompassing comprehensive and standardized financial and ESG information. The sample includes publicly listed firms across various industries within the BRICS economies, selected based on the availability of ESG scores and financial performance data. The analysis spans the period from January 1, 2016, to December 31, 2023, resulting in 2.987 firm-year observations.

This consistent temporal coverage ensures robust longitudinal analysis of ESG measures and financial performance. To contextualize the corporate sustainability data, the study integrates Foreign Direct Investment (FDI) inflow data sourced from the World Bank website. The World Bank is recognized for its reliable, globally comparable macroeconomic data, collected and validated through stringent methodologies (Dollar & Kraay, 2003; Kaufmann, Kraay, & Mastruzzi, 2010). This combination of firm-level and macroeconomic data provides a holistic perspective on the dynamics of ESG and shareholder value creation within the BRICS economies.

Table 9 describes the data set encompassing ten industries, with the highest representation from Industrials (544 observations), Information Technology (535 observations), and Materials (447 observations). This distribution ensures coverage of diverse sectors is critical to the BRICS economies.

Table 9 - Industry Sample Distribution

Industry	Brazil	China	India	South Africa	Total
Healthcare	32	194	32	8	266
Consumer Discretionary	93	151	64	32	340

Energy	40	96	32	8	176
Real Estate	24	32	0	8	64
Industrials	72	432	32	8	544
Materials	79	280	48	40	447
Consumer Staples	96	135	40	24	295
Communication Services	16	64	8	16	104
Utilities	96	104	16	0	216
Information Technology	16	479	40	0	535
Total	564	1967	312	144	2987

Note: Elaborated by authors

Table 10 provides a detailed breakdown of the yearly distribution of firm-year observations for the sample used in this study. It includes data from the BRICS countries—Brazil, China, India, and South Africa—spanning an eight-year period from 2016 to 2023. However, Russia is notably absent from the dataset.

The Russia-Ukraine war began on February 24, 2022, when Russia launched a large-scale invasion of Ukraine. The conflict is ongoing, and the international community has imposed a range of economic and financial sanctions on Russia since the invasion started. These sanctions have significantly impacted data availability and financial reporting from Russian entities. For this reason, the financial and sustainability data was not available in Bloomberg's terminal and is not considered in this analysis.

Table 10 - Country Sample Distribution

Year	Brazil	China	India	South Africa	Total
2016	72	249	39	18	378
2017	71	249	39	18	377
2018	71	249	39	18	377
2019	71	249	39	18	377
2020	71	247	39	18	375
2021	70	243	39	18	370
2022	70	243	39	18	370
2023	68	238	39	18	363
Total	564	1967	312	144	2987

Note: Elaborated by authros

3.4.2 Measurements

This section outlines the key variables and metrics employed in the study, categorized into dependent variables, independent variables, and control variables. The selected measurements are designed to comprehensively capture corporate financial performance, ESG

metrics, and other relevant factors, ensuring a robust analysis of the relationships between these constructs. The variables and their definitions are supported by data sourced from reputable platforms such as Bloomberg and the World Bank, ensuring reliability and consistency.

Table 11 presents the descriptive statistics for the key variables used in the study, offering insights into their distribution, central tendencies, and variability.

Table 11 - Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
TSR	2925	15.002	18.352	-205.901	243.378
Tobins Q	2790	2.703	2.938	0.492	40.259
ROA	2954	7.399	8.641	-94.044	120.812
ROE	2930	15.174	19.741	-293.197	297.279
ESG	2545	2.771	1.185	0.69	7.63
ESG E	2217	2.51	1.787	0.01	9.37
ESG S	2410	2.364	1.678	0.04	10
ESG G	2582	4.94	0.777	2.35	9.7
ESG Dis	2686	45.82	12.273	20.36	98.662
ESG E Dis	2567	34.63	20.31	0.423	90.577
ESG S Dis	2686	28.505	13.422	3.204	88.09
ESG G Dis	2681	75.643	10.293	46.96	98.615
Sales	2968	168006.2	514050.04	7.843	8778350
MKT Cap	2790	272085.87	907342.87	286.358	17826703
MKT to Book	2796	4.498	9.299	0.076	369.386
FDI	2624	1.406e+11	6.834e+10	2.000e+09	2.125e+11

Note: Generated in Stata software. All variables are in raw format.

3.4.3 Variable construction and transformation

Dependent variables include Total Shareholder Return (TSR), Tobin's Q, Return on Assets (ROA), and Return on Equity (ROE). Independent variables cover agregated and pillar-level ESG performance and disclosure scores, extracted via Bloomberg mnemonics. Control variables include market capitalization, sales, firm age, and FDI inflow.

All variables were winsorized and standardized using z-scores, allowing for comparability across scales and enhancing model interpretability (Gelman, 2008; Aiken & West, 1991). Z-scores reduce multicollinearity and allow regression coefficients to be interpreted as the effect of a one-standard-deviation change, enabling meaningful comparisons of ESG and financial variables (Bring, 1994).

Table 12 provides an overview of the variables utilized in this study, categorized into dependent, independent, and control variables. These variables serve as the foundation for analyzing the relationship between ESG performance, financial performance, and control variables. This table includes the mnemonics from the Bloomberg Terminal, which are standardized identifiers used to extract each variable, ensuring consistency and traceability of the data. Additionally, the table incorporates detailed descriptions of each construct as defined by Bloomberg, offering clarity on how these metrics are calculated and their relevance to the study. This level of detail ensures the study's transparency and facilitates replication by other researchers.

Table 12 - Variable Description

Type	Variable	Code	Description	Source
DPV	Total Shareholder Return (TSR)	F1169	Return on equity (ROE), in percentage, measures a corporation's profitability by revealing how much adjusted profit a company generates with the money shareholders have invested.	Bloomberg
DPV	Tobin´s Q	RX242	Ratio of the market value of a firm to the replacement cost of the firm's assets.	Bloomberg
DPV	ROA	RR028	Indicator of a company's profitability relative to its total assets, in percentage.	Bloomberg
DPV	ROE	RR029	Corporation's profitability measure revealing how much profit a company generates with shareholders' money, in percentage.	Bloomberg
INDV	ESG Score	SR001	Bloomberg's score evaluating a company’s aggregated ESG performance based on financial materiality.	Bloomberg
INDV	ESG_E	SR002	Bloomberg's score evaluating a company’s aggregated environmental performance.	Bloomberg
INDV	ESG_S	SR003	Bloomberg's score evaluating a company’s aggregated social performance.	Bloomberg

INDV	ESG_G	SR004	Bloomberg's score evaluating a company's aggregated governance performance.	Bloomberg
INDV	ESG Disc	RX317	Bloomberg's proprietary score based on the extent of a company's ESG data disclosure.	Bloomberg
INDV	ESG_E_Disc	RX374	Bloomberg's proprietary score for environmental data disclosure extent.	Bloomberg
INDV	ESG_S_Disc	RX375	Bloomberg's proprietary score for social data disclosure extent.	Bloomberg
INDV	ESG_G_Disc	RX376	Bloomberg's proprietary score for governance data disclosure extent.	Bloomberg
CV	Market Capitalization	RR250	Total market value of all outstanding shares of a company at the end of the declared period in the company's reporting currency.	Bloomberg
CV	Market-to-book ratio	RR903	Share price-to-book value ratio, calculated as: Price/Book Value = Latest Price/ Book Value per Share.	Bloomberg
CV	Sales	IS800	Net sales value adjusted for revenues attributable to extraordinary items, such as restructuring, mergers and acquisitions, subsidiary sales gains, and other non-recurring activities.	Bloomberg
CV	FDI	no	Foreign Direct Investment	World Bank

Note: Elaborated by authors. DV: dependent variable; INDV: independent variable, CV: control variable.

3.4.4 Econometric model

Following Khatib (2025), we recognize the importance of explicitly diagnosing and addressing potential endogeneity issues in ESG–financial performance models. Endogeneity may arise from omitted variables, measurement errors, reverse causality, or simultaneity. In this study, we carefully examine these sources and adopt a structured approach consistent with the best practices outlined in the recent literature.

We first address potential sources of endogeneity through both theoretical reasoning and empirical diagnostics. Reverse causality is a central concern in ESG research, particularly in emerging markets, where ESG disclosure may respond to firm performance rather than proactively shape it. As highlighted by Hahn and Kühnen (2013), corporate financial performance can influence sustainability reporting—either by providing firms with the resources to invest in ESG communication or by triggering strategic responses to stakeholder

pressures. This underscores the possibility that ESG practices and disclosures are endogenous to financial outcomes.

To empirically assess this issue, we applied the Durbin–Wu–Hausman (DWH) test to the ESG performance and disclosure models. This procedure evaluates whether the potentially endogenous regressors (aggregate and disaggregated ESG dimensions, as well as ESG disclosure variables) are correlated with the error term, which would bias OLS estimates. The results, reported in Appendix A (Table A1), provide mixed evidence across specifications: in several models the null hypothesis of exogeneity is rejected, indicating significant endogeneity ($p < 0.05$), while in others the test suggests no endogeneity or borderline outcomes ($0.05 \leq p < 0.10$). In such cases, the application of instrumental variable estimations becomes necessary to obtain consistent coefficients, thereby reinforcing the econometric robustness of our analysis.

To test for reverse causality, we estimated a series of regressions in which ESG performance and disclosure scores were regressed on financial variables (TSR, ROA, ROE, Tobin's Q), including their lags. The results, presented in Appendix Table A2, indicate that several financial indicators significantly predict ESG metrics, confirming potential endogeneity—as also noted by Lueg and Pesheva (2023), although with different findings.

To mitigate omitted variable bias, we adopt the model specification proposed by Lueg and Pesheva (2023), which includes a comprehensive set of financial and institutional control variables, as well as sector dummies and year fixed effects. This peer-reviewed framework supports the assumption that relevant observable covariates are adequately accounted for.

To evaluate the time-series properties of the panel variables, we conducted Fisher-type Augmented Dickey-Fuller (ADF) tests using Pm statistics. As shown in Appendix A (Table A3), all variables reject the null hypothesis of non-stationarity at the 1% level, suggesting that they are stationary in panel form. This applies to both ESG metrics (aggregate and disaggregated) and financial performance indicators. The stationarity of ESG variables is particularly relevant, as they typically exhibit persistence over time. Nonetheless, the results indicate sufficient temporal and cross-sectional variation to support estimation in level form, mitigating concerns about spurious regression.

We also consider the possibility of cross-sectional dependence, which may arise from unobserved macroeconomic or industry-level shocks. To address this, we include sector and year fixed effects in all models and cluster standard errors at the firm level—thereby accounting for intra-firm autocorrelation and heteroskedasticity, as recommended by Petersen (2009). For robustness, we compare results estimated with and without clustered standard errors. As

expected, clustered standard errors are generally larger, but the main coefficients remain directionally consistent and statistically significant, reinforcing the validity of the results.

To address endogeneity, we implement Two-Stage Least Squares (2SLS) estimation using lagged ESG performance and disclosure scores as instruments. These lagged scores are commonly employed in literature to capture temporal precedence and presumed exogeneity with respect to current financial outcomes (e.g., Lu, Onuk, Xia, & Zhang, 2025; Yu & Xiao, 2022). To confirm instrument strength, we report Kleibergen–Paap Wald F-statistics, all of which exceed the critical threshold of 10. Hansen J-tests consistently fail to reject the null hypothesis of instrument exogeneity. Additional diagnostics—including Anderson–Rubin and Kleibergen–Paap LM tests—are reported in Tables 6 to 11. First-stage regressions are statistically significant at the 1% level in nearly all models, confirming the relevance and validity of the instruments. Table 13 resumes the instrumental variable diagnostics and validity assessment criteria to ensure the quality of regressions.

Table 13 - Instrumental Variable Diagnostics and Validity Assessment

Test	Description	Interpretation
First-Stage F-statistic	Tests the strength of the instrumental variables by measuring how well they explain the endogenous regressors.	$F > 10$ indicates strong instruments. Higher values suggest better predictive power.
Kleibergen-Paap LM Test (Underidentification)	Test whether the instrumental variables are relevant, meaning they are correlated with the endogenous regressors.	$p < 0.05$ suggests the instruments are relevant, rejecting underidentification.
Weak Identification Test (Stock-Yogo Critical Values)	Evaluates whether the instruments are weak, which could lead to biased estimates.	If F-stat > critical value, instruments are not weak. Lower values indicate potential weak instruments.
Hansen J-test (Overidentification Test)	Tests whether the instruments are valid (i.e., exogenous), meaning they are uncorrelated with the error term.	$p > 0.05$ suggests valid instruments. A high p-value indicates no violation of exogeneity.
Anderson-Rubin Wald Test (Weak IV-robust inference)	Provides a joint significance test of the endogenous regressors, accounting for weak instruments.	$p > 0.05$ means there is no evidence of weak instruments. If significant, instruments may be problematic.

Cragg-Donald F-statistic	Measures instrument strength in multiple endogenous variable models.	Higher values indicate stronger instruments. Compared to Stock-Yogo critical values for validity.

Note. This table summarizes the main diagnostic tests applied to assess the strength and validity of the instrumental variables used in the models. The First-Stage F-statistic evaluates instrument relevance, with values above 10 indicating sufficient strength. The Kleibergen-Paap LM test assesses underidentification, where $p < 0.05$ rejects the null of underidentification. The Weak Identification test compares the first-stage F-statistic against Stock–Yogo critical values to detect weak instruments. The Hansen J-test examines overidentification, with $p > 0.05$ suggesting that instruments are valid and uncorrelated with the error term. The Anderson–Rubin Wald test provides weak-IV robust inference for joint significance of endogenous regressors, while the Cragg–Donald F-statistic further evaluates instrument strength in multi-endogenous variable settings.

Overall, the combination of theoretical justification, empirical evidence of endogeneity, comprehensive diagnostics, and consistent results across specifications supports the use of 2SLS as a reliable approach to estimate the causal effect of ESG engagement on financial performance in emerging markets.

The first stage regresses ESG performance on instrumental variables and control variables:

$$(1)\ \mathrm{ESG}_{it} = \gamma_0 + \gamma_1\, \mathrm{IV}_{it} + \gamma_2\, \mathrm{Controls}_{it} + \mu_i + \nu_t + \varepsilon_{it}$$

Where,

ESG_{it}: Endogenous independent variable (e.g., ESG measurements).

IV_{it}: Instrumental variables, such as lagged ESG scores or industry characteristics, strongly correlated with ESG performance but uncorrelated with the error term.

$\mathrm{Controls}_{it}$: Control variables, including firm-level characteristics (e.g., market capitalization, sales, market-to-book index) and macroeconomic factors (e.g., FDI).

μ_i: Sector-specific fixed effects.

ν_t: Time-specific effects.

ε_{it}: Error term.

The second stage estimates the impact of predicted ESG performance on the dependent variable:

$$(2)\ Y_{it} = \beta_0 + \beta_1\, \hat{\mathrm{Y}}_\mathrm{ESG}_{it} + \beta_2\, \mathrm{Controls}_{it} + \mu_i + \nu_t + Y\eta_{it}$$

In this equation:

Y_{it}: Dependent variable representing financial performance (e.g., ROA, ROE, TSR, Tobin´s Q).

$\hat{\mathrm{Y}}_\mathrm{ESG}_{it}$: Predicted values of ESG performance obtained from the first stage.

Controls_{it}: Control variables as specified above.

μ_i: Sector-specific effects.

ν_t: Time-specific effects.

$Y\eta_{it}$: Error term.

3.5 RESULTS

3.5.1 ESG Performance and Financial Metrics

Table 14 presents 2SLS regression results evaluating the impact of ESG performance—measured using Bloomberg ESG Scores—on Total Shareholder Return (TSR), Tobin's Q, Return on Assets (ROA), and Return on Equity (ROE). The analysis is based on 2,231 firm-year observations and includes fixed effects for year and industry.

Holistic ESG performance does not exhibit a statistically significant association with any of the financial metrics. Although the coefficient for TSR is positive ($\beta = 0.0350$), it is not significant, suggesting a neutral effect. This finding diverges from studies such as Grishunin et al. (2023) and Strekalina et al. (2023), which report positive impacts of ESG performance among BRICS firms and within certain sectors. Similarly, no significant effects are found for ROA and ROE, aligning with prior research indicating that aggregated ESG scores may not influence short-term profitability.

By contrast, a negative and statistically significant relationship is observed between ESG performance and Tobin's Q ($\beta = -0.0374$, $p < 0.01$), suggesting that stronger ESG commitments may reduce market valuation. This may reflect investor concerns about excessive cost allocation or perceptions of symbolic signaling over substantive engagement.

Disaggregating ESG into its environmental (E), social (S), and governance (G) components reveals distinct effects. Environmental performance (ESG_E) is positively associated with TSR ($\beta = 0.0682$, $p < 0.05$) and ROE ($\beta = 0.181$, $p < 0.10$), indicating reputational and operational benefits. These results are consistent with Agranat (2023) and Umar et al. (2022), who emphasize the financial relevance of environmental initiatives.

Governance performance (ESG_G) also shows a significant positive association with Tobin's Q ($\beta = 0.0229$, $p < 0.05$), reinforcing prior evidence that strong governance practices enhance market confidence and reduce capital costs.

Conversely, social performance (ESG_S) shows no significant relationship with any of the financial indicators, reflecting the ambiguous role of social factors commonly reported in the literature. This dimension appears less influential in driving financial outcomes, potentially due to perceived implementation costs or limited investor responsiveness.

Table 14: Impact of Environmental, Social and Governance Score on Financial Performance - 2SLS Regression Results. (Model 1.1 to 1.8).

Variables	TSR (1.1)	TQ (1.2)	ROA (1.3)	ROE (1.4)	TSR (1.5)	TQ (1.6)	ROA (1.7)	ROE (1.8)
zw_ESG	0.0350	**-0.0374*** **	0.000673	0.0207				
	(0.0267)	**(0.0117)**	(0.0238)	(0.0259)				
zw_ESG_E					**0.0682** **	0.0289	0.140	**0.181***
					(0.0343)	(0.0189)	(0.0883)	**(0.101)**
zw_ESG_S					0.0200	-0.0203	-0.0446	0.00466
					(0.0288)	(0.0144)	(0.0469)	(0.0492)
zw_ESG_G					0.0213	**0.0229** **	0.0333	-0.0157
					(0.0337)	**(0.0112)**	(0.0379)	(0.0483)
zw_Sales	0.0746***	-0.0160	0.000198	0.0357*	0.0784***	-0.0138	0.0197	0.0461*
	(0.0254)	(0.0123)	(0.0198)	(0.0205)	(0.0256)	(0.0122)	(0.0227)	(0.0235)
zw_MKT_Cap	-0.0999***	0.0883***	0.0160	-0.0796**	-0.116***	0.0901***	-0.0296	-0.129***
	(0.0343)	(0.0315)	(0.0307)	(0.0318)	(0.0339)	(0.0313)	(0.0393)	(0.0445)
zw_MTB	0.302***	0.795***	0.149***	0.292***	0.288***	0.797***	0.159***	0.295***
	(0.0363)	(0.0890)	(0.0323)	(0.0352)	(0.0368)	(0.0901)	(0.0371)	(0.0420)
zw_FDI	-0.0623**	0.0680***	0.0123	-0.0596**	-0.0559**	0.0873***	0.0354	-0.0610
	(0.0278)	(0.0117)	(0.0247)	(0.0288)	(0.0277)	(0.0129)	(0.0323)	(0.0387)
L.zw_TSR	0.554***				0.557***			
	(0.0357)				(0.0360)			
L.zw_TQ		0.444***				0.446***		
		(0.0403)				(0.0407)		
L.zw_ROA			0.685***				0.636***	
			(0.0276)				(0.0365)	
L.zw_ROE				0.536***				0.522***
				(0.0363)				(0.0458)
Constant	-0.0799	0.218***	0.0359	-0.0904	-0.0660	0.237***	0.0235	-0.176**
	(0.0559)	(0.0506)	(0.0601)	(0.0600)	(0.0565)	(0.0514)	(0.0762)	(0.0781)
Year Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-Stage F-stat (ESG)	2303.98***	3157.62***	2255.49***	2299.76***				
First-Stage F-stat (ESG_E)					166.96***	151.14***	45.10***	45.56***
First-Stage F-stat (ESG_S)					347.88***	451.92***	249.44***	254.27***
First-Stage F-stat (ESG_G)					387.20***	693.90***	267.69***	271.69***
Kleibergen-Paap LM Test (Underident)	610.91***	539.12***	596.52***	610.07***	404.66***	316.58***	121.57***	121.22***
Weak Ident. Test (Stock-Yogo)	19.93	19.93	19.93	19.93	155.76***	115.46***	19.65**	19.45**
Hansen J-test (Overidentification)	1.315	0.157	1.695	1.717	5.206	2.277	6.145	4.881
Anderson-Rubin Wald Test	1.24	5.14**	0.87	1.02	1.96*	2.05*	1.48	1.73
Observations	2,231	2,038	2,231	2,231	2,231	2,038	1,477	1,477
R-squared	0.424	0.859	0.528	0.397	0.419	0.858	0.464	0.373

Note: The variables are TSR = total shareholder return, TQ = Tobin's Q, ROA = return on assets, ROE = return on equity, FDI = foreign direct investment, MKT_Cap = market capitalization, Sales = annual sales, MTB = market-to-book ratio, and ESG = corporate sustainability performance. The disaggregated ESG components include ESG_E = environmental score, ESG_S = social score, and ESG_G = governance score. All variables are in z-scored form. This table presents the 2SLS regression results, addressing potential endogeneity concerns. Year and industry fixed effects are included in all models, and lagged dependent variables account for performance persistence. Instrument validity is confirmed by the first-stage F-statistics, Kleibergen-Paap LM test, and Hansen J-test. Bold numbers indicate statistical significance. *p < 0.10; **p < 0.05; ***p < 0.01. Data source: Bloomberg.

The absence of significance may stem not only from institutional and regulatory weaknesses in BRICS economies, but also from limited stakeholder pressure. In many of these contexts, key social actors—such as employees, consumers, and civil society—lack the legal instruments or influence needed to shape corporate behavior or reward social initiatives.

As noted by Jamali and Karam (2018), in institutional environments characterized by weak civic engagement and limited stakeholder voice, social sustainability efforts often fail to translate into economic value. Similarly, Ioannou and Serafeim (2015) argue that social metrics are frequently undervalued by market analysts due to their perceived ambiguity and lack of financial materiality.

In sum, while environmental and governance performance are positively associated with firm performance, the social dimension remains financially neutral. Holistic ESG performance, as an aggregate construct, does not provide conclusive evidence of enhancing shareholder value or profitability. On the contrary, results suggest that excessive or unfocused ESG efforts may erode firm value in the long term, consistent with the premises of shareholder theory.

3.5.2 ESG Disclosure and Financial Performance

Table 15 presents 2SLS regression results assessing the relationship between ESG disclosure and financial performance. Holistic ESG disclosure (zw_ESG_Dis) exhibits a significant positive effect on TSR ($\beta = 0.165$, $p < 0.01$), ROA ($\beta = 0.106$, $p < 0.01$), and ROE ($\beta = 0.155$, $p < 0.01$), indicating that greater transparency enhances operational efficiency and investor confidence. These results are consistent with Lueg and Pesheva (2021), who find that integrated disclosure improves shareholder value.

However, ESG disclosure negatively affects Tobin's Q ($\beta = –0.0328$, $p < 0.05$), suggesting investor skepticism regarding the long-term value of sustainability reporting. This result reflects concerns that increased transparency may signal higher regulatory costs or resource reallocation—findings aligned with Fatemi et al. (2018) and Christensen et al. (2022).

Comparative evidence supports this dual effect: while disclosure may lower the cost of debt (Dua & Sharma, 2024), it can simultaneously raise the cost of equity and the overall weighted average cost of capital (WACC). This highlights a disconnect between internal efficiency gains and market-based valuation.

Disaggregated analysis reveals that governance disclosure (ESG_G_Dis) is the most influential component, showing positive associations with Tobin's Q ($\beta = 0.0405$, $p < 0.01$), ROA ($\beta = 0.125$, $p < 0.05$), and ROE ($\beta = 0.110$, $p < 0.10$). These findings underscore the role

Table 15 - Impact of Environmental, Social and Governance Disclosure on Financial Performance (Model 2.1 to 2.8).

Variables	TSR (2.1)	TQ (2.2)	ROA (2.3)	ROE (2.4)	TSR (2.5)	TQ (2.6)	ROA (2.7)	ROE (2.8)
zw_ESG_Dis	**0.165*** **	**-0.0328** **	**0.106** **	**0.155*** **				
	(0.0519)	**(0.0131)**	**(0.0474)**	**(0.0524)**				
zw_ESG_E_Dis					-0.0468	**0.0646** **	0.0303	-0.0241
					(0.0642)	**(0.0304)**	(0.0582)	(0.0675)
zw_ESG_S_Dis					**0.167** **	**-0.119*** **	0.0108	**0.122***
					(0.0692)	**(0.0287)**	(0.0611)	**(0.0709)**
zw_ESG_G_Dis					**0.106***	**0.0405*** **	**0.125** **	**0.110***
					(0.0608)	**(0.0157)**	**(0.0511)**	**(0.0611)**
zw_Sales	0.0855***	-0.0139	0.00766	0.0381*	0.0867***	-0.0232*	0.000612	0.0362
	(0.0272)	(0.0121)	(0.0215)	(0.0224)	(0.0289)	(0.0137)	(0.0225)	(0.0236)
zw_MKT_Cap	-0.156***	0.0948***	-0.0219	-0.127***	-0.176***	0.0871**	-0.0426	-0.144***
	(0.0385)	(0.0323)	(0.0332)	(0.0363)	(0.0380)	(0.0343)	(0.0341)	(0.0362)
zw_MTB	0.311***	0.790***	0.151***	0.289***	0.295***	0.752***	0.142***	0.275***
	(0.0411)	(0.0894)	(0.0352)	(0.0392)	(0.0425)	(0.0949)	(0.0359)	(0.0406)
zw_FDI	-0.0514	0.0726***	0.0363	-0.0385	-0.0448	0.0511***	0.00155	-0.0448
	(0.0351)	(0.0120)	(0.0292)	(0.0347)	(0.0392)	(0.0178)	(0.0348)	(0.0395)
L.zw_TSR	0.528***				0.533***			
	(0.0403)				(0.0411)			
L.zw_Tobins_Q		0.444***				0.470***		
		(0.0404)				(0.0461)		
L.zw_ROA			0.658***				0.662***	
			(0.0305)				(0.0306)	
L.zw_ROE				0.524***				0.531***
				(0.0394)				(0.0401)
Constant	-0.105*	0.224***	0.0442	-0.125*	-0.106*	0.321***	0.0387	-0.127*
	(0.0630)	(0.0506)	(0.0608)	(0.0637)	(0.0643)	(0.0503)	(0.0621)	(0.0649)
Year Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-Stage F-stat (ESG_Disc)	574.60***	1672.33***	561.02***	571.37***				
First-Stage F-stat (ESG_E_Disc)					250.86***	328.32***	244.77***	250.28***
First-Stage F-stat (ESG_S_Disc)					242.07***	326.80***	235.07***	241.81***
First-Stage F-stat (ESG_G_Disc)					129.45***	226.38***	124.93***	127.47***
Kleibergen-Paap LM Test (Underident)	375.94***	452.32***	369.40***	375.93***	152.73***	139.18***	156.43***	153.06***
Weak Ident. Test (Stock-Yogo)	19.93	19.93	19.93	19.93	44.37***	39.04***	45.64***	44.53***
Hansen J-test (Overidentification)	0.930	0.413	0.091	0.343	2.058	3.249	6.211	3.641
Anderson-Rubin Wald Test	5.82***	3.16**	2.62*	4.70***	2.96**	4.30***	2.61**	2.67**
Observations	1,854	2,038	1,854	1,854	1,854	1,669	1,854	1,854
R-squared	0.387	0.859	0.494	0.373	0.371	0.860	0.485	0.359

Note: The variables are TSR = total shareholder return, TQ = Tobin's Q, ROA = return on assets, ROE = return on equity, FDI = foreign direct investment, MKT_Cap = market capitalization, Sales = annual sales, MTB = market-to-book ratio, and ESG_Dis = corporate sustainability disclosure score. The disaggregated ESG disclosure components include ESG_E_Dis = environmental disclosure score, ESG_S_Dis = social disclosure score, and ESG_G_Dis = governance disclosure score. All variables are in z-scored form. This table presents the 2SLS regression results, assessing the impact of ESG disclosure on financial performance. Year and industry fixed effects are included, and lagged dependent variables account for persistence. Instrument validity is confirmed by the first-stage F-statistics, Kleibergen-Paap LM test, and Hansen J-test. Bold numbers indicate statistical significance. *p < 0.10; **p < 0.05; ***p < 0.01. Data source: Bloomberg.

of governance transparency in reducing agency costs and reinforcing market trust (Nuhu & Alam, 2024; Dua & Sharma, 2024).

Social disclosure (ESG_S_Dis) is positively associated with ROA and ROE, yet negatively affects Tobin's Q ($\beta = -0.119$, $p < 0.01$), suggesting that investors may remain cautious about the long-term financial implications of CSR-related expenditures. Environmental disclosure (ESG_E_Dis), meanwhile, shows no statistically significant effects across financial metrics, in line with the mixed evidence reported in previous studies (Meng et al., 2023).

Comparing the holistic and disaggregated results, governance disclosure clearly emerges as the primary driver of financial performance. Social and environmental disclosures demonstrate limited or inconsistent effects, reinforcing the strategic importance of prioritizing governance in ESG communication.

In summary, while holistic sustainability disclosure enhances short-term profitability and shareholder returns, it may also contribute to reduced market valuation in the long run—highlighting a potential trade-off between operational efficiency and investor perception.

3.5.3 Curvilinear Effect of ESG on Financial Metrics

Table 16 present the curvilinear effects of ESG disclosure and ESG performance on financial outcomes. Overall, the results confirm that the relationship between sustainability engagement and financial performance is not linear. Rather, both under- and over-engagement in ESG tend to produce suboptimal outcomes, supporting the Too-Much-of-a-Good-Thing (TMGT) hypothesis (Zhang et al., 2022).

For ESG disclosure, an inverted U-shaped pattern emerges across TSR, ROA, and ROE (Models 3.1, 3.3, and 3.4). Figure 19 suggests that moderate transparency enhances both market-based and accounting-based performance. However, excessive disclosure appears counterproductive, potentially due to increased complexity, stakeholder fatigue, or perceptions of greenwashing. These results align with theoretical expectations from signaling and legitimacy theory, in which disclosure adds value only when it is perceived as credible and consistent with underlying performance (Mahoney et al., 2013; Giese et al., 2019).

In contrast, Tobin's Q displays a U-shaped pattern (Model 3.2), revealing a unique sensitivity to ESG disclosure. Firms with moderate levels of transparency are penalized by the market, while those with either low or high disclosure are rewarded. This "stuck in the middle" effect suggests that partial disclosure may be interpreted as symbolic or insincere—costly enough to raise concerns but insufficient to establish trust (Delmas & Burbano, 2011; Krüger,

Table 16 - Curvilinear Effect of ESG Disclosure and ESG Performance on Financial Performance (Model 3.1 to 3.8).

Variables	TSR (3.1)	TQ (3.2)	ROA (3.3)	ROE (3.4)	TSR (3.5)	TQ (3.6)	ROA (3.7)	ROE (3.8)
zw_ESG_Dis	**0.225*** **	**-0.0384*** **	**0.137** **	**0.216*** **				
	(0.0653)	**(0.0141)**	**(0.0593)**	**(0.0661)**				
zw_ESG_Dis²	**-0.0972*** **	**0.0245** **	**-0.0509***	**-0.101*** **				
	(0.0293)	**(0.0109)**	**(0.0266)**	**(0.0302)**				
zw_ESG					**0.0641***	**-0.0683*** **	-0.00881	0.0436
					(0.0346)	**(0.0164)**	(0.0326)	(0.0352)
zw_ESG²					**-0.0385***	**0.0442*** **	0.0110	-0.0307
					(0.0218)	**(0.0111)**	(0.0212)	(0.0230)
zw_Sales	0.0746**	-0.0133	0.00142	0.0268	0.0717***	-0.0154	0.000908	0.0333
	(0.0290)	(0.0119)	(0.0230)	(0.0241)	(0.0259)	(0.0121)	(0.0199)	(0.0208)
zw_MKT_Cap	-0.145***	0.0991***	-0.0154	-0.116***	-0.101***	0.0989***	0.0165	-0.0802**
	(0.0387)	(0.0320)	(0.0332)	(0.0354)	(0.0345)	(0.0316)	(0.0307)	(0.0318)
zw_MTB	0.355***	0.786***	0.173***	0.334***	0.313***	0.790***	0.146***	0.300***
	(0.0412)	(0.0894)	(0.0365)	(0.0395)	(0.0371)	(0.0888)	(0.0328)	(0.0362)
zw_FDI	-0.0449	0.0733***	0.0404	-0.0314	-0.0578**	0.0649***	0.0107	-0.0561*
	(0.0358)	(0.0120)	(0.0301)	(0.0354)	(0.0278)	(0.0117)	(0.0248)	(0.0287)
L.zw_TSR	0.521***				0.554***			
	(0.0403)				(0.0357)			
L.zw_Tobins_Q		0.446***				0.444***		
		(0.0404)				(0.0402)		
L.zw_ROA			0.654***				0.685***	
			(0.0306)				(0.0276)	
L.zw_ROE				0.517***				0.536***
				(0.0394)				(0.0362)
Constant	-0.0235	0.206***	0.0885	-0.0405	-0.0400	0.177***	0.0243	-0.0587
	(0.0702)	(0.0510)	(0.0662)	(0.0709)	(0.0597)	(0.0516)	(0.0632)	(0.0650)
Year Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-Stage F-stat (zw_ESG_Dis)	295.87***	1122.72***	298.28***	295.82***				
First-Stage F-stat (zw_ESG)					657.26***	873.39***	651.95***	656.70***
Kleibergen-Paap LM Test (Underidentification)	328.20***	510.50***	334.73***	329.07***	565.18***	551.57***	576.87***	577.08***
Weak Identification Test (Stock-Yogo)	606.27***	2209.62***	597.97***	605.27***	1400.27***	1673.97***	1374.79***	1375.46***
Hansen J-test (Overidentification)	1.06*	0.19	0.08	0.42	1.59*	0.42	1.61*	1.61*
Anderson-Rubin Wald Test	7.03**	3.65*	2.85*	5.98**	2.20*	9.19**	0.90	0.91
Observations	1,854	2,038	1,854	1,854	2,231	2,038	2,231	2,231
R-squared	0.384	0.859	0.492	0.370	0.424	0.860	0.528	0.396

Note: The variables are TSR = total shareholder return, TQ = Tobin's Q, ROA = return on assets, ROE = return on equity, FDI = foreign direct investment, MKT_Cap = market capitalization, Sales = annual sales, MTB = market-to-book ratio, and ESG_Dis = ESG disclosure score. The squared terms ESG_Dis² and ESG² capture curvilinear effects of ESG disclosure and ESG performance on financial performance. This table presents the 2SLS regression results, controlling for potential endogeneity. Year and industry fixed effects are included, and lagged dependent variables account for performance persistence. Instrument validity is confirmed by the first-stage F-statistics, Kleibergen-Paap LM test, and Hansen J-test. Bold numbers indicate statistical significance. *p < 0.10; **p < 0.05; ***p < 0.01. Data source: Bloomberg.

2015). Only firms that fully embrace transparency or clearly opt out of ESG signaling appear to receive positive valuation effects.

ESG performance exhibits similarly non-linear dynamics. TSR and ROE follow inverted U-shaped curves (Models 3.5 and 3.8), consistent with the idea that moderate ESG implementation enhances profitability and investor returns, but overcommitment may reduce efficiency or dilute shareholder value. ROA and Tobin's Q, however, follow U-shaped patterns (Models 3.7 and 3.6), where firms with moderate ESG efforts are again penalized. These results indicate that the operational burden of ESG implementation may outweigh its benefits in the short term, especially when engagement is incomplete or perceived as symbolic.

A particularly noteworthy finding is the divergence between TSR and ROA in response to ESG performance. While TSR increases at moderate ESG levels, ROA declines—suggesting that investor expectations are rising in the absence of actual operational improvement. This mismatch may be indicative of reputational signaling without substantive efficiency gains, a pattern often associated with symbolic sustainability or greenwashing (Hummel & Schlick, 2016; Walker & Wan, 2012). It highlights the importance of analyzing market-based and accounting-based metrics jointly, as overreliance on TSR may obscure inefficiencies or inflated expectations.

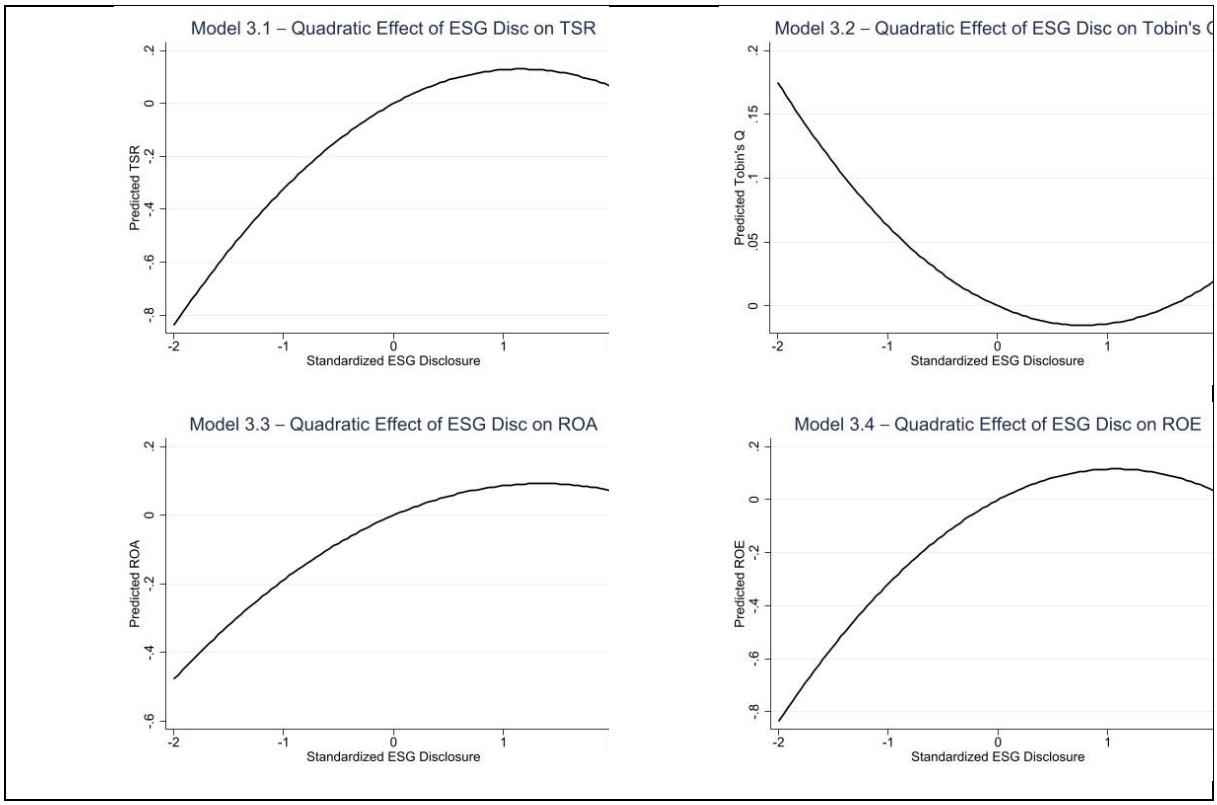
Taken together, the evidence suggests that the optimal level of ESG engagement is metric-dependent. TSR, ROA, and ROE show clear peaks at moderate ESG levels, while Tobin's Q penalizes firms in the middle of the ESG spectrum. This implies that different stakeholders (investors, analysts, operational managers) interpret ESG signals through distinct lenses, leading to heterogeneous reactions across financial indicators. The figures reinforce this complexity by visually confirming that the financial value of ESG is contingent not only on the level of engagement, but also on the balance between narrative (disclosure) and substance (performance).

From a theoretical perspective, these findings extend the TMGT framework by demonstrating that the interaction between signaling and operational credibility is central to determining financial outcomes. ESG adds value when it reaches a strategic "sweet spot"—where performance and disclosure are aligned, credible, and sufficient to generate stakeholder trust without incurring excessive cost. Firms that fall short or go too far in either direction may compromise profitability, damage credibility, or invite skepticism from the market.

These patterns provide important implications for managers and policymakers. Rather than maximizing ESG disclosure or performance indiscriminately, firms should seek strategic calibration—balancing reputational goals with operational capacity. In doing so, they can avoid

the risks of overexposure or symbolic compliance and position themselves to capture the long-term value of sustainability engagement.

Figure 19 - Curvilinear Relationship between ESG Disclosure and Financial Performance (Models 3.1–3.4)



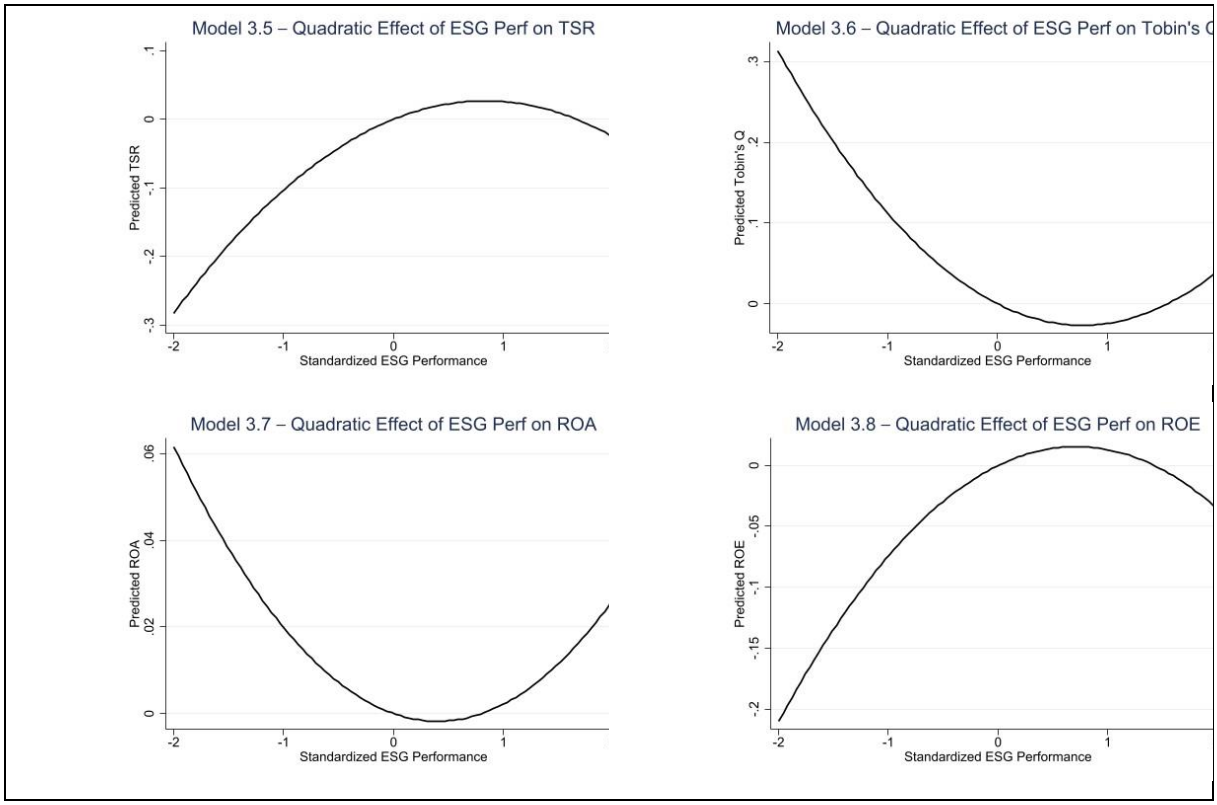
Note. This figure depicts the predicted curvilinear effects of standardized ESG disclosure on four financial performance metrics: (3.1) Total Shareholder Return (TSR), (3.2) Tobin's Q, (3.3) Return on Assets (ROA), and (3.4) Return on Equity (ROE). Predictions are based on the estimated coefficients from the 2SLS regressions reported in Table 3, using the full specification with controls. The x-axis represents the standardized ESG disclosure score, while the y-axis displays the predicted value of each performance metric.

Figure 20 exhibits a more complex curve for ESG performance. Initial positive effects on TSR reverse at moderate ESG levels and recover only at extreme engagement, indicating a non-linear trade-off between sustainability investment and short-term profitability (Yu et al., 2024; Bagh et al., 2024). Tobin's Q also follows this pattern, initially declining but rising at higher ESG levels, suggesting delayed investor recognition of long-term sustainability value (Lopez et al., 2007; Eccles et al., 2014).

Divergence between TSR and accounting metrics (ROA, ROE) suggests that investor optimism may outpace actual profitability, potentially due to greenwashing or inflated expectations (Krüger, 2015; Schaltegger & Burritt, 2010). In such cases, firms may benefit

reputationally from ESG signaling despite limited operational impact (Hummel & Schlick, 2016; Li et al., 2018).

Figure 20 - Curvilinear Relationship between ESG Performance and Financial Performance (Models 3.5–3.8)



Note. This figure illustrates the curvilinear effects of standardized ESG performance on the same set of financial performance indicators as in Figure 2. Based on Models 3.5 to 3.8, the predicted values are computed using the estimated coefficients from the full regressions.

While ESG can create long-term value, its financial benefits are not linear. Moderate engagement appears optimal; both under- and over-investment in ESG reduce efficiency or market credibility. These results support a strategic balance between ESG investment and disclosure to align financial performance with stakeholder trust (Lee & Li, 2022; Fatemi et al., 2018).

3.5.4 Selective ESG Dimensions and Financial Performance

Tables 17 and 18 present the 2SLS regression results assessing the financial consequences of selectively combining ESG dimensions in both disclosure and performance. The analysis provides robust evidence that fragmented ESG strategies—where firms emphasize

certain dimensions (e.g., environmental or social) while omitting others (e.g., governance)—tend to undermine financial outcomes.

Selective ESG disclosure exhibits consistently negative interaction effects on profitability and shareholder value. Notably, the interaction between environmental and social disclosure (ESG_E_Dis × ESG_S_Dis) significantly reduces both ROA ($\beta = -0.114$, $p < 0.01$) and ROE ($\beta = -0.165$, $p < 0.01$), while also harming TSR ($\beta = -0.161$, $p < 0.01$). Similar patterns are observed for ESG_E_Dis × ESG_G_Dis, which depresses ROA ($\beta = -0.106$, $p < 0.01$) and ROE ($\beta = -0.183$, $p < 0.01$), and ESG_S_Dis × ESG_G_Dis, which negatively affects ROA ($\beta = -0.0956$, $p < 0.01$) and ROE ($\beta = -0.155$, $p < 0.01$). These results support the notion that inconsistent or symbolic disclosure practices—where firms selectively report favorable information—erode stakeholder trust and reduce profitability, aligning with the theoretical arguments of Lueg and Pesheva (2021) on symbolic versus substantive ESG communication.

Interestingly, no statistically significant effect of selective disclosure interactions is observed on Tobin's Q, suggesting that market valuation remains unaffected in the short term by ESG disclosure fragmentation. This may indicate that equity markets either discount ESG narratives lacking performance backing or are slower to penalize superficial disclosure strategies.

Selective ESG performance interactions mirror the disclosure patterns. The ESG_E × ESG_S interaction significantly reduces ROA ($\beta = -0.140$, $p < 0.01$) and ROE ($\beta = -0.149$, $p < 0.01$), while also decreasing TSR ($\beta = -0.059$, $p < 0.01$). Similarly, ESG_E × ESG_G results in lower ROA ($\beta = -0.162$, $p < 0.01$) and ROE ($\beta = -0.169$, $p < 0.01$), and ESG_S × ESG_G leads to reduced ROA ($\beta = -0.107$, $p < 0.01$) and ROE ($\beta = -0.145$, $p < 0.01$). These findings confirm that disjointed ESG implementation—where companies invest in selected areas while neglecting others—can backfire, signaling inefficiencies, misalignment, or internal contradictions that compromise firm performance.

As with disclosure, Tobin's Q remains statistically unaffected by selective performance combinations, reinforcing that market-based measures may be less sensitive to ESG fragmentation in the short term. The consistency of negative results in TSR, ROA, and ROE, however, underscores that both capital returns and profitability are penalized when ESG efforts lack integration.

These results have both theoretical and practical implications. From a stakeholder theory perspective, they highlight that partial commitment to sustainability—focusing only on visible or less costly areas—may violate stakeholder expectations and damage corporate legitimacy

Table 17 - Impact of Selective Environmental, Social and Governance Disclosure on Financial Performance - 2SLS Regression Results. (Model 4.1 to 4.12).

Variables	TSR (4.1)	TQ (4.2)	ROA (4.3)	ROE (4.4)	TSR (4.5)	TQ (4.6)	ROA (4.7)	ROE (4.8)	TSR (4.9)	TQ (4.10)	ROA (4.11)	ROE (4.12)
zw_ESG_E_Dis	-0.0480	**0.0710****	0.0306	-0.0266	-0.0296	**0.0662****	**0.101****	-0.00686	0.0273	**0.0624****	**0.134*****	0.0517
	(0.0628)	**(0.0306)**	(0.0572)	(0.0660)	(0.0618)	**(0.0304)**	**(0.0473)**	(0.0643)	(0.0643)	**(0.0314)**	**(0.0504)**	(0.0670)
zw_ESG_S_Dis	**0.274*****	**-0.140*****	0.0830	**0.232*****	**0.250*****	**-0.129*****	-0.0295	**0.209*****	**0.183*****	**-0.120*****	-0.0668	**0.138***
	(0.0805)	**(0.0324)**	(0.0709)	**(0.0812)**	**(0.0765)**	**(0.0306)**	(0.0466)	**(0.0766)**	**(0.0709)**	**(0.0287)**	(0.0456)	**(0.0721)**
zw_ESG_G_Dis	0.0944	**0.0410*****	**0.118****	**0.102***	**0.114***	**0.0378****	**0.134*****	**0.121****	**0.143****	**0.0374****	**0.153*****	**0.151****
	(0.0581)	**(0.0154)**	**(0.0488)**	**(0.0583)**	**(0.0608)**	**(0.0162)**	**(0.0511)**	**(0.0609)**	**(0.0672)**	**(0.0182)**	**(0.0568)**	**(0.0670)**
zw_ESG_E_Dis_zw_ESG_S_Dis	**-0.161*****	**0.0317****	**-0.114*****	**-0.165*****								
	(0.0374)	**(0.0160)**	**(0.0315)**	**(0.0368)**								
zw_ESG_E_Dis_zw_ESG_G_Dis					**-0.175*****	0.0204	**-0.106*****	**-0.183*****				
					(0.0409)	(0.0159)	**(0.0306)**	**(0.0406)**				
zw_ESG_S_Dis_zw_ESG_G_Dis									**-0.152*****	0.00609	**-0.0956*****	**-0.155*****
									(0.0408)	(0.0154)	**(0.0329)**	**(0.0403)**
zw_Sales	0.0625**	-0.0209	-0.0180	0.0111	0.0608**	-0.0215	-0.0222	0.00894	0.0717**	-0.0229*	-0.0159	0.0207
	(0.0308)	(0.0136)	(0.0235)	(0.0245)	(0.0309)	(0.0137)	(0.0234)	(0.0245)	(0.0311)	(0.0137)	(0.0242)	(0.0254)
zw_MKT_Cap	-0.153***	0.0917***	-0.0238	-0.122***	-0.145***	0.0888***	-0.0111	-0.113***	-0.157***	0.0882**	-0.0179	-0.126***
	(0.0376)	(0.0342)	(0.0330)	(0.0340)	(0.0373)	(0.0342)	(0.0326)	(0.0333)	(0.0375)	(0.0344)	(0.0331)	(0.0339)
zw_MTB	0.352***	0.749***	0.180***	0.332***	0.358***	0.749***	0.184***	0.339***	0.360***	0.750***	0.187***	0.340***
	(0.0408)	(0.0951)	(0.0368)	(0.0391)	(0.0417)	(0.0948)	(0.0368)	(0.0401)	(0.0415)	(0.0946)	(0.0370)	(0.0399)
zw_FDI	-0.00424	0.0456**	0.0310	-0.00259	3.38e-05	0.0473***	0.00386	0.00232	-0.0181	0.0507***	-0.00576	-0.0175
	(0.0409)	(0.0181)	(0.0358)	(0.0410)	(0.0408)	(0.0179)	(0.0327)	(0.0407)	(0.0405)	(0.0177)	(0.0333)	(0.0405)
L.zw_TSR	0.519***				0.525***				0.524***			
	(0.0409)				(0.0405)				(0.0408)			
L.zw_Tobins_Q		0.472***				0.472***				0.471***		
		(0.0462)				(0.0461)				(0.0460)		
L.zw_ROA			0.652***				0.648***				0.648***	
			(0.0305)				(0.0307)				(0.0309)	
L.zw_ROE				0.519***				0.524***				0.522***
				(0.0398)				(0.0393)				(0.0396)
Constant	-0.00564	0.305***	0.113*	-0.0238	-0.0368	0.315***	0.0824	-0.0547	-0.0356	0.319***	0.0849	-0.0552
	(0.0707)	(0.0506)	(0.0660)	(0.0707)	(0.0680)	(0.0501)	(0.0636)	(0.0684)	(0.0687)	(0.0501)	(0.0644)	(0.0691)
Year Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-Stage F-stat (ESG_E_Disc)	157.33***	234.85***	157.05***	157.47***	170.62***	246.37***	166.99***	170.80***	177.49***	248.14***	171.11***	177.75***
First-Stage F-stat (ESG_S_Disc)	139.73***	197.64***	140.31***	139.94***	152.59***	209.36***	186.68***	153.16***	160.06***	218.55***	211.08***	161.33***
First-Stage F-stat (ESG_G_Disc)	132.25***	169.48***	132.33***	131.75***	117.99***	145.18***	116.04***	117.60***	97.83***	118.15***	97.84***	97.50***
Kleibergen-Paap LM Test (Underident)	219.90***	196.26***	222.70***	219.68***	200.04***	173.02***	372.00***	200.42***	134.17***	141.28***	378.79***	133.67***
Weak Ident. Test (Stock-Yogo)	63.32***	52.76***	64.63***	63.28***	59.74***	48.30***	107.09***	59.88***	40.45***	39.24***	91.31***	40.46***
Hansen J-test (Overidentification)	2.557	2.257	6.117	3.213	2.427	2.732	5.235	3.259	2.494	3.233	5.272	3.617
Anderson-Rubin Wald Test	3.92***	4.92***	2.93**	3.64***	3.67***	4.64***	2.91***	3.47***	3.49***	4.34***	3.05**	3.35**
Observations	1,854	1,669	1,854	1,854	1,854	1,669	1,854	1,854	1,854	1,669	1,854	1,854
R-squared	0.370	0.860	0.486	0.359	0.371	0.860	0.490	0.361	0.369	0.860	0.486	0.358

Note: The variables are TSR = total shareholder return, TQ = Tobin's Q, ROA = return on assets, ROE = return on equity, FDI = foreign direct investment, MKT_Cap = market capitalization, Sales = annual sales, MTB = market-to-book ratio, and ESG_E_Dis, ESG_S_Dis, ESG_G_Dis = environmental, social, and governance disclosure scores, respectively. Interaction terms (ESG_E_Dis × ESG_S_Dis, ESG_E_Dis × ESG_G_Dis, ESG_S_Dis × ESG_G_Dis) assess the combined effect of ESG components. This table presents the 2SLS regression results, addressing endogeneity concerns. Year and industry fixed effects are included, and lagged dependent variables account for performance persistence. Instrument validity is confirmed by the first-stage F-statistics, Kleibergen-Paap LM test, and Hansen J-test. Bold numbers indicate statistical significance. *p < 0.10; **p < 0.05; ***p < 0.01. Data source: Bloomberg.

Table 18 - Impact of Selective Environmental, Social and Governance Performance on Financial Performance - 2SLS Regression Results. (Model 5.1 to 5.12).

Variables	TSR (5.1)	TQ (5.2)	ROA (5.3)	ROE (5.4)	TSR (5.5)	TQ (5.6)	ROA (5.7)	ROE (5.8)	TSR (5.9)	TQ (5.10)	ROA (5.11)	ROE (5.12)
zw_ESG_E	**0.0810****	0.0285	**0.184***	**0.234****	**0.0800****	0.0303	**0.184***	**0.257****	**0.0807****	0.0289	**0.189***	**0.249****
	(0.0354)	(0.0192)	**(0.0957)**	**(0.112)**	**(0.0365)**	(0.0193)	**(0.0984)**	**(0.121)**	**(0.0350)**	(0.0191)	**(0.0998)**	**(0.113)**
zw_ESG_S	0.0191	-0.0202	-0.0553	0.0209	0.0199	-0.0209	-0.0220	0.00866	0.0352	-0.0204	-0.0216	0.0360
	(0.0288)	(0.0144)	(0.0474)	(0.0475)	(0.0287)	(0.0143)	(0.0537)	(0.0492)	(0.0301)	(0.0153)	(0.0471)	(0.0494)
zw_ESG_G	0.0297	**0.0227****	**0.103****	0.000501	0.0386	**0.0266****	**0.134*****	0.0371	0.0516	**0.0228***	**0.0827***	0.0511
	(0.0342)	**(0.0111)**	**(0.0479)**	(0.0479)	(0.0340)	**(0.0112)**	**(0.0519)**	(0.0477)	(0.0377)	**(0.0118)**	**(0.0459)**	(0.0564)
zw_ESG_E_zw_ESG_S	**-0.059*****	-0.000281	**-0.140*****	**-0.149*****								
	(0.0181)	(0.0105)	**(0.0369)**	**(0.0397)**								
zw_ESG_E_zw_ESG_G					**-0.061*****	-0.0169	**-0.162*****	**-0.169*****				
					(0.0212)	(0.0108)	**(0.0445)**	**(0.0483)**				
zw_ESG_S_zw_ESG_G									**-0.072*****	0.00111	**-0.107*****	**-0.145*****
									(0.0241)	(0.0115)	**(0.0403)**	**(0.0460)**
zw_Sales	0.0722***	-0.0138	0.0126	0.0325	0.0701***	-0.0148	0.00940	0.0269	0.0759***	-0.0138	0.0170	0.0427*
	(0.0258)	(0.0122)	(0.0217)	(0.0229)	(0.0260)	(0.0124)	(0.0217)	(0.0232)	(0.0260)	(0.0122)	(0.0236)	(0.0242)
zw_MKT_Cap	-0.108***	0.0900***	-0.0225	-0.118***	-0.102***	0.0891***	-0.0138	-0.107***	-0.115***	0.0903***	-0.0368	-0.139***
	(0.0340)	(0.0313)	(0.0377)	(0.0423)	(0.0338)	(0.0313)	(0.0373)	(0.0410)	(0.0341)	(0.0316)	(0.0405)	(0.0456)
zw_MTB	0.309***	0.797***	0.199***	0.344***	0.309***	0.798***	0.201***	0.349***	0.308***	0.797***	0.184***	0.328***
	(0.0368)	(0.0899)	(0.0390)	(0.0441)	(0.0371)	(0.0897)	(0.0392)	(0.0453)	(0.0368)	(0.0902)	(0.0382)	(0.0436)
zw_FDI	-0.0471*	0.0872***	0.0575*	-0.0412	-0.0495*	0.0876***	0.0553	-0.0455	-0.0525*	0.0874***	0.0487	-0.0431
	(0.0282)	(0.0129)	(0.0337)	(0.0393)	(0.0277)	(0.0130)	(0.0344)	(0.0389)	(0.0278)	(0.0129)	(0.0336)	(0.0401)
L.zw_TSR	0.554***				0.554***				0.554***			
	(0.0360)				(0.0360)				(0.0361)			
L.zw_Tobins_Q		0.446***				0.445***				0.446***		
		(0.0406)				(0.0405)				(0.0407)		
L.zw_ROA			0.630***				0.628***				0.635***	
			(0.0362)				(0.0363)				(0.0367)	
L.zw_ROE				0.516***				0.517***				0.519***
				(0.0456)				(0.0453)				(0.0459)
Constant	-0.0362	0.237***	0.0761	-0.118	-0.0471	0.240***	0.0698	-0.137*	-0.0317	0.237***	0.0616	-0.124
	(0.0573)	(0.0511)	(0.0775)	(0.0786)	(0.0566)	(0.0513)	(0.0777)	(0.0778)	(0.0574)	(0.0517)	(0.0769)	(0.0796)
Year Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	-0.0362	0.237***	0.0740	-0.118	-0.0471	0.240***	0.0562	-0.137*	-0.0317	0.237***	0.0616	-0.124
	(0.0573)	(0.0511)	(0.0765)	(0.0786)	(0.0566)	(0.0513)	(0.0758)	(0.0778)	(0.0574)	(0.0517)	(0.0769)	(0.0796)
First-Stage F-stat (ESG_E)	142.86***	143.99***	36.27***	36.20***	131.25***	132.95***	25.68***	33.22***	139.37***	139.21***	27.06***	27.17***
First-Stage F-stat (ESG_S)	319.13***	432.28***	177.85***	177.44***	288.51***	383.73***	83.83***	172.32***	215.90***	273.69***	126.19***	126.88***
First-Stage F-stat (ESG_G)	319.99***	558.69***	133.49***	177.52***	241.51***	415.11***	92.69***	132.07***	173.18***	249.91***	97.07***	98.03***
Kleibergen-Paap LM Test (Underident)	373.82***	309.02***	111.83***	111.22***	412.57***	306.75***	106.69***	99.83***	435.95***	316.61***	103.87***	103.35***
Weak Ident. Test (Stock-Yogo)	134.75***	383.96***	42.39***	41.69***	365.91***	382.76***	17.24***	15.63***	168.37***	113.01***	16.88***	16.72***
Hansen J-test (Overidentification)	5.645	2.28	3.44	5.01	6.24	2.92	3.47	5.70	4.31	2.31	5.529	3.878
Anderson-Rubin Wald Test	2.40**	1.98**	1.75	2.33**	2.43**	2.30**	2.19**	2.46**	2.34**	1.93*	1.72	2.19**
Observations	2,231	2,038	1,477	1,477	2,231	2,038	1,477	1,477	2,231	2,038	1,477	1,477
R-squared	0.419	0.858	0.463	0.375	0.419	0.858	0.461	0.375	0.417	0.858	0.457	0.362

Note: The variables are TSR = total shareholder return, TQ = Tobin's Q, ROA = return on assets, ROE = return on equity, FDI = foreign direct investment, MKT_Cap = market capitalization, Sales = annual sales, MTB = market-to-book ratio, and ESG_E, ESG_S, ESG_G = environmental, social, and governance performance scores, respectively. Interaction terms (ESG_E × ESG_S, ESG_E × ESG_G, ESG_S × ESG_G) assess the combined effect of ESG components. This table presents the 2SLS regression results, controlling for potential endogeneity. Year and industry fixed effects are included, and lagged dependent variables account for performance persistence. Instrument validity is confirmed by the first-stage F-statistics, Kleibergen-Paap LM test, and Hansen J-test. Bold numbers indicate statistical significance. *p < 0.10; **p < 0.05; ***p < 0.01. Data source: Bloomberg.

(Eccles et al., 2014). From a strategic viewpoint, the findings suggest that ESG components operate synergistically: isolating one dimension can negate the potential benefits of others.

Finally, the divergence in sensitivity across financial metrics warrants attention. While ROA and ROE respond strongly to selective ESG actions, Tobin's Q appears insensitive, and TSR responds only to performance fragmentation. This variation implies that market-based and accounting-based metrics capture different ESG effects—short-term perceptions versus internal efficiency—offering a nuanced view of how ESG coherence (or lack thereof) translates into financial consequences.

3.5.5 Interaction Between ESG Disclosure and Performance

Table 19 explores the joint effects of ESG performance and disclosure on financial outcomes. Results show that misalignment between disclosure and performance undermines financial performance. Specifically, the interaction term (zw_ESG × zw_ESG_Dis) is negative and significant for TSR ($\beta = -0.0459$, $p < 0.05$) and ROE ($\beta = -0.0494$, $p < 0.05$), suggesting that firms with high disclosure but weak ESG performance risk investor backlash and lower profitability—an effect consistent with greenwashing dynamics (Kathan et al., 2025; Yu et al., 2020). Conversely, the positive interaction with Tobin's Q ($\beta = 0.0225$, $p < 0.10$) indicates that strategically communicating ESG enhances market valuation (Banke et al., 2022), even when profitability gains are modest.

The graphical evidence—particularly from Figure 21—reveals a compelling pattern that advances the understanding of how ESG disclosure and performance jointly affect financial outcomes. Most notably, moderate levels of both ESG disclosure and performance correspond to a financial "gray zone". In this zone, firms that adopt a lukewarm ESG engagement—neither negligent nor intensive—suffer the greatest value erosion across all performance metrics, including ROA, ROE, TSR, and Tobin's Q. This pattern aligns with prior studies that emphasize the risks of symbolic ESG adoption or weak integration of sustainability into core operations (Delmas & Burbano, 2011; Margolis & Walsh, 2003). Such firms appear to incur ESG-related costs without generating operational or reputational benefits, leading to a breakdown in efficiency and a loss of market trust (Orlitzky et al., 2003).

This phenomenon is particularly severe in accounting-based indicators. ROA and ROE show significant declines in the gray zone, suggesting that ESG investments—when not strategically implemented—fail to produce internal productivity or profitability gains (Fatemi,

Table 19 - Interact Effect (Linear and Curvilinear) of ESG Disclosure and ESG Performance on Financial Performance (Model 6.1 to 6.8).

Variables	TSR (6.1)	TQ (6.2)	ROA (6.3)	ROE (6.4)	TSR (6.5)	TQ (6.6)	ROA (6.7)	ROE (6.8)
zw_ESG	-0.0103	**-0.0316****	-0.0315	-0.0210	-0.0105	**-0.0803*****	**-0.0869***	-0.0377
	(0.0339)	**(0.0153)**	(0.0269)	(0.0302)	(0.0540)	**(0.0284)**	**(0.0460)**	(0.0528)
zw_ESG_Dis	**0.107****	-0.0193	**0.0699***	**0.103****	**0.141****	0.0163	**0.133****	**0.151****
	(0.0489)	(0.0189)	**(0.0417)**	**(0.0480)**	**(0.0615)**	(0.0269)	**(0.0521)**	**(0.0609)**
zw_ESG_Dis_zw_ESG	**-0.0459****	**0.0225***	-0.0213	**-0.0494****	-0.0133	-0.0206	-0.0475	-0.0181
	(0.0204)	**(0.0115)**	(0.0191)	**(0.0213)**	(0.0308)	(0.0169)	(0.0296)	(0.0320)
zw_ESG²					0.0535	**0.0486****	**0.0937*****	**0.0770****
					(0.0369)	**(0.0200)**	**(0.0313)**	**(0.0353)**
zw_ESG_Dis²					**0.116*****	0.0214	**0.111*****	**0.115****
					(0.0436)	(0.0174)	**(0.0416)**	**(0.0465)**
zw_ESG²_zw_ESG_Dis²					**-0.0629*****	-0.00123	**-0.0480*****	**-0.0678*****
					(0.0172)	(0.00749)	**(0.0155)**	**(0.0177)**
zw_Sales	0.0689***	-0.0155	-0.00269	0.0295	0.0670**	-0.0155	-0.00367	0.0274
	(0.0265)	(0.0120)	(0.0206)	(0.0215)	(0.0265)	(0.0120)	(0.0205)	(0.0213)
zw_MKT_Cap	-0.117***	0.0953***	0.00292	-0.0948***	-0.116***	0.0957***	0.00249	-0.0925***
	(0.0336)	(0.0320)	(0.0300)	(0.0315)	(0.0338)	(0.0322)	(0.0299)	(0.0313)
zw_MTB	0.324***	0.789***	0.158***	0.314***	0.343***	0.792***	0.172***	0.336***
	(0.0362)	(0.0890)	(0.0329)	(0.0356)	(0.0368)	(0.0888)	(0.0335)	(0.0362)
zw_FDI	-0.0487*	0.0644***	0.0193	-0.0455	-0.0448	0.0677***	0.0254	-0.0413
	(0.0288)	(0.0117)	(0.0257)	(0.0297)	(0.0287)	(0.0120)	(0.0255)	(0.0295)
L.zw_TSR	0.553***				0.550***			
	(0.0357)				(0.0357)			
L.zw_Tobins_Q		0.444***				0.445***		
		(0.0403)				(0.0403)		
L.zw_ROA			0.688***				0.682***	
			(0.0275)				(0.0276)	
L.zw_ROE				0.536***				0.532***
				(0.0361)				(0.0360)
Constant	-0.0442	0.204***	0.0521	-0.0521	-0.130**	0.169***	-0.0505	-0.153**
	(0.0581)	(0.0507)	(0.0614)	(0.0623)	(0.0647)	(0.0531)	(0.0671)	(0.0697)
Year Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-Stage F-stat (zw_ESG_Dis)	511.46***	806.68***	510.52***	510.28***	432.48***	674.29***	433.27***	431.42***
First-Stage F-stat (zw_ESG)	1343.51***	1906.93***	1341.46***	1343.45***	718.32***	967.11***	719.53***	718.71***
Kleibergen-Paap LM Test (Underidentification)	594.49***	410.86***	601.81***	594.49***	555.42***	402.52***	563.41***	556.30***
Weak Identification Test (Stock-Yogo)	410.39***	493.09***	410.92***	407.94***	398.08***	534.62***	398.85***	395.05***
Hansen J-test (Overidentification)	1.506	0.434	1.699	2.102	1.268	0.280	1.598	1.795
Anderson-Rubin Wald Test	2.56**	4.91***	1.48	2.23	2.99**	5.34***	1.31	2.51
Observations	2,231	2,038	2,231	2,231	2,231	2,038	2,231	2,231
R-squared	0.419	0.859	0.525	0.392	0.420	0.859	0.525	0.393

Note: The variables are TSR = total shareholder return, TQ = Tobin's Q, ROA = return on assets, ROE = return on equity, FDI = foreign direct investment, MKT_Cap = market capitalization, Sales = annual sales, MTB = market-to-book ratio, and ESG = overall ESG performance. ESG_Dis represents ESG disclosure, while interaction terms (ESG × ESG_Dis, ESG², ESG_Dis², ESG² × ESG_Dis²) capture non-linear and interactive effects between ESG performance and disclosure on financial outcomes. This table presents the 2SLS regression results, controlling for endogeneity. Year and industry fixed effects are included, and lagged dependent variables account for performance persistence. Instrument validity is confirmed by the first-stage F-statistics, Kleibergen-Paap LM test, and Hansen J-test. Bold numbers indicate statistical significance. *p < 0.10; **p < 0.05; ***p < 0.01. Data source: Bloomberg.

Glaum, & Kaiser, 2018). These outcomes reinforce the notion that ESG requires focused execution to convert reputational capital into operational efficiency (Cheng et al., 2014).

Interestingly, even firms with modest ESG performance can improve their long-term valuation—as captured by Tobin's Q—and increasingly improve financial performance at high levels of disclosure. This highlights the market's preference for transparency and perceived commitment, even when operational implementation remains limited (Dhaliwal et al., 2011, 2012). From a signaling perspective, consistent disclosure may serve as a proxy for future strategic alignment and risk management capabilities, increasing perceived firm value in capital markets (Spence, 1973; Connelly et al., 2011).

The data also provide evidence of the Too Much of a Good Thing (TMGT) effect, though in a interact form. When both ESG performance and disclosure are simultaneously high, financial returns do not reverse, but their marginal gains diminish—particularly in Tobin's Q, ROA, and ROE. This saturation pattern is consistent with recent studies that identify non-linear ESG-financial performance relationships, in which additional investments yield decreasing returns beyond a certain point (Barnett et al., 2020).

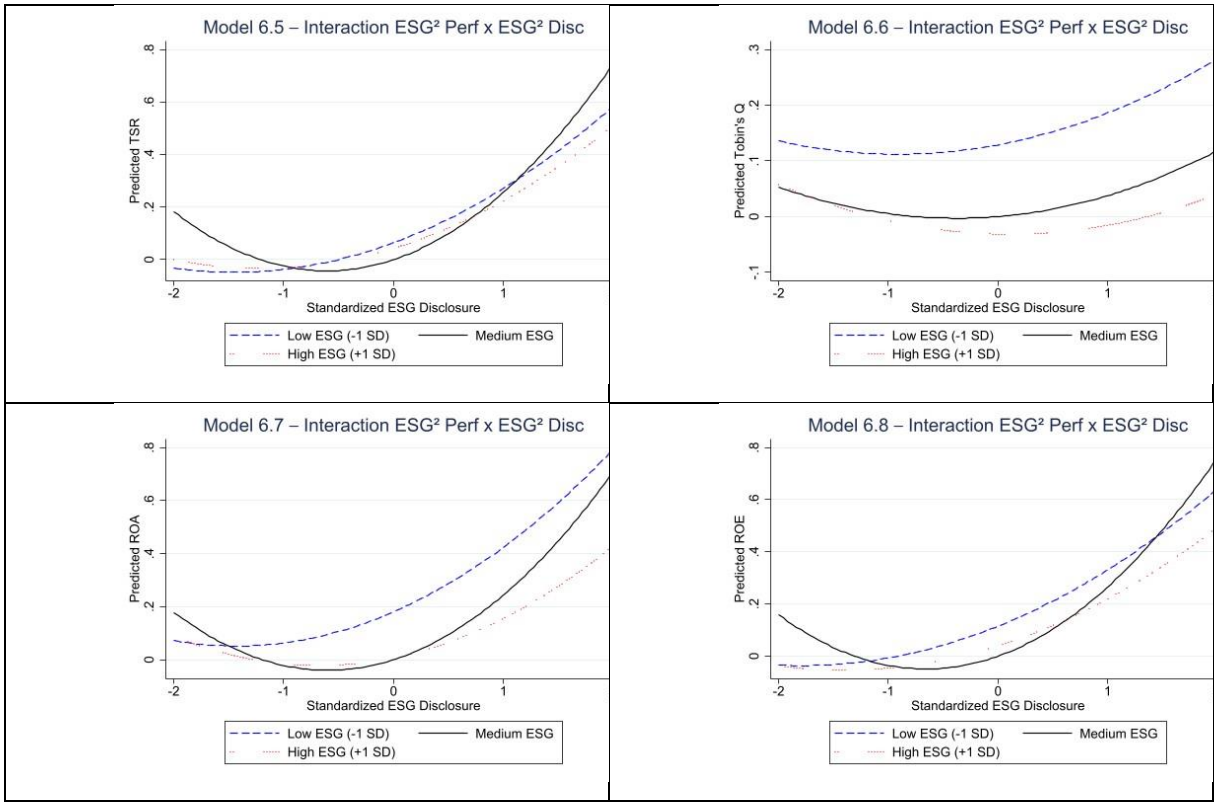
Notably, the best financial outcomes arise when firms combine high levels of disclosure with only moderate ESG performance. This configuration appears to offer the optimal balance: signaling credibility and strategic commitment to investors through transparency, while maintaining operational efficiency by avoiding excessive or misaligned ESG expenditures (Mio et al., 2020). This result is particularly relevant in emerging markets, where institutional voids and stakeholder skepticism may penalize overinvestment or perceived greenwashing (Li et al., 2023).

Finally, the results underscore a systematic divergence between market-based and accounting-based financial indicators. While ROA and ROE are tightly linked to internal productivity and cost control, TSR and Tobin's Q are more responsive to ESG narrative, signaling, and market perception (Fatemi et al., 2018; Ioannou & Serafeim, 2015). This divergence highlights the dual nature of ESG value creation: one rooted in efficiency and performance, the other in communication, reputation, and institutional signaling—particularly relevant in environments marked by regulatory fragility and information asymmetry, such as the BRICS context.

These findings offer strong theoretical implications aligned with signaling and legitimacy theory. The positive market response to high ESG disclosure—even amid moderate performance—highlights the importance of credible narrative as a signal of strategic intent (Spence, 1973). However, when such disclosure is not substantiated by operational ESG

engagement, stakeholders may perceive it as symbolic or disingenuous, reinforcing legitimacy theory's concern with decoupling between talk and action (Meyer & Rowan, 1977; Delmas & Burbano, 2011). The curvilinear joint effects identified also validate the Too Much of a Good Thing (TMGT) perspective, demonstrating that excessive ESG investment and disclosure may yield diminishing financial returns—particularly in accounting-based metrics. In the BRICS context, where institutional voids and stakeholder enforcement are uneven, firms that achieve alignment between ESG narrative and substance—without overextension—are best positioned to balance legitimacy, efficiency, and long-term value creation.

Figure 21 - Curvilinear Interaction between ESG Performance² and ESG Disclosure² (Models 6.5–6.8)



Note. The models include quadratic terms for ESG Performance and ESG Disclosure, as well as their interaction (ESG² × ESG² Disclosure). Estimates are based on the coefficients reported in Table 6, Models 6.5 to 6.8. The x-axis represents standardized ESG disclosure levels, while each curve reflects predicted financial performance at low (–1 SD), medium (mean), and high (+1 SD) levels of ESG performance. Y-axis values represent predicted standardized financial outcomes (TSR, ROA, ROE, and Tobin's Q).

3.6 DISCUSSION AND CONCLUSION

3.6.1 Synthesis of the Findings

The findings reveal no evidence that holistic ESG performance consistently translates into improved financial performance in BRICS markets. No significant relationship was found with TSR, ROA, or ROE, and a negative association emerged with Tobin's Q. This supports the shareholder theory view that broad ESG investment may divert resources from core operations, reducing short-term returns (Barnea & Rubin, 2010; Servaes & Tamayo, 2013). BRICS economies pose unique challenges due to fragmented regulation and governance. ESG strategies may reflect international investor pressures rather than intrinsic corporate values (Amel-Zadeh & Serafeim, 2018; Krueger et al., 2020), weakening the link between ESG and financial outcomes (Liang & Renneboog, 2017).

When disaggregated, environmental performance is positively associated with TSR and ROE—particularly relevant in sectors with sustainability-driven competitive advantages (Busch et al., 2016)—but not with ROA or Tobin's Q, which may reflect cost concerns in high-emission industries (Clarkson et al., 2008). Governance performance improves Tobin's Q, highlighting investor confidence in oversight structures (Gompers et al., 2003; Klapper & Love, 2004), though its impact on profitability is limited (Bhagat & Bolton, 2008). Social performance shows no significant financial effect, underscoring its limited market influence in BRICS settings (Margolis et al., 2009; Visser, 2008).

This finding deserves careful consideration. The absence of significant financial effects for the social pillar (ESG_S) may result from both methodological constraints and contextual limitations specific to emerging markets. From a methodological perspective, S scores tend to aggregate diverse and often loosely connected indicators, such as employee satisfaction, diversity policies, supply chain labor standards, and community engagement. These dimensions are not only difficult to measure consistently across firms and countries but also subject to considerable managerial discretion. As Mattingly (2015) notes in his comprehensive review of empirical studies using KLD data, the operationalization of social performance suffers from definitional ambiguity, inconsistent coding practices, and limited comparability over time, which constrains its reliability and explanatory power in financial models.

From a contextual standpoint, the BRICS countries share structural characteristics that may undermine the materiality of the social pillar. These include weaker legal enforcement, low levels of civic engagement, and institutional voids in areas such as labor protections and stakeholder pressure (Marquis & Qian, 2014; Jamali & Karam, 2018). In such environments,

corporate social initiatives may fail to generate reputational or relational capital capable of enhancing firm value. Furthermore, ESG integration by investors in BRICS remains incipient, with environmental and governance concerns attracting greater attention than social dimensions (Friede et al., 2015). This confluence of weak stakeholder pressure and underdeveloped market incentives helps explain the muted financial relevance of social performance in the sample. Similar findings in other meta-analyses reinforce the view that social ESG components, while normatively important, may lack short-term financial salience in emerging economies (Velte, 2021).

The results reinforce the findings of Lueg and Pesheva (2021), indicating that holistic ESG disclosure exerts a consistent and positive effect on financial metrics such as TSR, ROA, and ROE. Among the individual ESG dimensions, governance disclosure emerges as the most financially beneficial, with significant positive associations across all four performance indicators. This evidence is aligned with legitimacy theory, suggesting that transparent governance practices can compensate for weak institutional environments, enhancing stakeholder trust and reducing information asymmetries (Suchman, 1995; Deegan, 2002).

Social disclosure positively influences TSR and ROE, yet presents a negative association with Tobin's Q, a result that may reflect investor concerns regarding potential inefficiencies or perceptions of greenwashing (Marquis & Qian, 2014; Mahoney et al., 2013). These findings are echoed in studies from Turkey, Jordan, and Southeast Asia, where ESG disclosure improves financial outcomes only when supported by credible performance (Aliyev & Alishov, 2023; Al Amosh & Khatib, 2023b; Syarkani et al., 2024).

In turn, environmental disclosure exhibits a positive relationship with ROA but lacks broader significance across other financial metrics. This pattern supports the view that environmental reporting is often perceived more as a mechanism of regulatory compliance than a driver of sustained value creation (Clarkson et al., 2008; Delmas & Burbano, 2011).

Interaction effects between ESG dimensions (E×S, E×G, S×G) show consistent negative impacts, indicating that selective ESG efforts erode financial performance as found in Lueg and Pesheva, 2021. This contradicts stakeholder theory's assumption of value creation through targeted engagement (Harrison & Wicks, 2013) and instead supports shareholder theory arguments about misallocated resources (Friedman, 1970; Barnea & Rubin, 2010).

Finally, the negative interaction between ESG disclosure and performance—especially under high intensity—suggests that excessive ESG alignment may backfire, consistent with the TMGT effect. Disclosure without substantive action can be perceived as symbolic, eroding trust and value (Cho et al., 2015; Fatemi et al., 2018). These patterns reveal the presence of a

curvilinear threshold in ESG–financial performance relationships, aligned with the Too-Much-of-a-Good-Thing (TMGT) effect. While ESG engagement enhances firm value up to a point, excessive or poorly aligned efforts—especially when both disclosure and performance are high but lack coherence—may generate investor skepticism and reputational risk, ultimately diminishing financial returns. In BRICS economies, where enforcement is weak, firms often benefit more from signaling ESG commitment than from delivering tangible outcomes, exposing them to reputational risk (Lyon & Montgomery, 2015; Kim et al., 2014).

Additionally, the analysis uncovers a "gray zone" where firms exhibit intermediate levels of both ESG performance and disclosure but fail to generate significant financial outcomes. In this zone, companies lack the reputational credibility of high-disclosure leaders and the operational excellence of high-performance firms—resulting in the least favorable financial profiles. This finding highlights the strategic risk of ESG ambiguity and reinforces the importance of clear alignment between disclosure quality and substantive sustainability practices (Delmas & Burbano, 2011).

Notably, the best financial outcomes across BRICS firms arise not from full ESG commitment but from a strategic balance—moderate ESG performance coupled with high-quality disclosure. This configuration appears to offer the optimal trade-off: signaling credibility and long-term orientation to investors while avoiding excessive ESG expenditures that may dilute operational efficiency. In institutional environments marked by regulatory gaps and cost sensitivity, such as BRICS economies, investors seem to reward transparency more than substance, especially when firms signal intent and risk management capacity through structured ESG communication. This finding reinforces the role of disclosure as a symbolic asset that can generate market-based benefits even when internal ESG implementation is partial or evolving (Dhaliwal et al., 2011; Mio et al., 2020).

Moreover, the findings highlight a systematic divergence between market-based indicators (TSR, Tobin's Q) and accounting-based metrics (ROA, ROE). While profitability measures are tightly linked to operational efficiency and cost discipline—making them more sensitive to the internal implementation of ESG practices—market-based indicators respond more readily to the signaling effect of ESG disclosure. This disparity suggests that investors may reward firms for transparency and future-oriented ESG commitments, even in the absence of immediate productivity gains. Such a pattern is consistent with signaling and legitimacy theories, particularly in contexts marked by institutional fragility and information asymmetry, where stakeholders rely more on disclosed narratives than on verifiable operational outcomes (Spence, 1973; Connelly et al., 2011; Fatemi et al., 2018). This divergence reinforces the dual

nature of ESG value creation—operational versus symbolic—and underscores the need for alignment between ESG communication and strategic execution to sustain both internal and market-based financial performance.

3.6.2 Contributions to Theory

This study offers several key theoretical contributions to the Environmental, Social, and Governance (ESG) and corporate sustainability literature, particularly within the context of emerging markets.

First, it repositions Total Shareholder Return (TSR) as a central metric in the ESG–financial performance debate, despite its underrepresentation in the sustainable finance literature (Cardillo & Basso, 2025). While most studies privilege accounting-based indicators (e.g., ROA, ROE) or market valuation proxies (e.g., Tobin's Q), TSR captures both capital gains and dividends, reflecting comprehensive investor returns. In BRICS countries—characterized by volatility and institutional heterogeneity—TSR serves as a forward-looking proxy for market confidence and is especially relevant in assessing the financial credibility of ESG strategies.

Second, by distinguishing between holistic and selective ESG approaches, the study builds upon the conceptual framework of Hahn and Kühnen (2013) and empirically extends Lueg and Pesheva's (2021) typology. By contrasting emerging BRICS markets with evidence from developed economies, such as the Nordic countries, the findings demonstrate that fragmented engagement with environmental, social, or governance issues—rather than integrated ESG adoption—tends to erode profitability and TSR. This supports stakeholder theory's emphasis on coherence and depth in sustainability practices as key drivers of stakeholder trust and long-term value creation.

Third, the study empirically confirms the Too-Much-of-a-Good-Thing (TMGT) effect in the ESG-finance nexus. It identifies curvilinear relationships in which moderate levels of ESG disclosure and performance enhance financial outcomes, but excessive engagement diminishes or reverses these gains. This insight deepens discussions within agency and signaling theories, illustrating that beyond an optimal threshold, sustainability signaling may generate investor skepticism, perceptions of inefficiency, and reduced valuation credibility.

Fourth, the findings address a paradox observed in BRICS economies: while holistic ESG disclosure positively influences financial performance, holistic ESG performance does not. This divergence is especially pronounced in institutional environments marked by weak enforcement, fragmented regulation, and high information asymmetry (Jamali & Karam, 2018; Ioannou & Serafeim, 2012). In such settings, disclosure often serves as a strategic signal rather

than a reflection of substantive sustainability implementation (Amel-Zadeh & Serafeim, 2018). As suggested by signaling and legitimacy theories, firms may leverage standardized ESG communication to build investor trust and access capital, even when operational practices remain misaligned (Cho et al., 2015).

This behavior resonates with greenwashing literature, which emphasizes the growing disconnect between ESG reporting and execution in markets with low regulatory pressure (Delmas & Burbano, 2011; Lyon & Montgomery, 2015). Consequently, in BRICS countries, the financial value of ESG appears more tightly linked to firms' ability to project commitment through disclosure than to the actual implementation of sustainability practices (Eccles et al., 2014; Fatemi et al., 2018).

Similar findings have been reported in other emerging economies, such as Jordan, Mexico, and Indonesia, reinforcing the notion that signaling via disclosure can temporarily substitute for substantive performance (Dempere et al., 2024; Al Amosh & Khatib, 2023a; Syarkani et al., 2024).

This interpretation aligns with institutional theory, which highlights how informal governance mechanisms, institutional voids, and legitimacy pressures shape ESG behavior in emerging markets (North, 1990; Khanna & Palepu, 2010; Marquis & Raynard, 2015). Collectively, these contributions advance ESG theory by emphasizing that sustainability strategies are not universally beneficial, but contingent upon strategic alignment, institutional context, and stakeholder perception. By adopting a multi-theoretical lens—encompassing stakeholder, agency, legitimacy, signaling, and institutional theories—this study offers an integrative framework to explain the complex, non-linear, and context-dependent financial impacts of ESG practices. This perspective is particularly relevant in emerging economies, where institutional voids often amplify the gap between ESG communication and execution.

Overall, these contributions refine our understanding of how ESG strategies operate across institutional contexts, enhance theory on ESG value creation thresholds, and underscore the importance of integrated, credible, and performance-driven approaches to sustainability.

3.6.3 Contributions to Practice

This study provides several practical implications for corporate leaders, investors, and policymakers engaged in ESG integration within emerging markets, particularly in BRICS economies.

First, the findings highlight the financial advantages of adopting a holistic ESG disclosure strategy. Firms that integrate environmental, social, and governance principles into

their core strategy—rather than selectively emphasizing individual dimensions—tend to achieve superior outcomes in TSR, ROA, and ROE. This underscores the need for strategic alignment between ESG initiatives and operational objectives, encouraging executives to move beyond compliance or symbolic reporting toward embedding sustainability within decision-making and performance management systems.

Second, the results caution against excessive or unbalanced ESG disclosure. While transparency can enhance investor confidence, disclosure that is not matched by substantive ESG performance may elicit skepticism (Dhaliwal et al., 2011; Amel-Zadeh & Serafeim, 2018) and expose firms to greenwashing allegations (Delmas & Burbano, 2011; Cho et al., 2015). Particularly concerning is the "gray zone" of moderate ESG engagement, which neither delivers strong operational outcomes nor generates sufficient signaling credibility. In such cases, disclosure may appear symbolic or misaligned with actual practices, ultimately eroding investor trust. This dynamic is amplified in emerging markets where regulatory oversight is weaker. Firms must ensure that ESG communication is credible, measurable, and reflective of real performance, especially in industries facing heightened social or environmental scrutiny.

Third, by emphasizing Total Shareholder Return (TSR) as a central metric, the study introduces a forward-looking benchmark for executives and investors. TSR captures long-term value creation more comprehensively than traditional accounting ratios, as it reflects both capital gains and dividends. Managers should evaluate how ESG strategies influence not only internal operational indicators but also shareholder expectations and market-based returns, particularly within volatile and capital-constrained environments like BRICS.

Fourth, the evidence identifies governance as the most consistently value-generating ESG dimension. Enhanced governance practices—including board independence, ethical leadership, and transparency—strengthen internal accountability mechanisms and reduce investor risk. In contexts marked by institutional fragility, robust corporate governance can compensate for weak external enforcement, functioning as a trust-building mechanism for capital providers and other stakeholders.

Fifth, the findings offer clear guidance for public policy and regulatory development. Policymakers in BRICS economies should promote ESG frameworks that emphasize not only disclosure but also performance integrity. Regulatory mechanisms that include standardization, third-party verification, and enforcement against symbolic reporting can reduce opportunistic behavior and enhance the credibility and legitimacy of sustainability initiatives. These institutional safeguards are essential to attract foreign capital and improve firms' access to global ESG-aligned financing mechanisms.

In sum, the study supports a strategic, credible, and performance-oriented ESG posture. Firms in emerging markets must move beyond ESG as a reputational tool and treat it as a mechanism for sustainable value creation—balancing stakeholder expectations, investor trust, and operational performance in complex institutional settings. Executives should therefore monitor not only the volume but also the coherence and perceived authenticity of their ESG initiatives, ensuring that disclosure and performance evolve in tandem to avoid counterproductive outcomes and maximize long-term value.

3.6.4 Research Limitations and Future Research Agenda

While this study offers relevant insights into ESG dynamics in emerging markets, several limitations open promising avenues for future research.

First, the analysis is limited to publicly traded firms in BRICS countries, with Russia excluded due to data unavailability following international sanctions. Future research could extend the temporal scope and include non-listed or medium-sized enterprises, providing a more comprehensive understanding of ESG practices across varying organizational profiles and ownership structures.

Second, although two-stage least squares (2SLS) panel regressions were employed to mitigate endogeneity concerns, the analysis remains subject to the inherent limitations of ESG data quality. Bloomberg ESG scores, despite their widespread use, may not fully capture qualitative aspects of sustainability practices or account for the reporting heterogeneity across sectors and jurisdictions. Future studies should consider alternative or complementary data sources—such as MSCI, Refinitiv, or firm-level ESG disclosures—and adopt mixed-method approaches that integrate quantitative rigor with qualitative insights.

Third, while this study controls for firm and year fixed effects, it does not explicitly account for the impact of the COVID-19 pandemic on ESG–performance relationships. Future research could apply a difference-in-differences (DiD) design to isolate the causal effects of exogenous shocks and investigate the role of ESG engagement in enhancing corporate resilience and recovery capacity under crisis conditions.

Together, these avenues provide meaningful opportunities to deepen theoretical contributions and expand empirical knowledge on ESG strategies in emerging economies. By addressing contextual, methodological, and temporal dimensions, future research can clarify how sustainability practices evolve across diverse institutional environments and contribute to long-term value creation.

Additionally, scholars should explore whether the curvilinear ESG–performance relationships and the existence of gray zones identified in BRICS are also observable in other institutional contexts. Comparative studies across developed and emerging markets would help validate the generalizability of the Too-Much-of-a-Good-Thing (TMGT) effect, enhance understanding of strategic ESG alignment, and refine theoretical models on the optimal scope and depth of sustainability engagement.

Moreover, future research should consider the influence of macroeconomic conditions—such as GDP growth, inflation rates, and country risk—on the ESG–financial performance relationship. Including these contextual variables would help disentangle firm-level effects from broader systemic forces. It is also essential to explore the concept of ESG sovereignty, which reflects how countries internalize sustainability frameworks according to their institutional and developmental trajectories rather than adhering passively to global standards. Finally, expanding the scope of comparative studies to include emerging economies outside the BRICS—such as Mexico, Indonesia, or Turkey—would enrich the external validity of ESG–performance models and uncover region-specific dynamics in ESG value creation.

4 FINAL CONSIDERATIONS

This thesis investigated the complex and context-dependent relationship between Environmental, Social, and Governance (ESG) and Corporate Social Responsibility (CSR) initiatives and corporate financial performance (CFP). By combining a systematic bibliometric review with an empirical analysis of firms in emerging markets, particularly within the BRICS countries, the research offers both theoretical insights and practical implications for sustainability-driven financial strategies.

The first article conducted a comprehensive bibliometric and systematic review of 108 peer-reviewed studies published between 2019 and 2023, identifying key moderating variables—such as governance structures, cultural context, technological readiness, and market maturity—that significantly shape the ESG/CSR–CFP nexus. The analysis revealed substantial inconsistencies in the literature, largely due to diverse methodological approaches, inconsistent definitions of ESG and financial performance, and a lack of standardization in measurement frameworks. Moreover, several critical moderating variables—such as social norms, innovation capacity, and regional regulatory environments—remain underexplored, limiting the generalizability of current findings. This review emphasized the need for multi-theoretical models and more refined methodological strategies, such as longitudinal studies and qualitative

designs, to deepen understanding of how ESG and CSR initiatives influence financial outcomes.

The second article empirically assessed the financial impact of ESG disclosure and performance in a panel of nearly 3,000 firm-year observations from BRICS economies between 2016 and 2023. Using a Two-Stage Least Squares (2SLS) regression approach to address endogeneity, the study found that integrated ESG disclosure—as opposed to selective or symbolic engagement—was positively associated with profitability (ROA, ROE) and market performance (TSR). However, holistic ESG performance alone did not improve financial outcomes, and in some cases, particularly under conditions of excessive engagement, diminished firm value. These results underscore a critical tension in emerging markets: while sustainability communication (disclosure) serves as an effective signaling mechanism to investors, its financial benefits often exceed those of substantive operational implementation.

From a theoretical standpoint, the thesis advances the ESG-finance literature by emphasizing the role of moderating variables and introducing Total Shareholder Return (TSR) as a central, yet underutilized, metric in the evaluation of ESG strategies. The empirical confirmation of the Too-Much-of-a-Good-Thing (TMGT) effect—wherein moderate ESG engagement enhances value but excessive efforts lead to diminishing returns—adds depth to agency, stakeholder, and signaling theories. Moreover, the study reinforces the relevance of legitimacy theory, especially in institutional environments with weak regulatory enforcement, where ESG disclosure operates more as a reputational signal than a reflection of actual sustainability performance.

In practical terms, this thesis provides actionable insights for corporate leaders, policymakers, and investors operating in or engaging with emerging markets. First, it affirms the importance of strategic ESG disclosure, especially when aligned with governance transparency and long-term business objectives. Second, it warns against symbolic or unbalanced sustainability engagement, which may trigger investor skepticism and reputational risk, particularly in markets sensitive to greenwashing. Third, the identification of governance as the most value-generating ESG dimension suggests that firms should prioritize leadership structures, board independence, and ethical oversight to improve performance credibility. For policymakers, the findings highlight the need to strengthen regulatory frameworks, promote standardized ESG reporting, and incentivize substantive—not merely symbolic—sustainability action.

Nonetheless, the research is subject to several limitations. The systematic review was limited to a five-year publication window and two databases (Web of Science and Scopus),

potentially omitting foundational or regionally focused studies. The empirical analysis excluded non-listed firms and focused primarily on BRICS economies, limiting generalizability to other emerging markets. Moreover, ESG scores were sourced from a single provider (Bloomberg), which may not fully capture the qualitative dimensions or sectoral nuances of sustainability performance. Future studies should consider integrating data from multiple sources (e.g., Refinitiv, MSCI) and adopting mixed-method approaches to better understand the social, cultural, and organizational dynamics of ESG implementation.

In summary, this thesis demonstrates that the financial outcomes of ESG and CSR strategies depend not only on what firms do but also on how they communicate, who they are, and where they operate. The integration of bibliometric review and econometric evidence reveals that ESG value creation is contingent upon both internal factors—like governance quality and strategic alignment—and external conditions—such as institutional robustness and investor expectations. By bridging theoretical gaps and offering empirically grounded recommendations, this research contributes to a more comprehensive and context-sensitive understanding of sustainability in corporate finance and strategic management.

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APPENDIX A – ADDITIONAL STATISTICAL TABLES

Table A1. - Results of the Durbin–Wu–Hausman Endogeneity Test for ESG Disclosure and Performance Models

Model	Y (Dependent)	Endógenou(s)	χ^2(df)	p-value	Endogeneity
1.1	TSR	ESG	12,941	0,0048	Yes
1.2	Tobins_Q	ESG	1,65	0,199	No
1.3	ROA	ESG	0,693	0,4053	No
1.4	ROE	ESG	6,74	0,0094	Yes
1.5	TSR	ESG_E, ESG_S, ESG_G	20,306	0,0001	Yes
1.6	Tobins_Q	ESG_E, ESG_S, ESG_G	6,634	0,0845	Marginal
1.7	ROA	ESG_E, ESG_S, ESG_G	7,011	0,0715	Marginal
1.8	ROE	ESG_E, ESG_S, ESG_G	12,941	0,0048	Yes
2.1	TSR	ESG_Dis	11,257	0,0008	Yes
2.2	Tobin's Q	ESG_Dis	2,415	0,1202	No
2.3	ROA	ESG_Dis	0,982	0,3216	No
2.4	ROE	ESG_Dis	6,404	0,0114	Yes
2.5	TSR	ESG_E_Dis, ESG_S_Dis, ESG_G_Dis	16,326	0,001	Yes
2.6	Tobin's Q	ESG_E_Dis, ESG_S_Dis, ESG_G_Dis	8,124	0,0437	Yes
2.7	ROA	ESG_E_Dis, ESG_S_Dis, ESG_G_Dis	16,255	0,001	Yes
2.8	ROE	ESG_E_Dis, ESG_S_Dis, ESG_G_Dis	25,123	0,0000	Yes

Note. This table reports panel regressions with fixed effects, designed to test reverse causality by assessing whether financial performance influences ESG performance (columns 1–4) and ESG disclosure (columns 5–8). The models include both contemporaneous and one-period lagged values of financial indicators — Total Shareholder Return (TSR), Return on Assets (ROA), Return on Equity (ROE), and Tobin's Q — to capture potential reverse causality effects. All regressions control for firm size (Sales), market capitalization (MKT_Cap), market-to-book ratio (MTB), and exposure to foreign direct investment (FDI), in addition to year and industry fixed effects. Robust standard errors are reported in parentheses. Variance Inflation Factor (VIF) diagnostics were performed and no value exceeded the critical threshold of 5, indicating that multicollinearity is not a concern in these models. Significance levels: $p < 0.10$, $p < 0.05$, $p < 0.01$.

Table A2 - Reverse Causality Regressions: Financial Performance → ESG Performance and ESG Disclosure

VARIABLES	ESG (R.1)	ESG (R.2)	ESG (R.3)	ESG (R.4)	ESG_Dis (R.5)	ESG_Dis (R.6)	ESG_Dis (R.7)	ESG_Dis (R.8)
L.zw_ESG_Dis					0.428***	0.560***	0.427***	0.429***
					(0.0292)	(0.0270)	(0.0299)	(0.0292)
L.zw_ESG	0.449***	0.513***	0.448***	0.449***				

	(0.0236)	(0.0237)	(0.0239)	(0.0237)				
zw_Tobins_Q		-0.0290				-0.0572		
		(0.0355)				(0.0539)		
L.zw_Tobins_Q		0.0313				0.0432**		
		(0.0231)				(0.0177)		
zw_TSR	-0.03***				-0.0639***			
	(0.0112)				(0.0149)			
L.zw_TSR	0.0160				0.0208			
	(0.0120)				(0.0142)			
zw_ROA			-0.0165				-0.0523***	
			(0.0129)				(0.0144)	
L.zw_ROA			-0.000450				0.0134	
			(0.0155)				(0.0161)	
zw_ROE				-0.0328***				-0.0678***
				(0.0107)				(0.0139)
L.zw_ROE				0.0139				0.0140
				(0.0114)				(0.0125)
zw_Sales	-0.079***	-0.122***	-0.0870***	-0.0850***	-0.0962**	-0.165***	-0.114***	-0.108***
	(0.0301)	(0.0266)	(0.0290)	(0.0288)	(0.0379)	(0.0359)	(0.0355)	(0.0346)
zw_MKT_Cap	0.135***	0.0783*	0.136***	0.134***	0.170***	0.0276	0.172***	0.168***
	(0.0381)	(0.040)	(0.0382)	(0.0381)	(0.0403)	(0.0386)	(0.0401)	(0.0405)
zw_MTB	0.0264	0.0459	0.0229	0.0277	0.0675*	0.0926	0.0616*	0.0698*
	(0.0354)	(0.051)	(0.0355)	(0.0353)	(0.0357)	(0.0986)	(0.0350)	(0.0357)
zw_FDI	0.0783*	0.0784*	0.0808*	0.0792*	0.172***	0.205***	0.176***	0.175***
	(0.0446)	(0.047)	(0.0448)	(0.0447)	(0.0476)	(0.0517)	(0.0478)	(0.0473)
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	-0.166***	-0.161***	-0.167***	-0.166***	-0.139***	-0.0795**	-0.139***	-0.137***
	(0.0310)	(0.0338)	(0.0312)	(0.0311)	(0.0291)	(0.0328)	(0.0293)	(0.0291)
Observations	2,608	2,412	2,608	2,608	2,608	2,412	2,608	2,608
R-squared	0.619	0.569	0.619	0.620	0.701	0.651	0.700	0.72
Nº of Firms	377	374	377	377	377	374	377	377

Note. This table reports fixed-effects panel regressions designed to test for potential reverse causality, assessing whether prior financial performance influences subsequent ESG performance (columns 1–4) and ESG disclosure (columns 5–8). The models include both contemporaneous and one-period lagged values of financial indicators—Total Shareholder Return (TSR), Return on Assets (ROA), Return on Equity (ROE), and Tobin's Q—to capture possible feedback effects. All regressions control for firm size (Sales), market capitalization (MKT_Cap), market-to-book ratio (MTB), and foreign direct investment exposure (FDI), along with year and sector fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. Variance Inflation Factor (VIF) diagnostics were performed, and no value exceeded the critical threshold of 5, indicating that multicollinearity is not a concern in these models. The significance levels are denoted as follows: $p < 0.10$, $p < 0.05$, $p < 0.01$.

Table A2 - Results of Fisher-Type Panel Unit Root Tests (ADF) Using Pm Statistic.

Variable	Pm Statistic	Stationary
ESG	13.5437***	Yes
ESG_D	10.6725***	Yes
ESG_E	19.3761***	Yes
ESG_S	8.8934***	Yes
ESG_G	23.3369***	Yes
ESG_E_Dis	14.3974***	Yes
ESG_S_Dis	10.6718***	Yes
ESG_G_Dis	10.7791***	Yes
TSR	38.8606***	Yes
Tobin's Q	24.1726***	Yes
ROA	23.8020***	Yes
ROE	34.2454***	Yes

Note: This table reports the Pm statistic from Fisher-type Augmented Dickey-Fuller unit root tests for all key variables used in the study. The null hypothesis assumes the presence of a unit root in all panels (non-stationarity). A Pm statistic with a p-value < 0.01 indicates rejection of the null hypothesis at the 1% level, supporting panel-level stationarity for the variable.